



2026
QHackathon

Quantum Joa

DYNAMIC CIRCUIT

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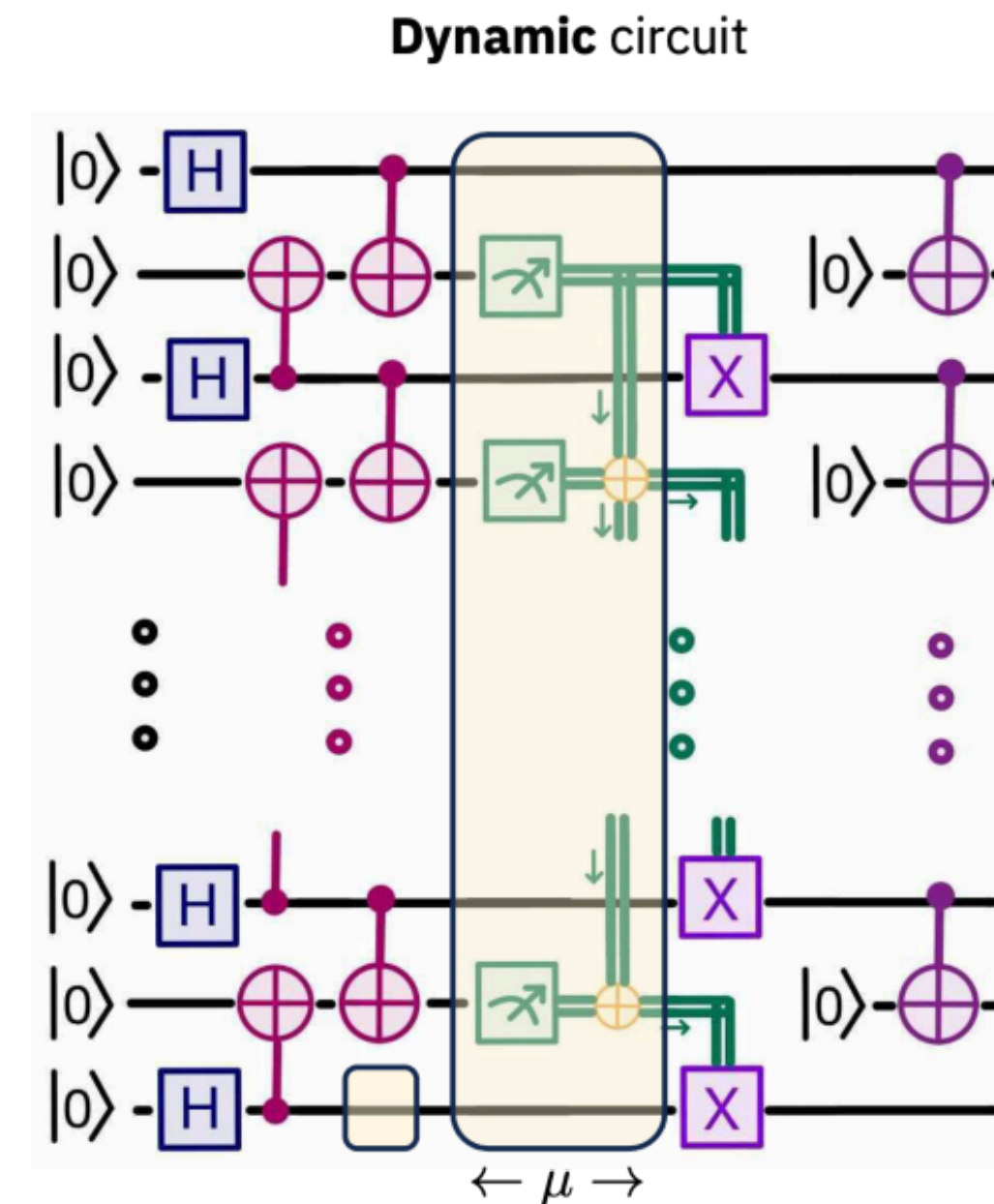
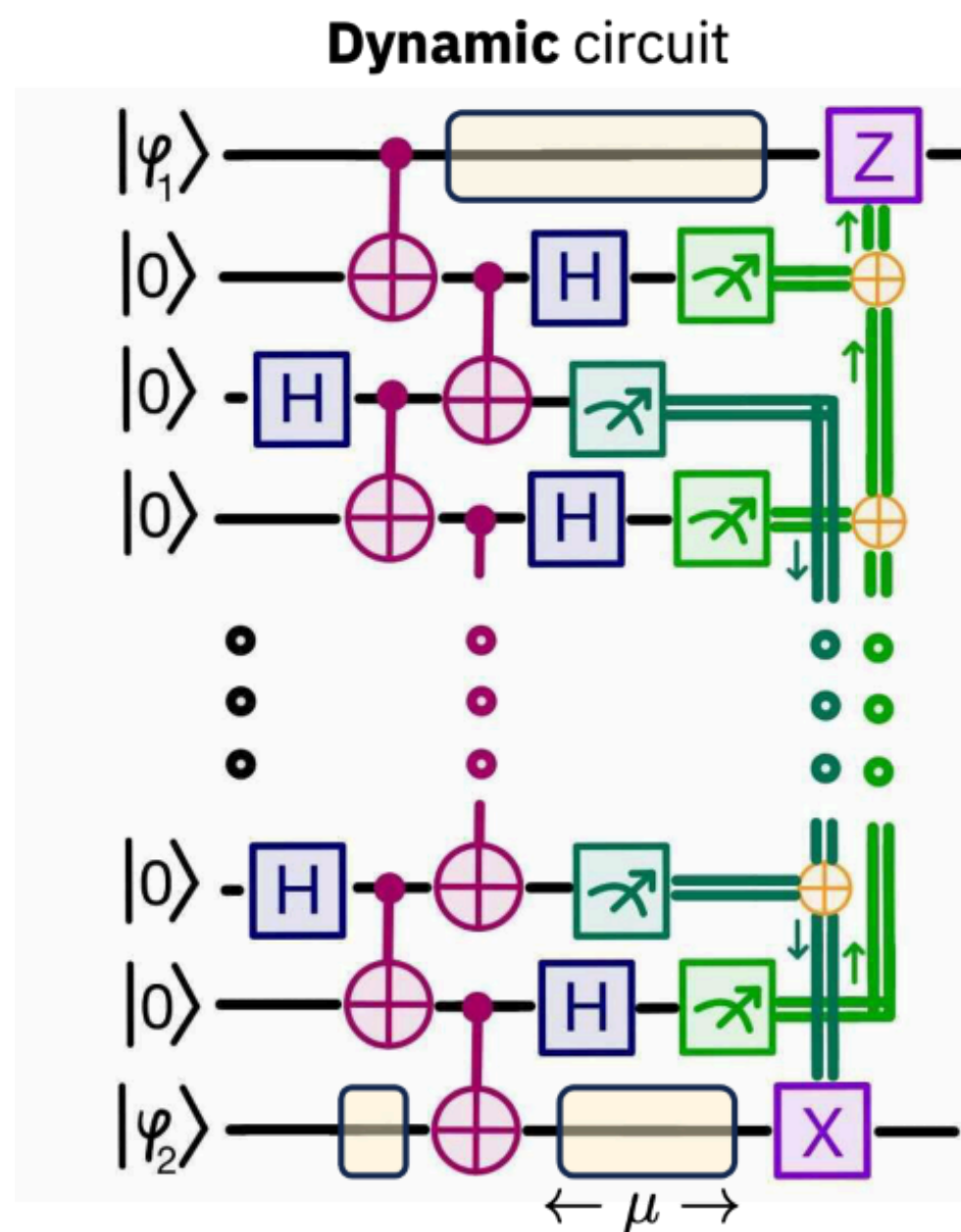
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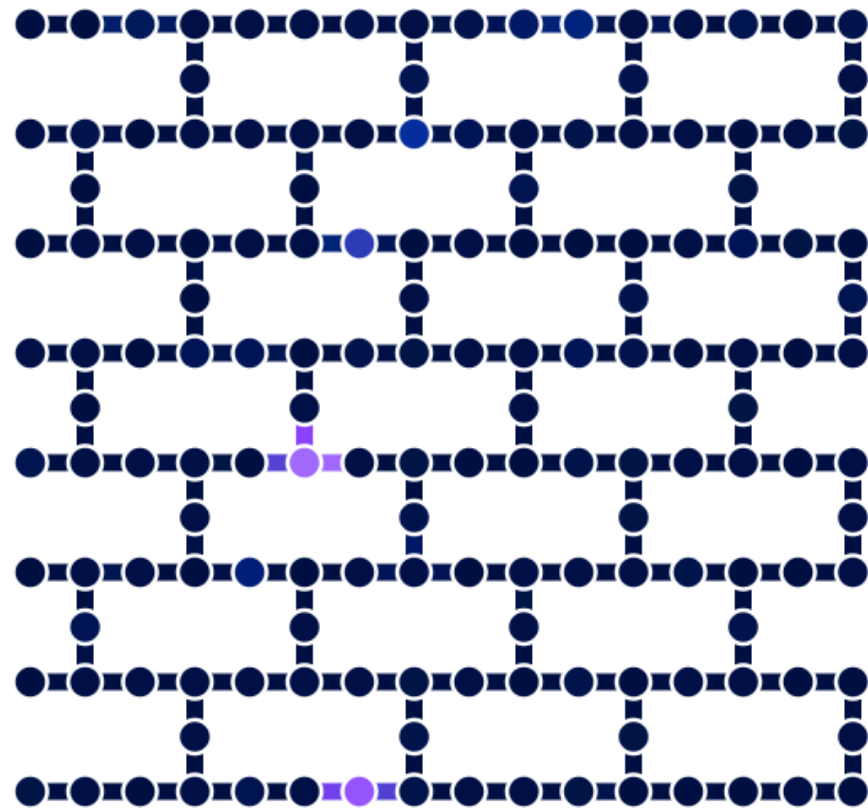
Introduction

Dynamic circuits are an optimization technique for reducing circuit depth or improving gate fidelity by using measurement-related circuit equivalence relations.

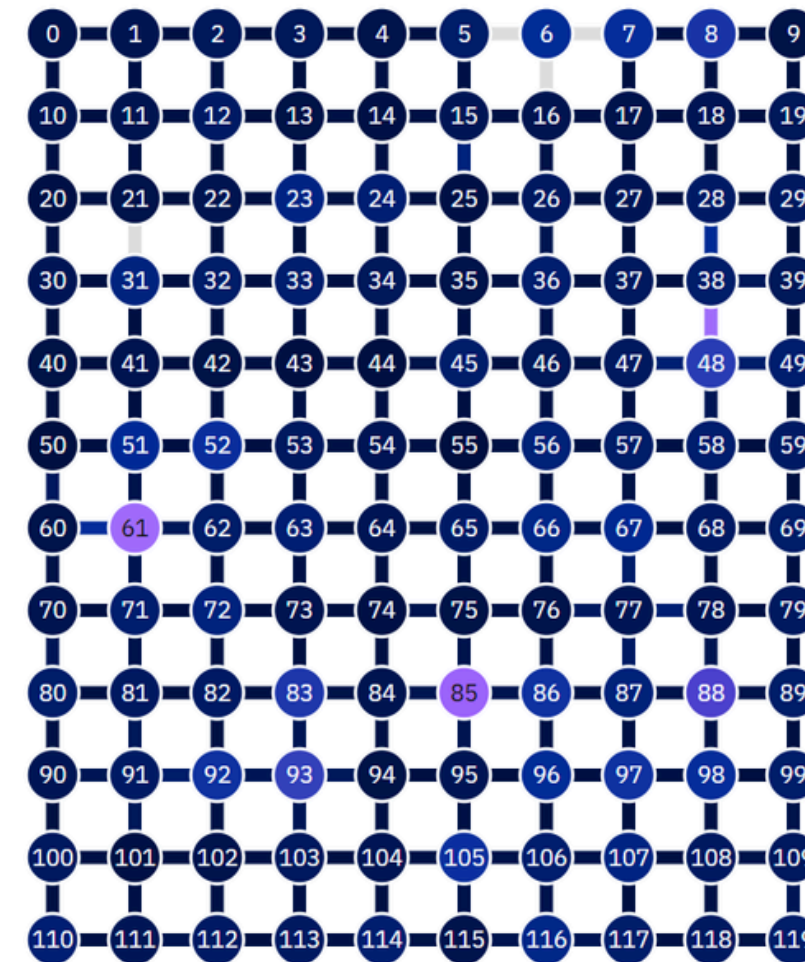


Why do we need Dynamic Circuit?

In superconducting quantum computer, the connectivity of two qubit gate is limited. To operate two qubit gate between far qubits, we should attend SWAP operations.



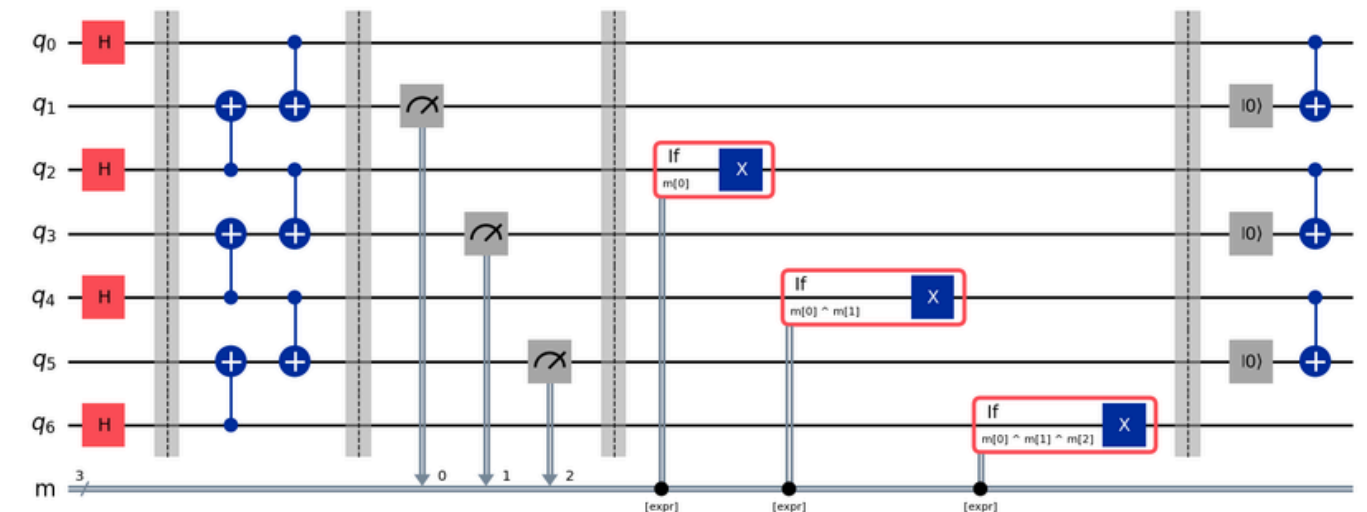
Connectivity of ibm_boston
(Heron r3)



Connectivity of ibm_miami
(Nighthawk r1)

Problem 1

Preparing Long-Range Entanglement with Dynamic Circuits



Core Principle: The Lieb--Robinson bound forces any purely unitary entangling protocol at distance n to have depth $O(n)$; dynamic circuits achieve the same task in $O(1)$ quantum depth via mid-circuit measurement and classical feed-forward.

1.1. Circuit Implementations

- **Task:** Prepare an n -qubit GHZ state on a one-dimensional qubit chain: $|GHZ_n\rangle = \frac{|0\rangle^{\otimes n} + |1\rangle^{\otimes n}}{\sqrt{2}}$.
- **Analysis:** Derive the equivalence between the unitary and dynamic circuits (Appendix A of Ref. [1] may be used as a guide).

1.2. Protocol Variants

Implement two GHZ-preparation circuits on the same 1D chain:

- (i) **Unitary Approach:** Single Hadamard + CNOT ladder; two-qubit depth $O(n)$.
- (ii) **Dynamic Approach:** Constant quantum depth using $\frac{n}{2} - 1$ MCM and classically conditioned Pauli corrections.

Required Metrics to Report:

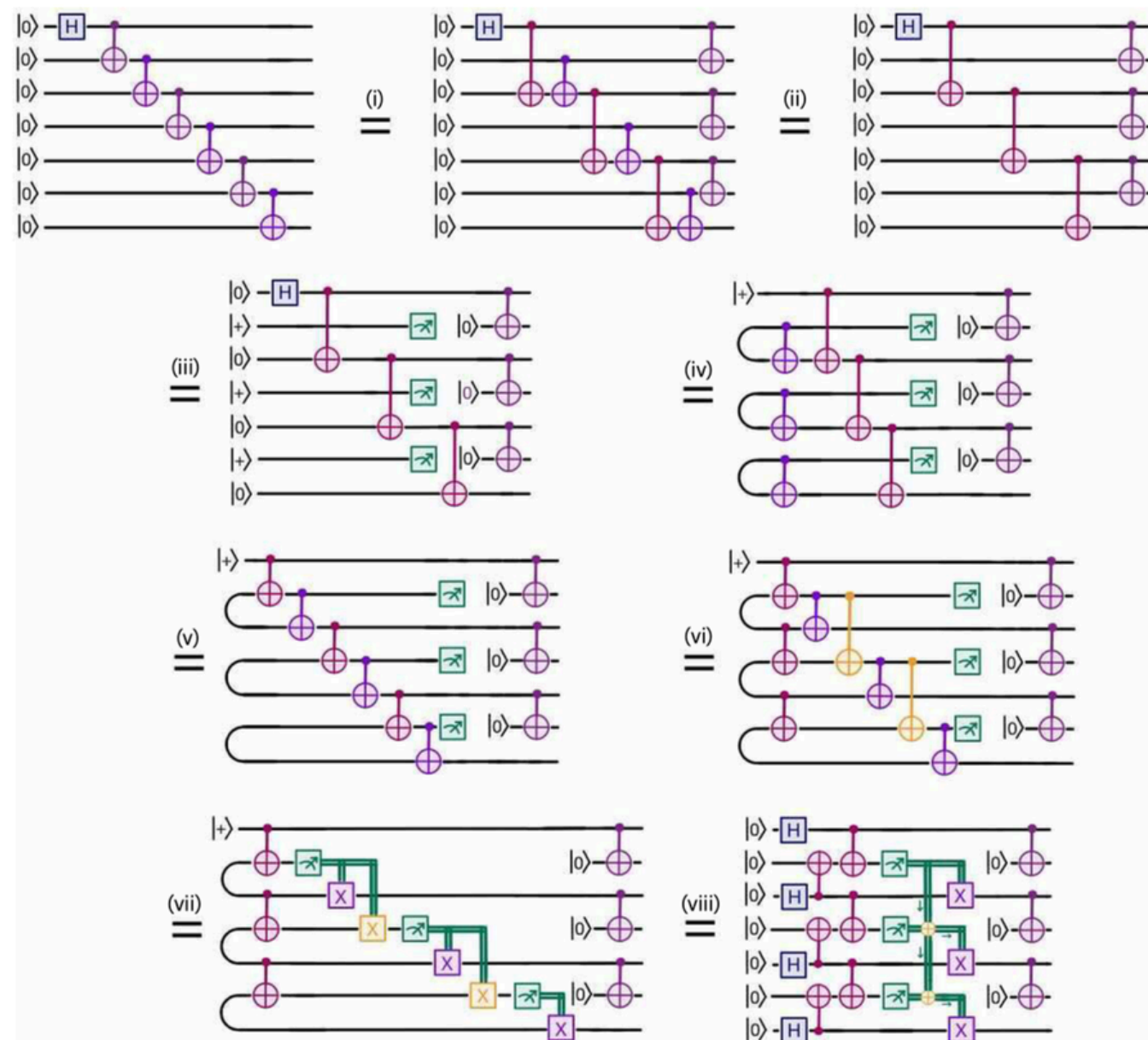
$$\rightarrow 2Q \text{ Depth} \mid N_{\text{CNOT}} \mid t_{\text{idle}} \mid N_{\text{meas}} \mid F_{\text{GHZ}}$$

Problem 1

$$|GHZ\rangle_n = \frac{|0\rangle^{\otimes n} + |1\rangle^{\otimes n}}{\sqrt{2}}$$

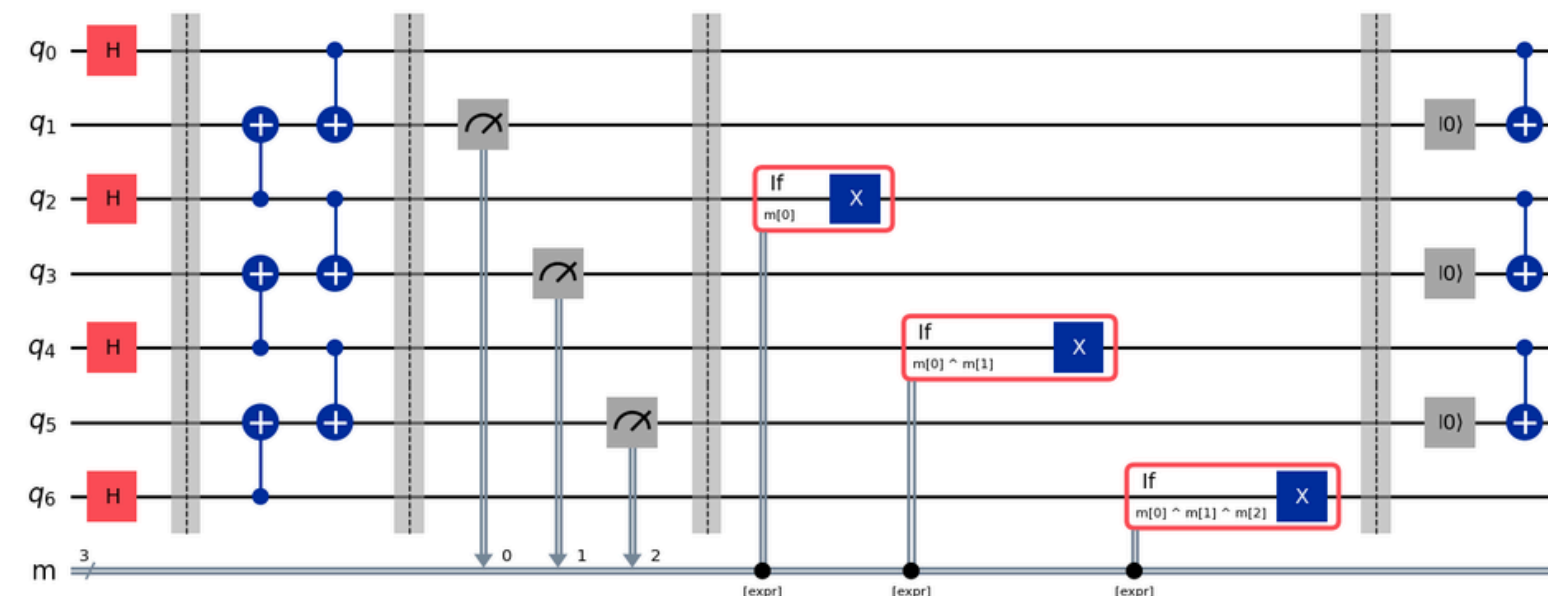
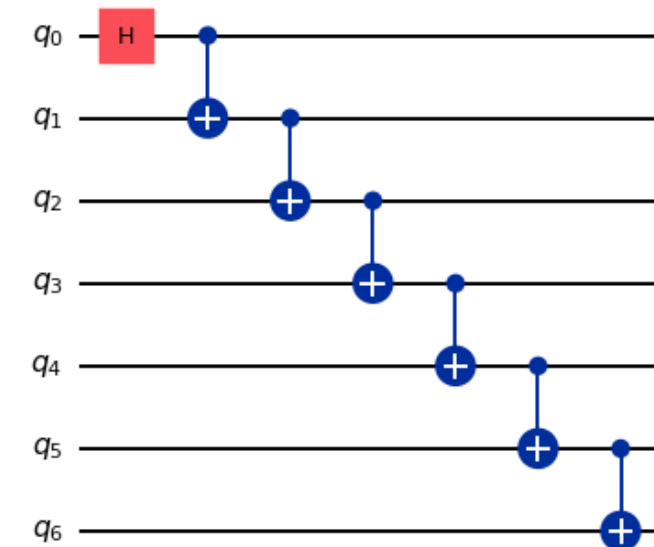
Preparing GHZ State with Dynamic Circuits.

Implement two circuits that prepare an n-qubit GHZ state on a one-dimensional qubit chain:



Unitary implementation

- depth $O(n)$



Dynamic Circuit

- const. depth $O(1)$
- mid-circuit measure

Mathematical Equivalence

Unitary circuit

The initial state is

$$|\psi_0\rangle = |000\rangle = [1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]^T.$$

After H_0 ,

$$|\psi_1^{(U)}\rangle = \frac{|000\rangle + |001\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} [1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]^T.$$

After $CX_{0 \rightarrow 1}$,

$$|\psi_2^{(U)}\rangle = \frac{|000\rangle + |011\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} [1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0]^T.$$

After $CX_{1 \rightarrow 2}$,

$$|\psi_{\text{final}}^{(U)}\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} [1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1]^T.$$

Dynamic circuit

After $H_0 H_2 = H \otimes I \otimes H$,

$$|\psi_1^{(D)}\rangle = \frac{|000\rangle + |001\rangle + |100\rangle + |101\rangle}{2} = \frac{1}{2} [1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0]^T.$$

After $CX_{2 \rightarrow 1}$,

$$|\psi_2^{(D)}\rangle = \frac{|000\rangle + |001\rangle + |110\rangle + |111\rangle}{2} = \frac{1}{2} [1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1]^T.$$

After $CX_{0 \rightarrow 1}$,

$$|\psi_3^{(D)}\rangle = \frac{|000\rangle + |011\rangle + |101\rangle + |110\rangle}{2} = \frac{1}{2} [1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0]^T.$$

Measuring q_1 gives $m = q_0 \oplus q_2$. Each outcome occurs with probability $1/2$:

$$|\psi_{m=0}\rangle = \frac{|000\rangle + |101\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} [1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0]^T,$$

$$|\psi_{m=1}\rangle = \frac{|011\rangle + |110\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} [0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 0]^T.$$

For $m = 1$, the feed-forward correction X_2 gives

$$X_2 |\psi_{m=1}\rangle = \frac{|010\rangle + |111\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} [0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1]^T.$$

Resetting q_1 maps both measurement branches to the same state:

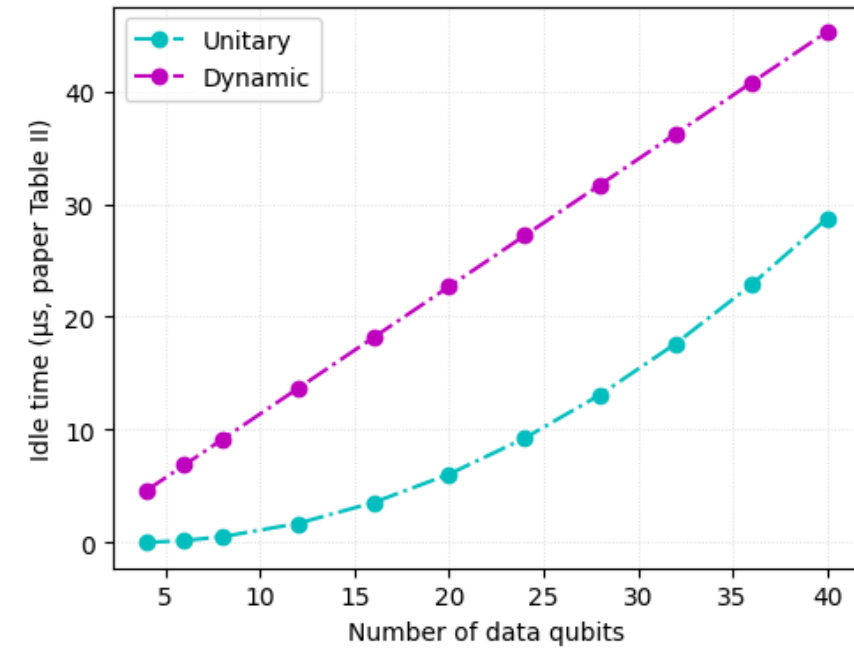
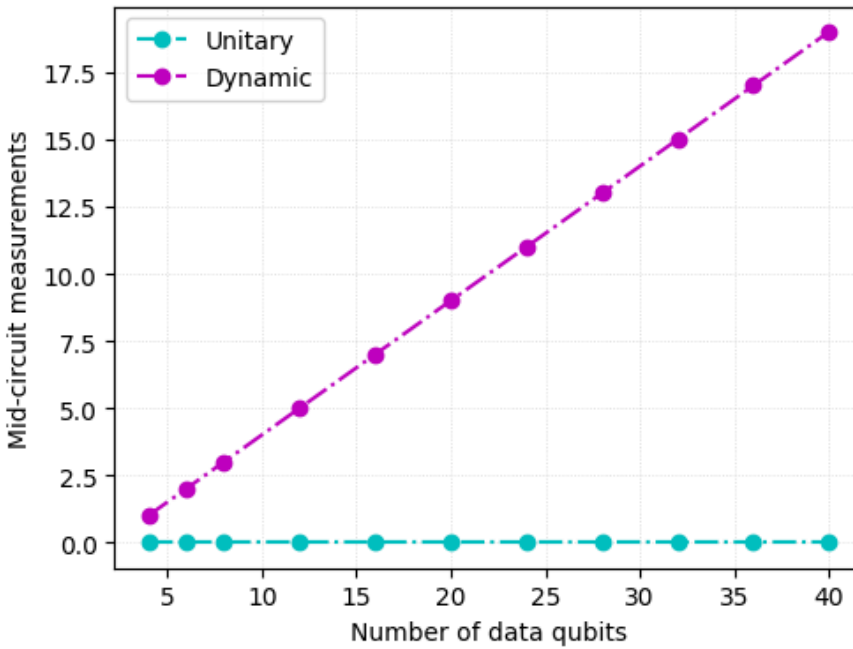
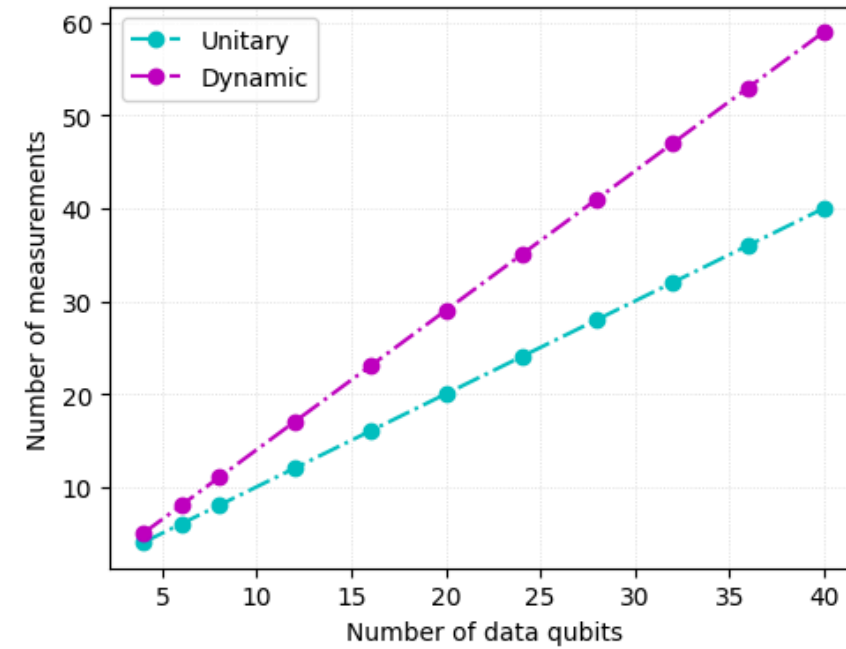
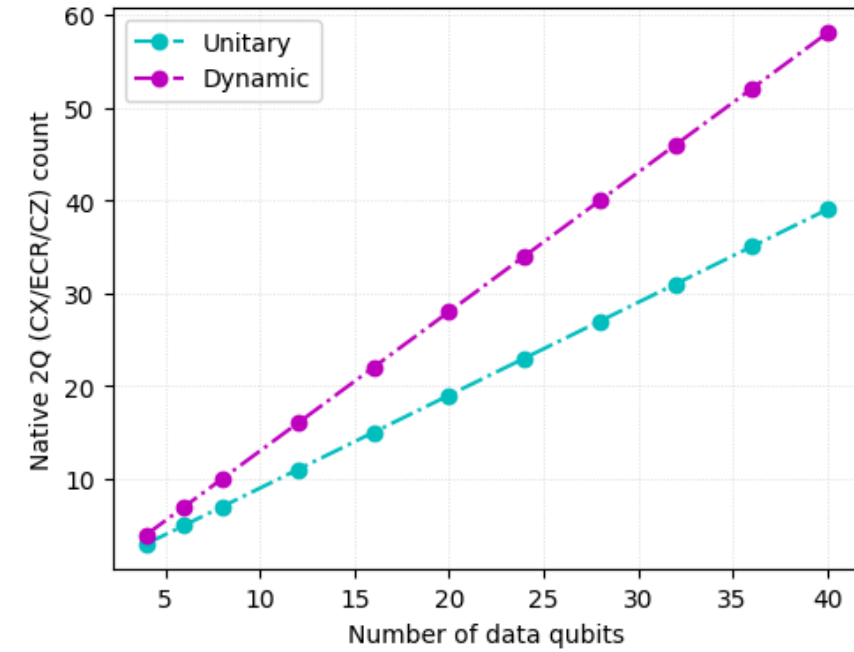
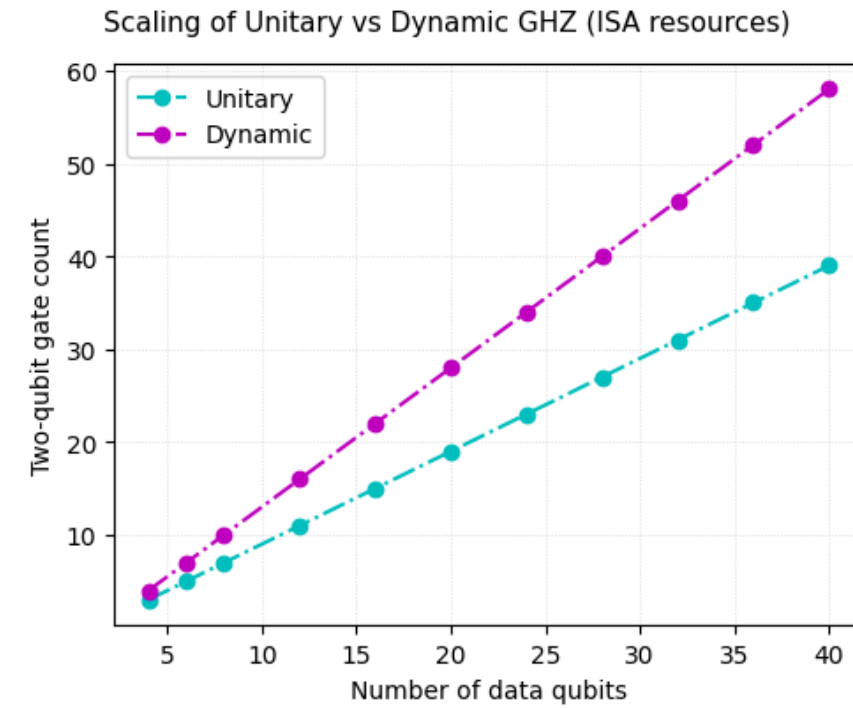
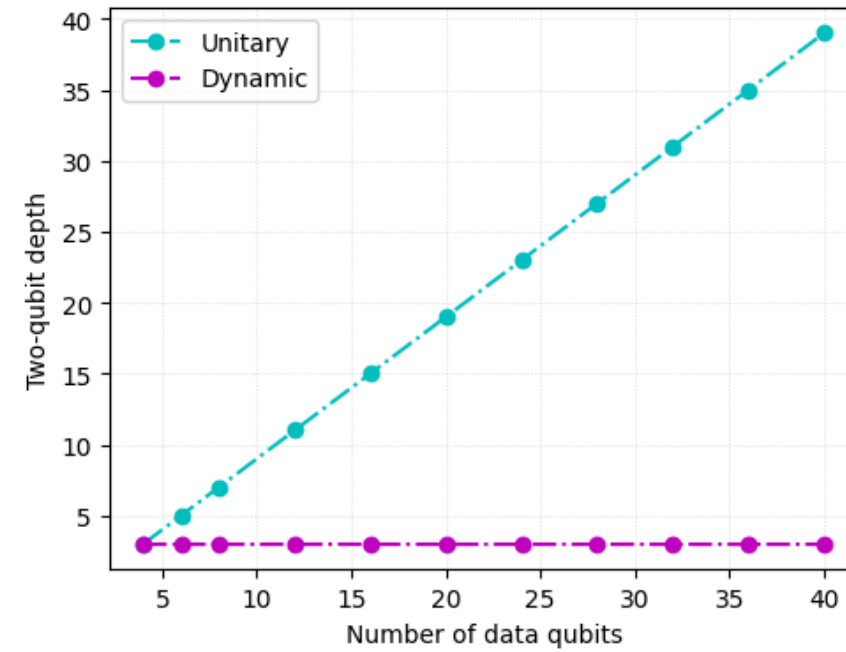
$$|\psi_4^{(D)}\rangle = \frac{|000\rangle + |101\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} [1 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0]^T.$$

Finally, $CX_{0 \rightarrow 1}$ re-entangles the reset qubit:

$$|\psi_{\text{final}}^{(D)}\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} [1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1]^T = |\psi_{\text{final}}^{(U)}\rangle.$$

Resource counts

GHZ Circuit	Idle time error (per CNOT gate time)	CNOTs	Mid-circuit measurements
Unitary	$\frac{n^2}{4} - \frac{3n}{2} + 2$	$n - 1$	0
Post-Processing	1	$\frac{3n}{2} - 2$	$\frac{n}{2} - 1$
Dynamic	$1 + \frac{n}{2}\mu$	$\frac{3n}{2} - 2$	$\frac{n}{2} - 1$



GHZ state and its fidelity

n-qubit GHZ State:

$$|GHZ\rangle_n = \frac{|0\rangle^{\otimes n} + |1\rangle^{\otimes n}}{\sqrt{2}}$$

Definition of Stabilizer

$$S|\psi\rangle = |\psi\rangle.$$

GHZ State Stabilizer:

$$g_0 = X_0 X_1 \cdots X_{n-1},$$

One X stabilizer, n-1 Z stabilizer

$$g_i = Z_{i-1} Z_i \quad (i = 1, \dots, n-1)$$

GHZ Stabilizer group

$$\mathcal{S}_{\text{GHZ}} = \langle g_0, \dots, g_{n-1} \rangle, \quad |\mathcal{S}_{\text{GHZ}}| = 2^n$$

GHZ state and its fidelity

GHZ fidelity: $F_{\text{GHZ}} = \text{Tr}(\rho |\text{GHZ}_n\rangle \langle \text{GHZ}_n|) = \langle \text{GHZ}_n | \rho | \text{GHZ}_n \rangle$

GHZ Projector: $|\text{GHZ}_n\rangle \langle \text{GHZ}_n| = \frac{1}{2^n} \sum_{S \in \mathcal{S}_{\text{GHZ}}} S$

$$F_{\text{GHZ}} = \frac{1}{2^n} \sum_{S \in \mathcal{S}_{\text{GHZ}}} \text{Tr}(\rho S)$$
$$= \frac{1}{2^n} \sum_{S \in \mathcal{S}_{\text{GHZ}}} \langle S \rangle_\rho$$



Monte Carlo
Sampling

$$\tilde{F}_{\text{GHZ}} = \frac{1}{M} \sum_{k=1}^M \langle S_k \rangle_\rho + \mathcal{O}\left(\frac{1}{\sqrt{m}}\right)$$

Get expectation value via
stabilizer measurement !!

Noiseless Simulation

Noiseless Sumulation to Verify that our circuit is correct

n=7 qubits

sampled 32 stabilizers

2048 shots on AerSimulator sampler

```
Unitary GHZ fidelity (n=7): 1.000000  
Dynamic GHZ fidelity (n=7): 1.000000
```

⇒ Our circuit is corrected 😊

Noisy Simulation

Swept over 4-40 qubits on a noisy simulator

Applied M3 measurement mitigation and Pauli twirling.

DD is not used since there is no idle error in the noise model we used.

Noise Model Error Rate:

one_qubit_error=0.001,

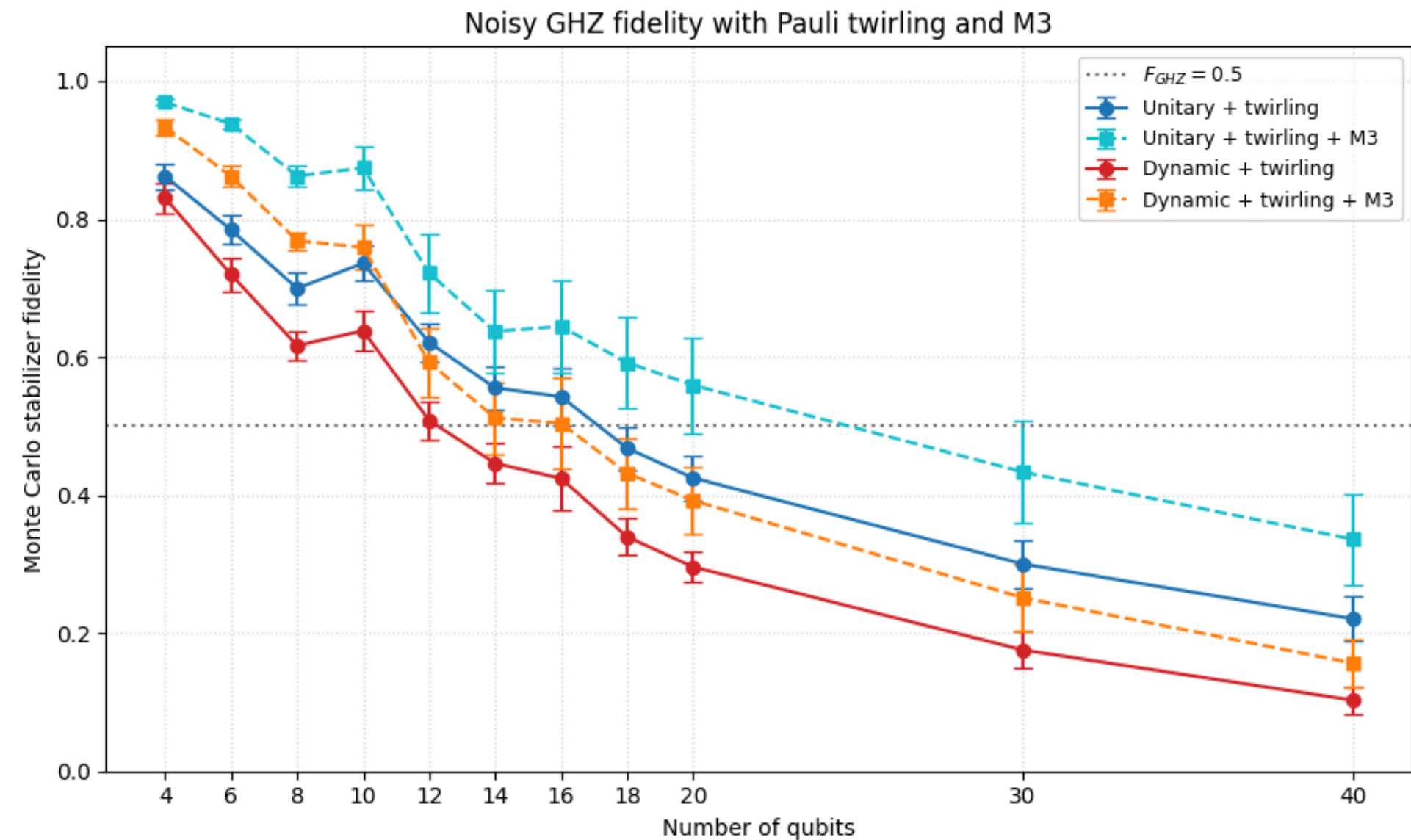
two_qubit_error=0.01,

reset_error=0.01,

readout_error=0.02

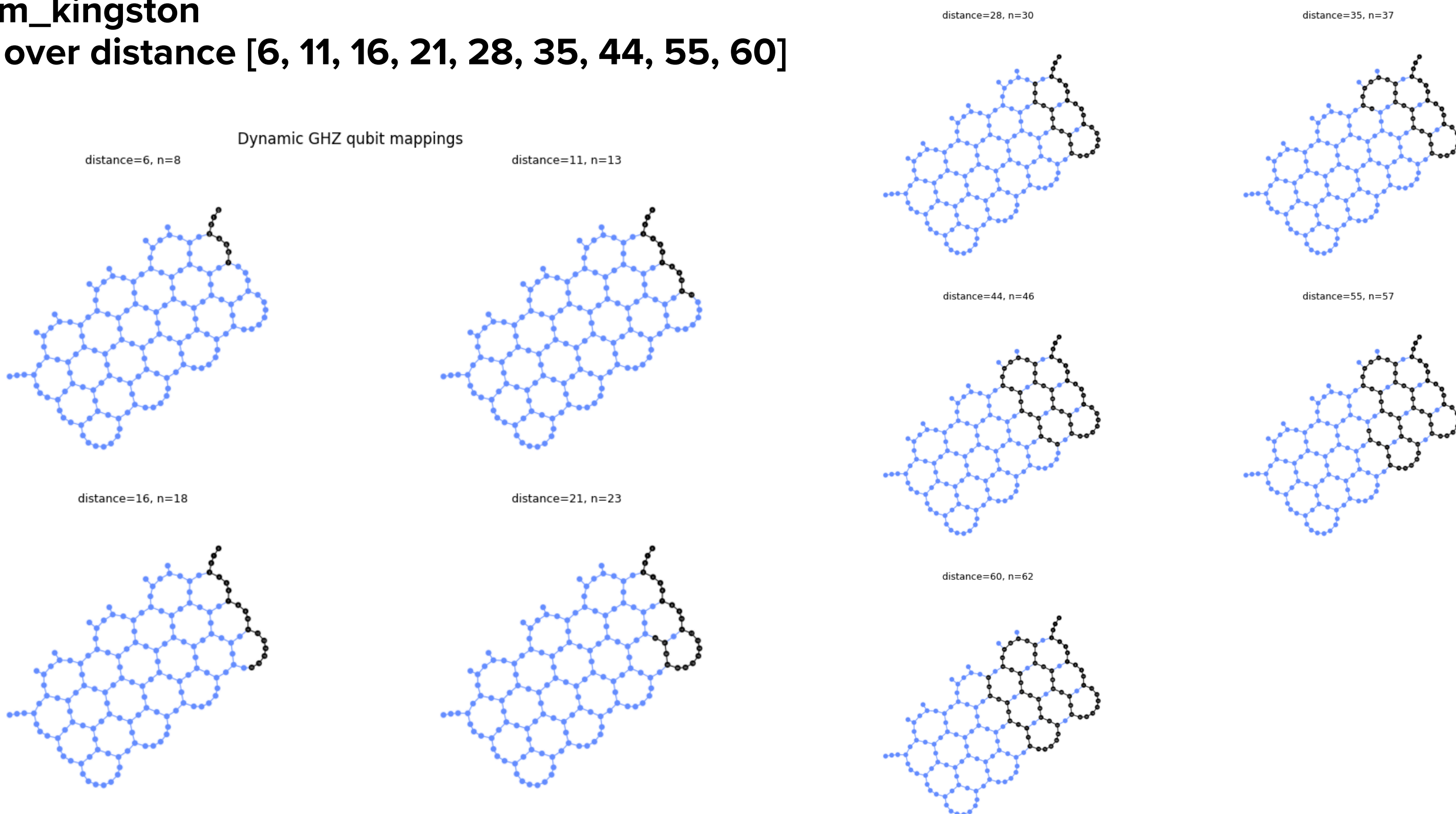
Sampled 12 stabilizers

1024 shots on Sampler



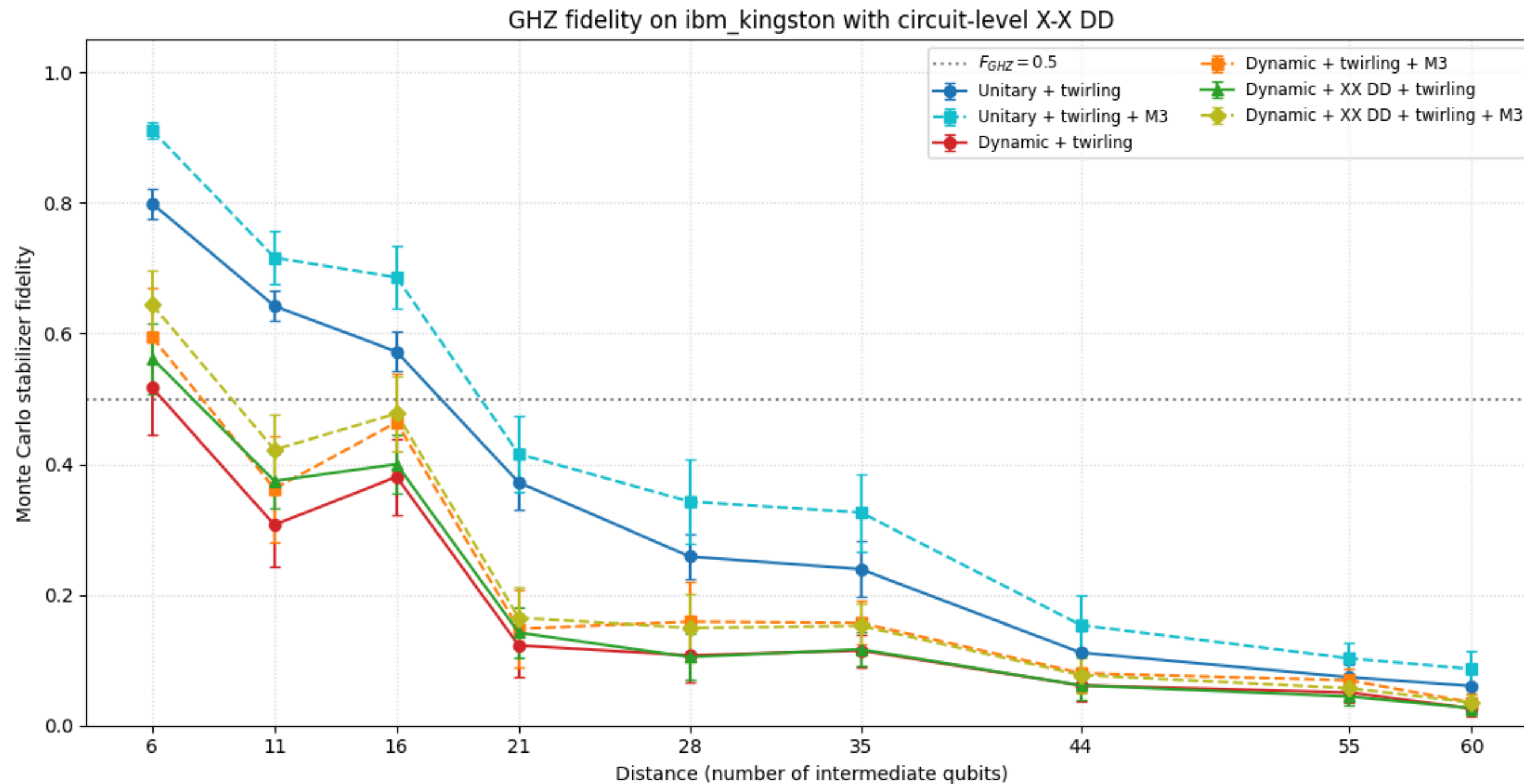
Qubit mapping of real hardware

Ran on ibm_kingston
Swept over distance [6, 11, 16, 21, 28, 35, 44, 55, 60]



Real Hardware execution

Applied M3 measurement mitigation, Pauli twirling, dynamical decoupling

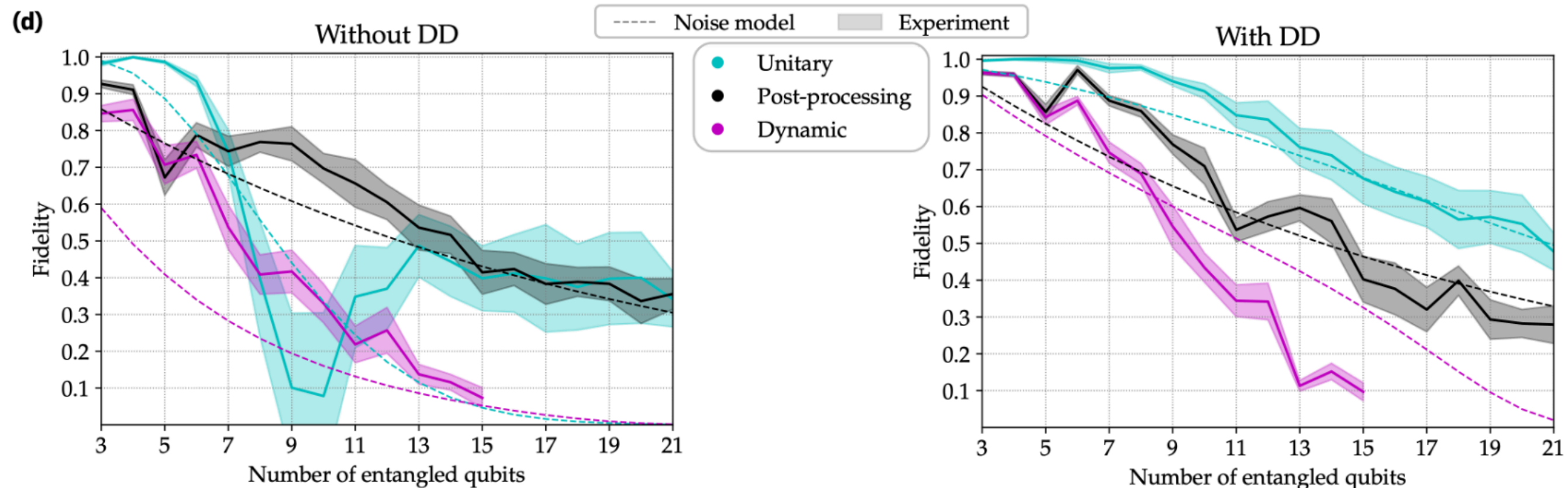


of qubits = distance + 2

Sampled 12 stabilizers
1024 shots on Sampler

Conclusion of Problem 1

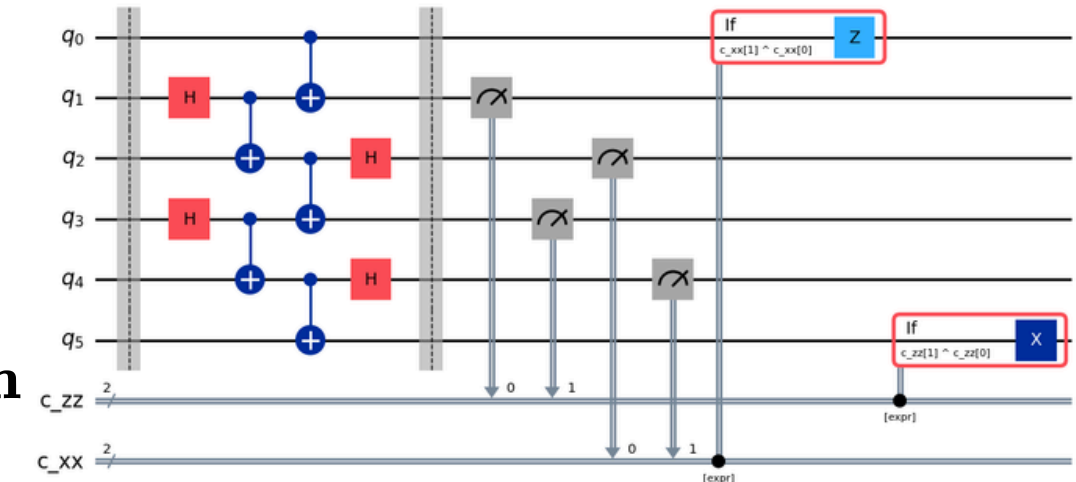
- The fidelity is over 1/2 for up to 18 qubits for unitary and 8 and 18 for dynamic.
- No crossover observed -> consistent with the paper.
("Efficient Long-Range Entanglement Using Dynamic Circuits", E. Baumer, et al.)
- This is because the measurement and CX overhead is larger in dynamic circuit.



Ref: "Efficient Long-Range Entanglement Using Dynamic Circuits", E. Baumer, et al.

Problem 2

Hardware-Aware Circuit Design for Long-Range CNOT Teleportation



Core Principle: The efficiency of a dynamic circuit depends strongly on the underlying chip topology.

2.1. Topology & Mapping

- Task:** Design protocols for two distinct layouts (2D Square vs. IBM Heavy-Hex).
- Analysis:** Explain system qubit vs. "sacrificial" ancilla bus assignment.

2.2. Protocol Variants

Implement a long-range CNOT separated by n intermediate qubits:

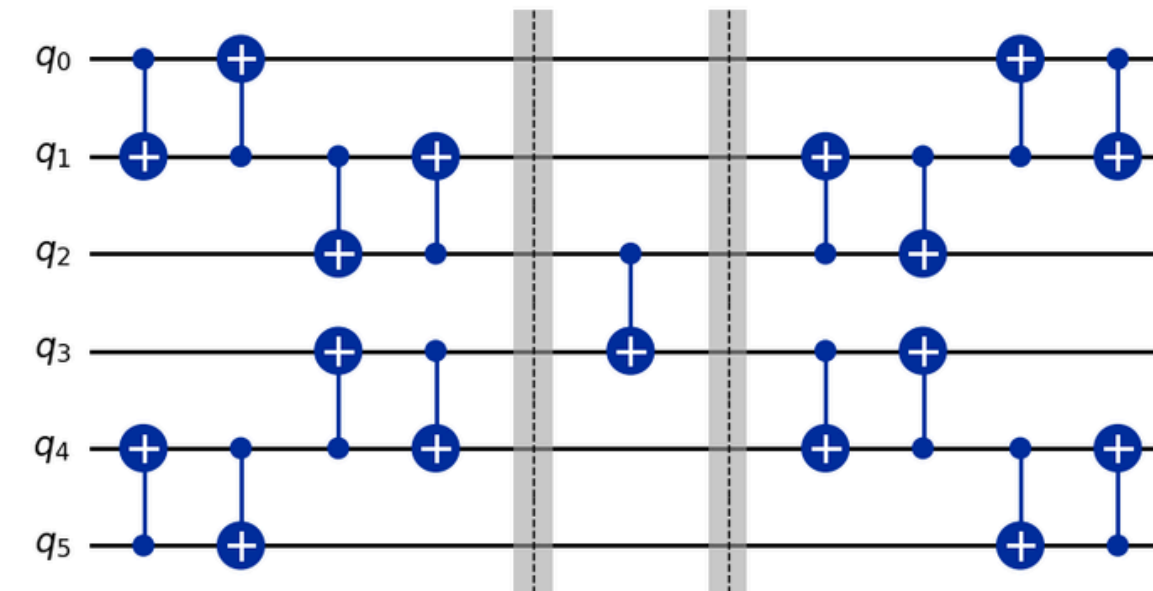
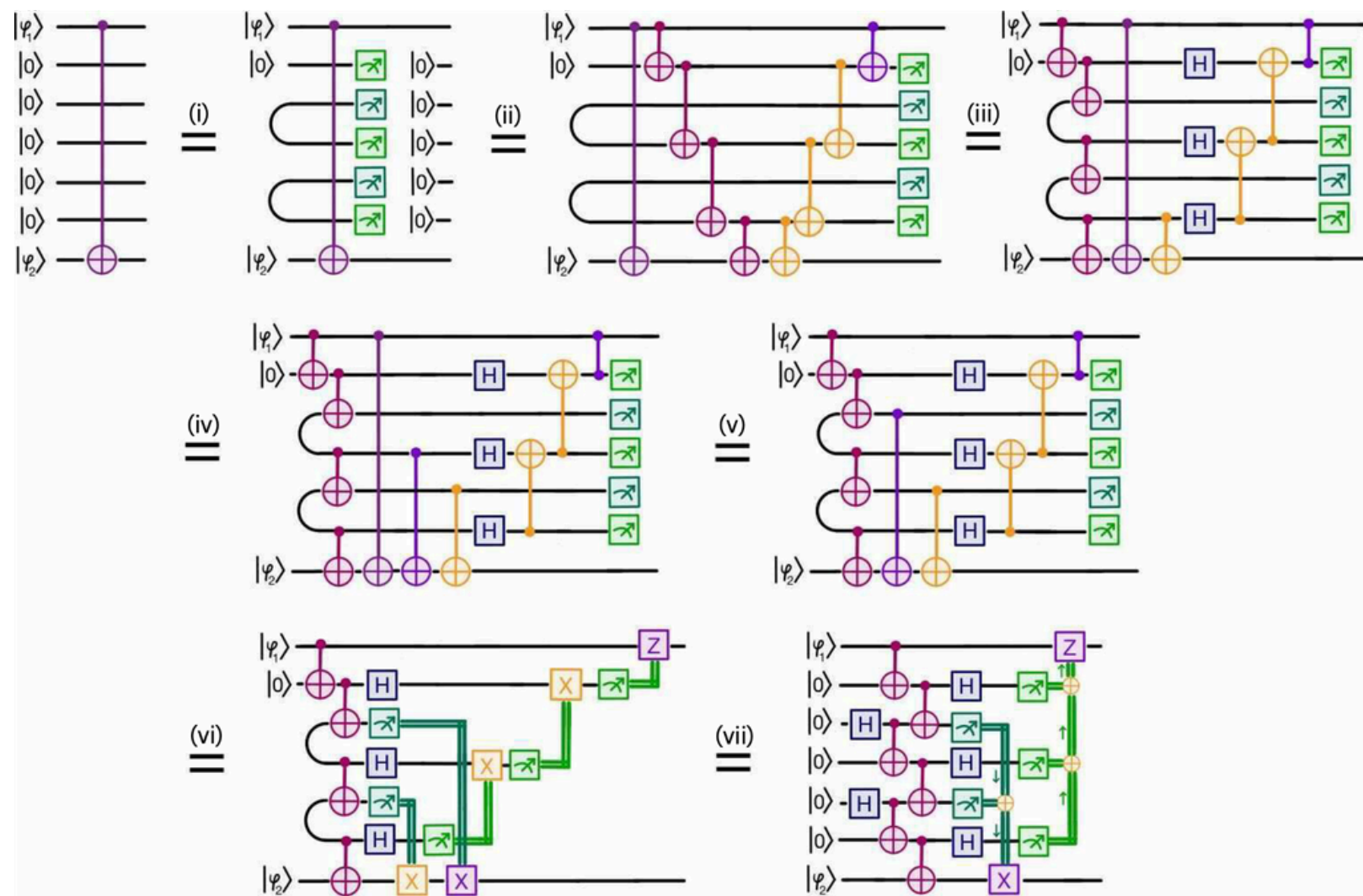
- Unitary Approach:** Ancilla-based SWAP (App. B) & Default transpiler baseline
- Dynamic Approach:** Mid-circuit measurement (MCM) with real-time feed-forward

Required Metrics to Report:

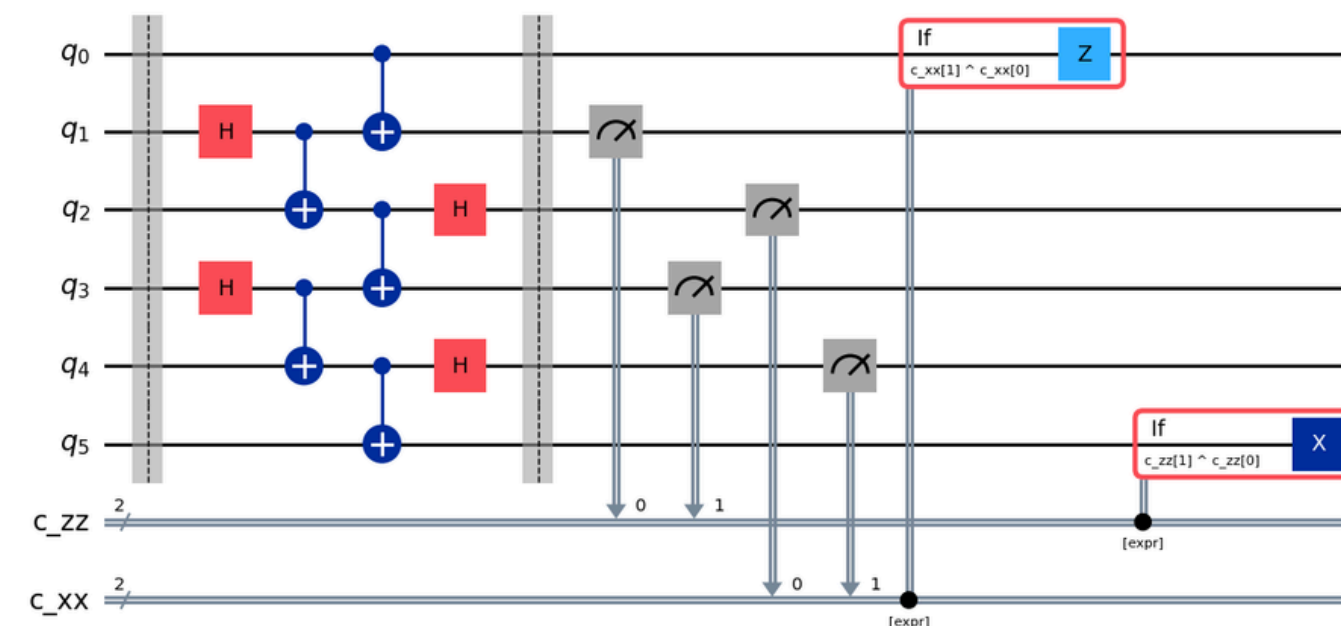
→ Gate Counts (N_{CNOT}) | Circuit Depth (2Q depth) | MCM Counts (N_{meas})

Problem 2

Hardware-Aware Circuit Design for Long-Range CNOT Gate Teleportation



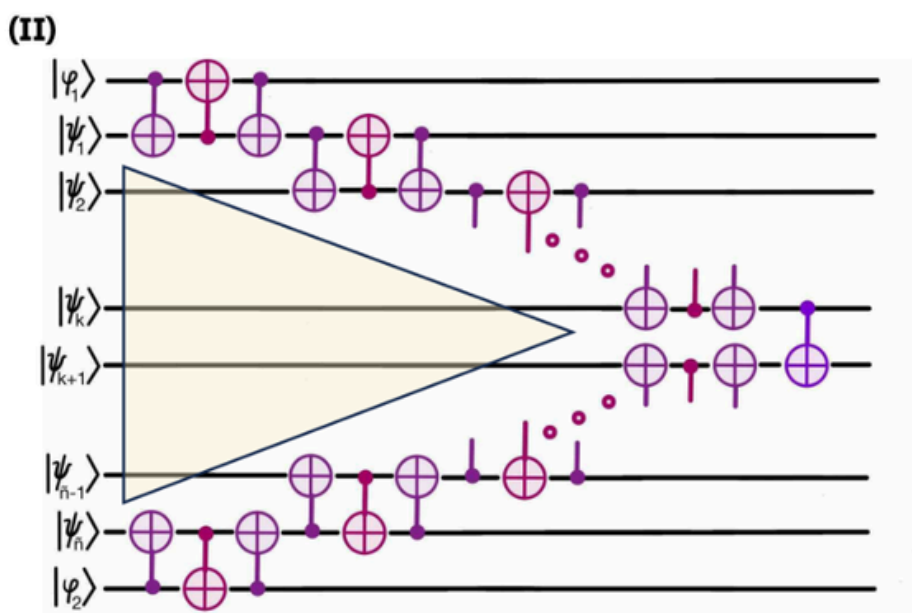
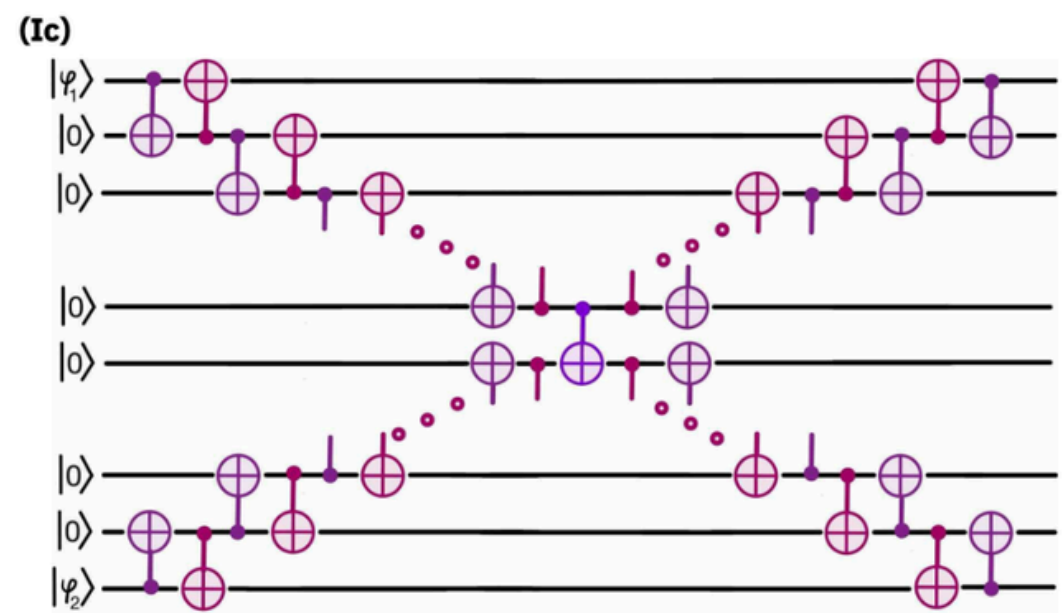
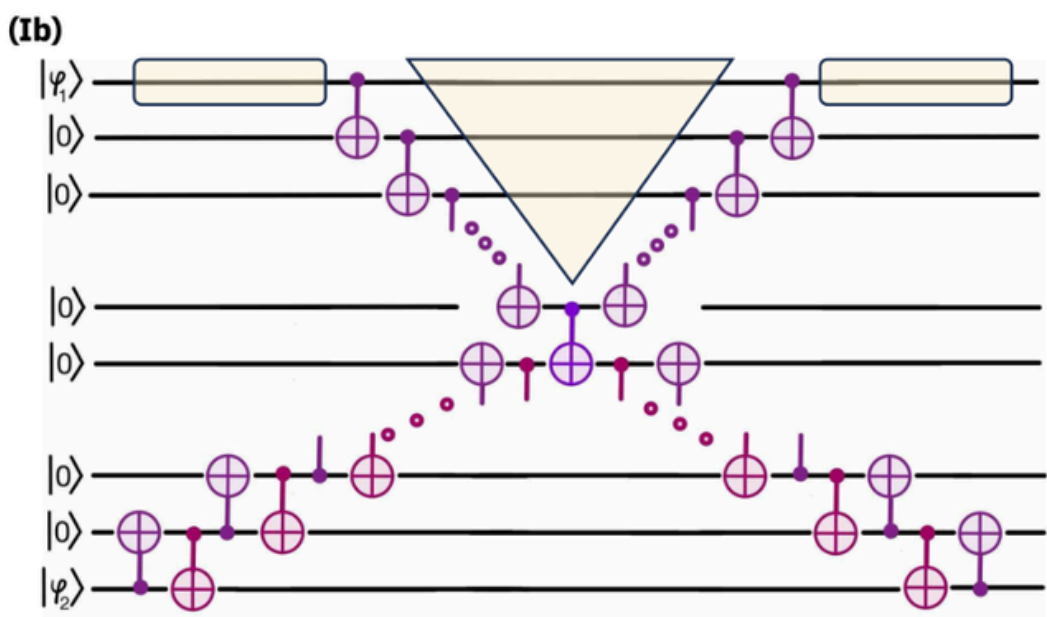
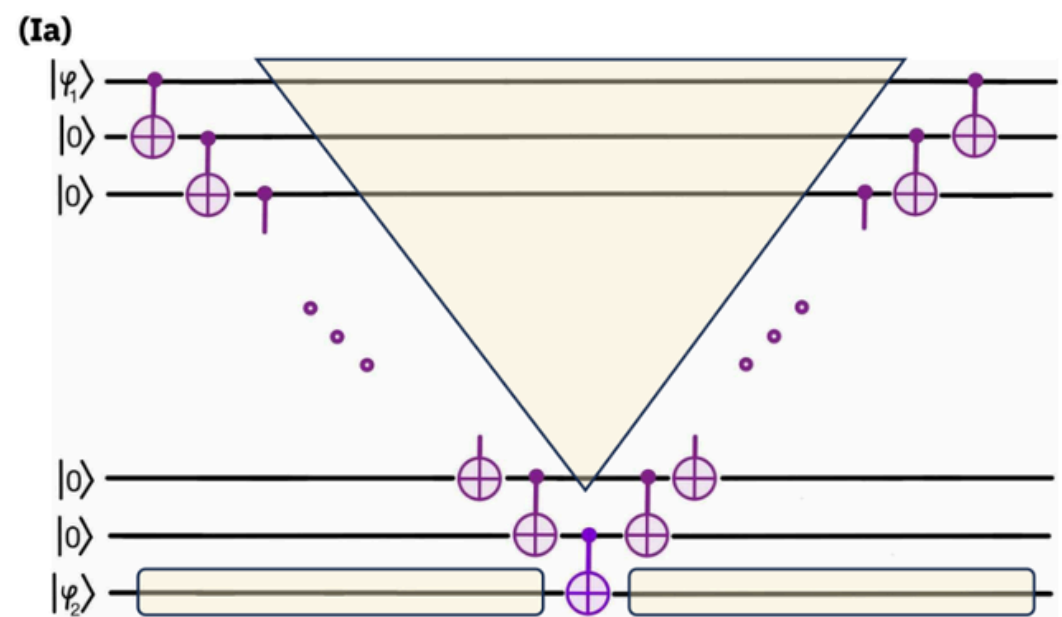
Unitary Implementation



Dynamic Circuit

Problem 2

To minimize the idle time (which is quadratic in others), we chose **Strategy Ic** among **Unitary circuits**.



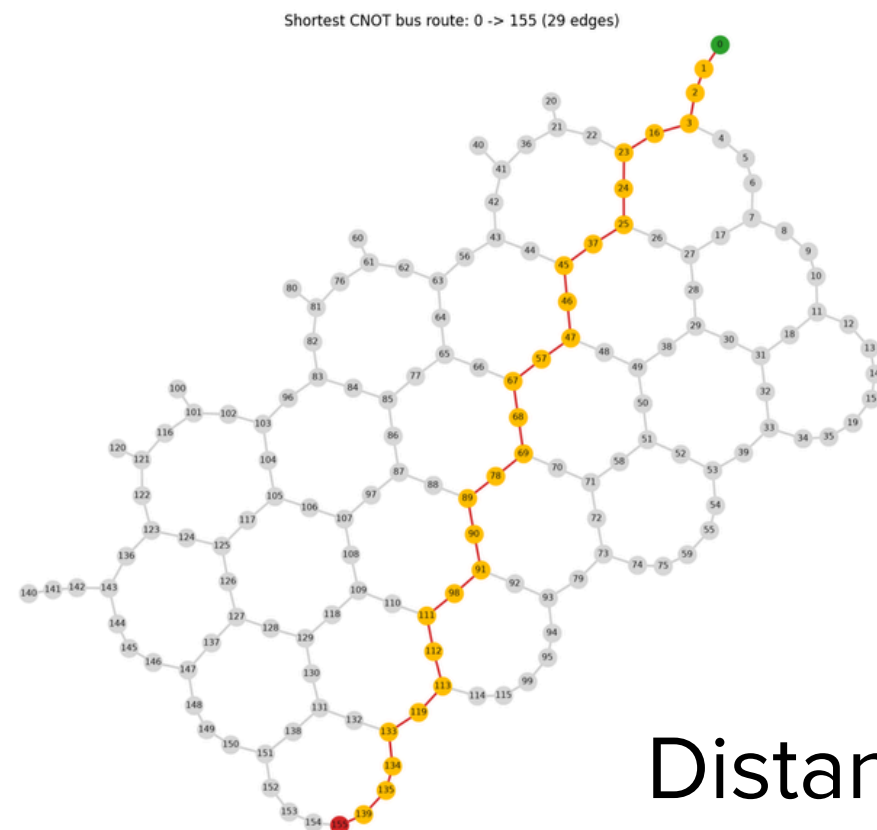
Case	t_{idle}
Unitary Ia)	$n^2 + 2n$
Unitary Ib)	$\frac{n^2}{4} + n$
Unitary Ic)	0
Unitary II)	$\frac{3}{4}\tilde{n}^2 - \frac{3}{2}\tilde{n}$
Unitary II) with normed n	$\approx \frac{3}{16}n^2 - \frac{15}{8}n + \frac{45}{16}$
Dynamic circuits	$2\mu + 2$

Problem 2

Pseudocode for shortest bus route finding

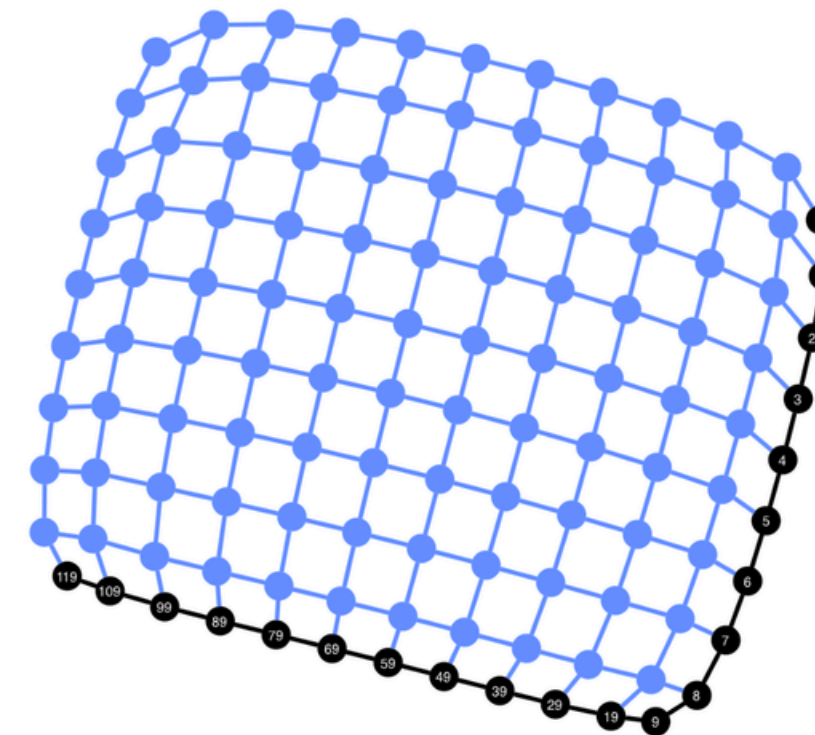
1. Run BFS on G from c to t , obtain shortest path $P = [v_0, \dots, v_{L-1}]$
2. Set $\text{chain} \leftarrow P, \quad n \leftarrow L - 2$
3. Assign logical qubits as:
 - q_0 (system control) $\mapsto v_0$,
 - q_i (sacrificial ancilla bus) $\mapsto v_i, \quad i = 1, \dots, L - 2$,
 - q_{n+1} (system target) $\mapsto v_{L-1}$
4. Use this mapping as the transpiler initial layout.

Shortest Bus Route - IBM_yonsei (Heavy-hex)



Distance: 28

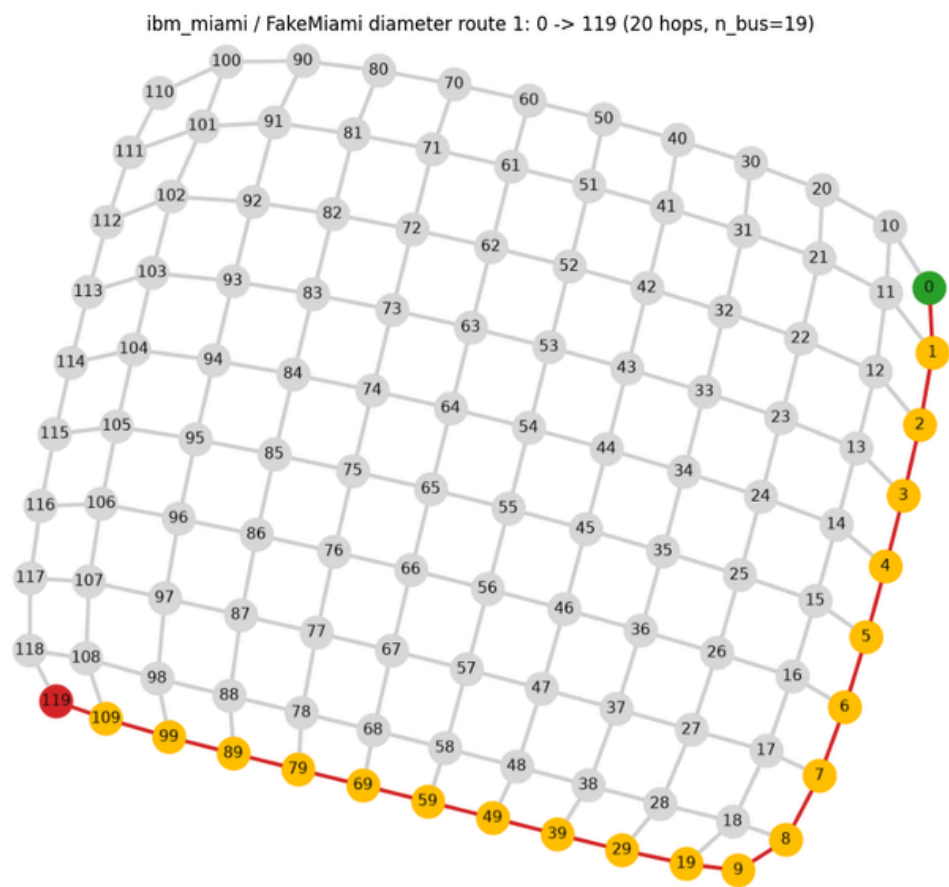
Shortest Bus Route - IBM_miami(2D lattice)



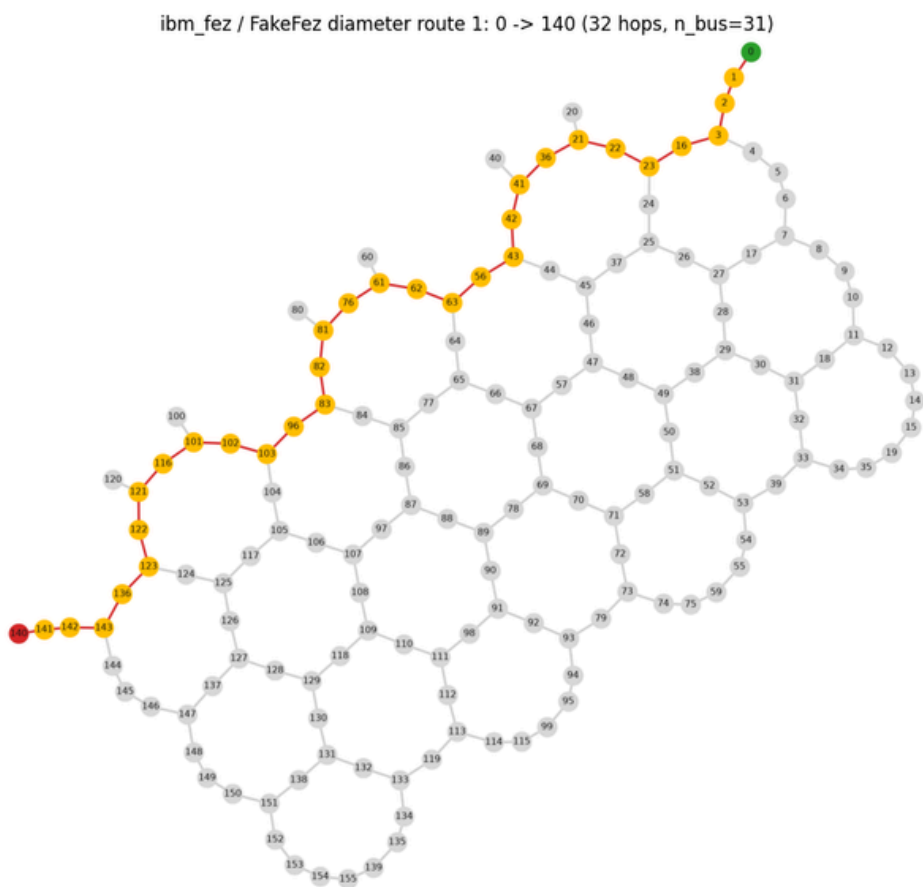
Distance: 19

Problem 2

IBM_miami(2D lattice)



IBM_yonsei (Heavy-hex)



Summary Metrics for Backend Topologies

Backend	Mean	Var	Std	Diam.
FakeMiami	7.33	13.33	3.65	20
FakeYonsei	12.64	41.75	6.46	32

Topology Difference Analysis (Pseudocode)

1. Initialize:
 $\mathcal{B} \leftarrow \{\text{FakeMiami}, \text{FakeFez}\}$
 $\text{results} \leftarrow \emptyset$
2. For each backend $B \in \mathcal{B}$ do:
 $n \leftarrow$ number of qubits in B
 $P \leftarrow \{(i, j) \mid 0 \leq i < j < n\}$
For each pair $(i, j) \in P$ do:
path $\leftarrow$ Shortest path between i and j on G_B
hops $\leftarrow |\text{path}| - 1$
bus_n $\leftarrow$ hops $- 1$
Record $(B, i, j, \text{hops}, \text{bus_n}, \text{path})$
End For
3. Statistical Aggregation:
 $\mu \leftarrow \text{mean}(\text{hops})$
 $\sigma^2 \leftarrow \text{variance}(\text{hops})$
 $\sigma \leftarrow \sqrt{\sigma^2}$
 $\mathcal{D} \leftarrow \max(\text{hops})$ // Network Diameter
Append $(B, \mu, \sigma^2, \sigma, \mathcal{D})$ to results
End For
4. Output:
Display summary metrics table (results)

Unitary vs. Dynamic

Theoritical Resource Comparison of Long-range CNOT Methods

Implementation	t_{idle}	N_{CNOT}	N_{meas}	2Q depth
Unitary lc	0	$4n + 1$	0	$2n + 1$
Dynamic	$2\mu + 2$	$n + 1$	n	2

		2Q gate count		2Q depth		mid-circuit measurements	
		dynamic	unitary	dynamic	unitary	dynamic	unitary
topology	distance						
2D square (ibm_miami)	0	1	1	1	3	0	0
	1	2	5	2	5	1	0
	2	3	9	2	5	2	0
	3	4	13	2	9	3	0
	6	7	25	2	13	6	0
	11	12	45	2	25	11	0
	16	17	65	2	33	16	0
heavy-hex (ibm_yonsei)	0	1	1	1	3	0	0
	1	2	5	2	5	1	0
	2	3	9	2	5	2	0
	3	4	13	2	9	3	0
	6	7	25	2	13	6	0
	11	12	45	2	25	11	0
	16	17	65	2	33	16	0

Table: Resource Comparison of Long-range CNOT Methods

Problem 3

Impact of Noise on the Dynamic Long-Range CNOT

Objective: Apply depolarizing noise, readout error, and thermal relaxation (T_1/T_2).

Analysis: Observe how each noise source degrades CNOT fidelity as n increases.

Key Question : Which noise channel is the dominant limiting factor?

Error Budget Equation:

$$\lambda_{\text{tot}} = t_{\text{idle}}\lambda_{\text{idle}} + N_{\text{CNOT}}\lambda_{\text{CNOT}} + N_{\text{meas}}\lambda_{\text{meas}}$$

Discussion Task:

- Verify consistency between simulation results and the theoretical error budget.
- Plug in the resource counts ($t_{\text{idle}}, N_{\text{CNOT}}, N_{\text{meas}}$) obtained from problem 2.

Problem 3

By reference [1], the state fidelity of quantum state is

Since $\sigma = |\phi\rangle\langle\phi|$, we can simplify the general expression as

$$\begin{aligned} F(\rho, \sigma) &:= \left[\text{Tr} \left(\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right) \right]^2 \\ &= \langle \phi | \rho | \phi \rangle \\ &= \text{Tr}[\rho \sigma]. \end{aligned}$$

Thus, we can measure GHZ state fidelity as

$$\tilde{F}_{\text{GHZ}} = \frac{1}{M} \sum_{k=1}^M \langle S_k \rangle_{\rho} + \mathcal{O} \left(\frac{1}{\sqrt{m}} \right)$$

Problem 3

However, the process fidelity of the channel is

$$\begin{aligned} F(\mathcal{U}, \tilde{\mathcal{U}}; \rho) &= \left[\text{Tr} \left(\sqrt{\sqrt{\mathcal{U}(\rho)} \tilde{\mathcal{U}}(\rho) \sqrt{\mathcal{U}(\rho)}} \right) \right]^2 \\ &= \text{Tr} [\mathcal{U}(\rho) \tilde{\mathcal{U}}(\rho)] \\ &= \text{Tr} [\rho \Lambda(\rho)]. \end{aligned}$$

Using Choi-Jamiolkowski isomorphism, the representative channel state is

$$\rho_{\Lambda} = (I \otimes \Lambda) (|\phi\rangle\langle\phi|), \quad |\phi\rangle = \frac{1}{\sqrt{d}} \sum_{i=1}^d |i\rangle |i\rangle.$$

$$F_{\text{proc}}(\mathcal{U}, \tilde{\mathcal{U}}) := F(\rho_{\mathcal{U}}, \rho_{\tilde{\mathcal{U}}}).$$

Problem 3

Type of qubit physical noises

- Thermal relaxation (T1/T2)
- Depolarizing noise
- Readout error

The combined process fidelity can be expressed as

$$\mathcal{F}_{\text{proc}} = \mathcal{F}_{\text{idle}} \mathcal{F}_{\text{CNOT}} \mathcal{F}_{\text{meas}}$$

By reference [1], the lower bound of combined process fidelity is

Lemma 1. *Consider a channel $\Lambda = \prod_P \Lambda_P^{\lambda_P}$ for some rates λ_P . Then $\mathcal{F}_{\text{proc}}(\Lambda, \mathcal{I}) \geq \exp \left(- \sum_P \lambda_P \right)$.*

Problem 3

Type of qubit physical noises

- Thermal relaxation (T1/T2) (λ_{idle})
- Depolarizing noise (λ_{CNOT})
- Readout error (λ_{meas})

Thus, the lower bound of the combined process fidelity is

$$\mathcal{F}_{\text{proc}} \geq e^{-\lambda_{\text{tot}}}$$

where $\lambda_{\text{tot}} = t_{\text{idle}}\lambda_{\text{idle}} + N_{\text{CNOT}}\lambda_{\text{CNOT}} + N_{\text{meas}}\lambda_{\text{meas}}$

Thermal relaxation

Using Pauli-Lindbladian model, depolarizing / dephasing channel is

$$\Lambda_{\text{noise}}^t = \exp \left(\frac{t}{T_1} \mathcal{L}_{\text{damp}} + \frac{t}{T_2} \mathcal{L}_{\phi} \right)$$

$$\tilde{\Lambda}_{\text{noise}}^t = \Gamma_X^{\frac{t}{4T_1}} \Gamma_Y^{\frac{t}{4T_1}} \Gamma_Z^{\frac{t}{2T_2}}.$$

By Lemma 1, for any pauli operator P , the lower bound of process fidelity of idling is

$$\lambda_{\text{idle}} = t_{CNOT, sec} \left(\frac{1}{4T_1} + \frac{1}{4T_1} + \frac{1}{2T_1} \right)$$

Depolarizing noise

By reference [1], the relation between gate fidelity and the process fidelity is

$$\mathcal{F}_{\text{gate}}(\mathcal{U}, \tilde{\mathcal{U}}) = \frac{d \mathcal{F}_{\text{proc}}(\rho_{\mathcal{U}}, \rho_{\tilde{\mathcal{U}}}) + 1}{d + 1}.$$

If we know the hardware gate fidelity of CNOT, the identity holds

$$\mathcal{F}_{\text{proc}}(\rho_{\mathcal{U}}, \rho_{\tilde{\mathcal{U}}}) = \frac{(d + 1)\mathcal{F}_{\text{gate}}(\mathcal{U}, \tilde{\mathcal{U}}) - 1}{d} = e^{-\lambda_{\text{CNOT}}}.$$

Thus, where the dimension of gate $d = 2^2 = 4$,

$$\lambda_{\text{CNOT}} = -\ln \left[\frac{(d + 1)\mathcal{F}_{\text{gate}}(\mathcal{U}, \tilde{\mathcal{U}}) - 1}{d} \right].$$

Readout error

Using classical error model of measurement error,

$$\mathcal{F}_{\text{meas}} = (1 - p)^{N_{\text{meas}}} = e^{-\lambda_{\text{meas}} N_{\text{meas}}}$$

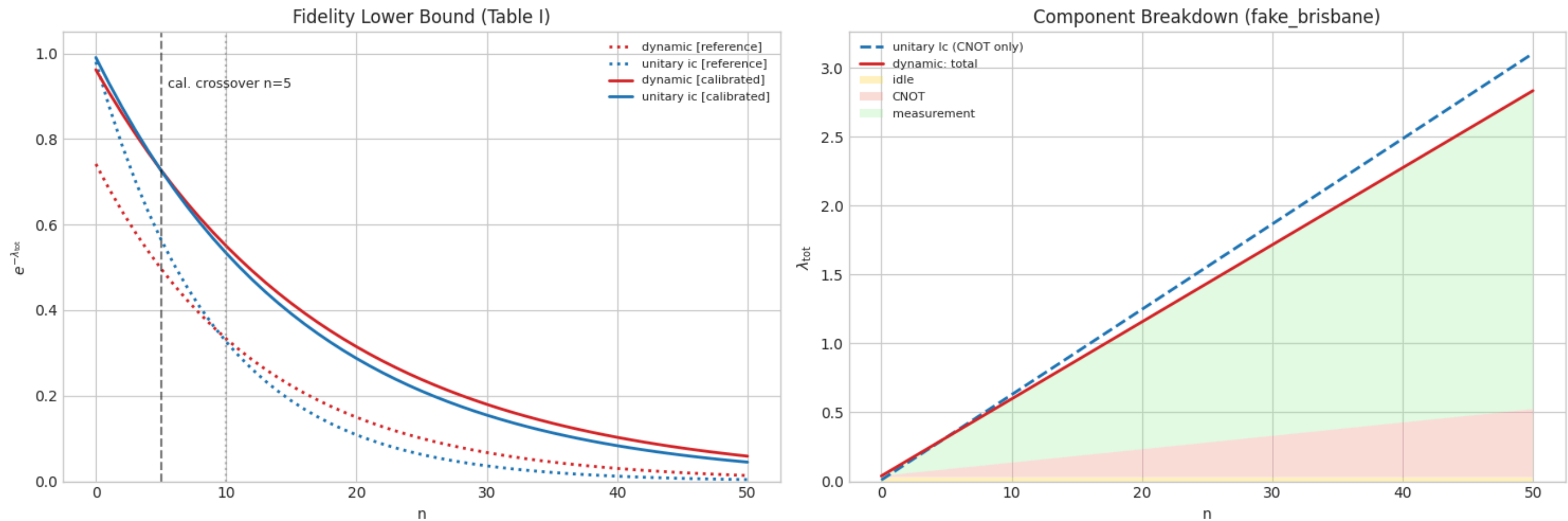
where p is the readout error probability.

Thus, the readout coefficient is

$$\lambda_{\text{idle}} = -\ln(1 - p)$$

Parameter calculation

Analytical Error Budget — fake_brisbane



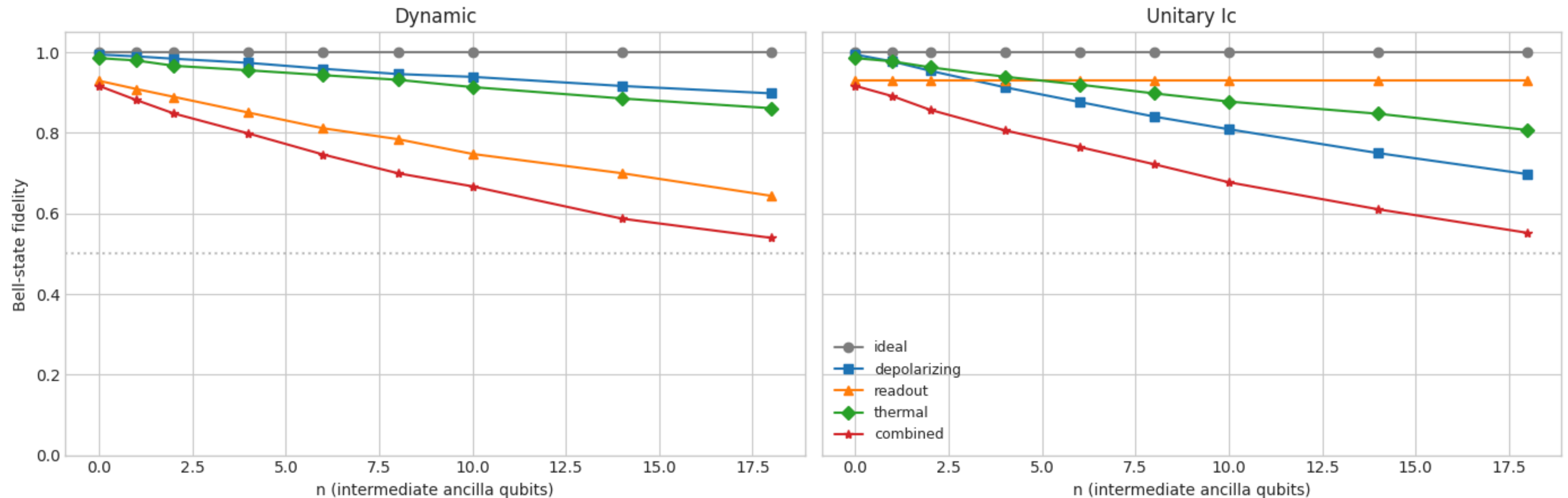
	Parameter	Reference (paper)	Calibrated (fake_brisbane)	Actual hardware calibration (ibm_aachen)
0	lambda_idle	0.0300	0.003622	0.000294
1	lambda_CNOT	0.0200	0.009696	0.002055
2	lambda_meas	0.0300	0.023092	0.004220
3	mu	3.65	3.030	42.353
4	3*lambda_CNOT	0.0600	0.029088	0.006164
5	crossover?	True	True	True

Cross over detected!!
: Over distance 10, dynamic circuit is better than unitary

Dominant error analyze

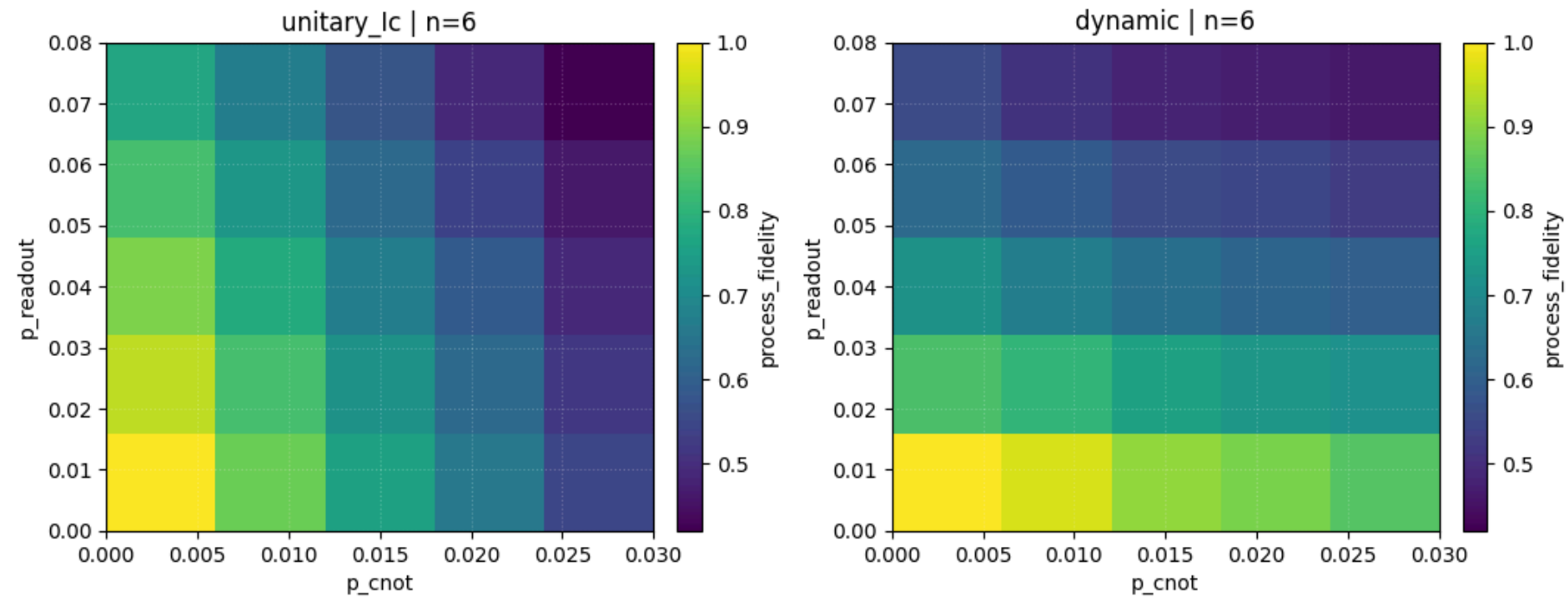
- To analyze dominant error, fake_brisbane used for noisy circuit simulation.
- For dynaic, readout error is dominant. For unitary, depolarizing & thermal dominant.

Aer Noise Sweep (fake_brisbane calibration)

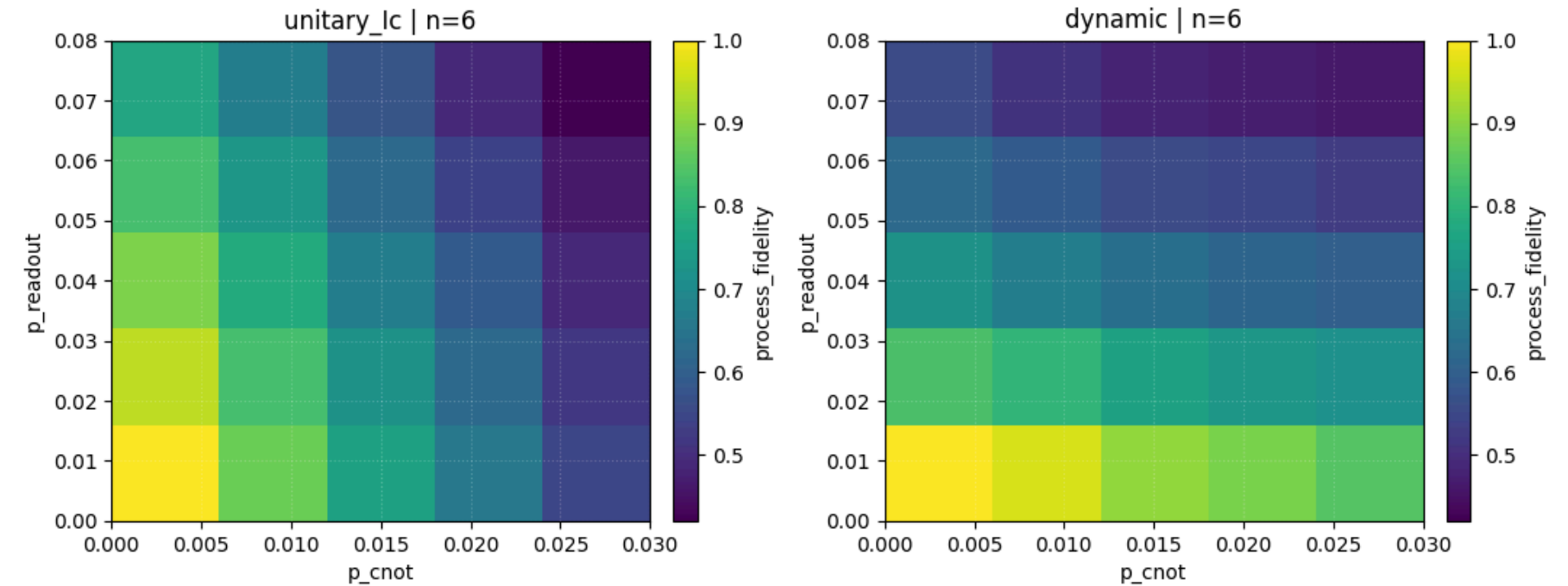


Dominant error analyze

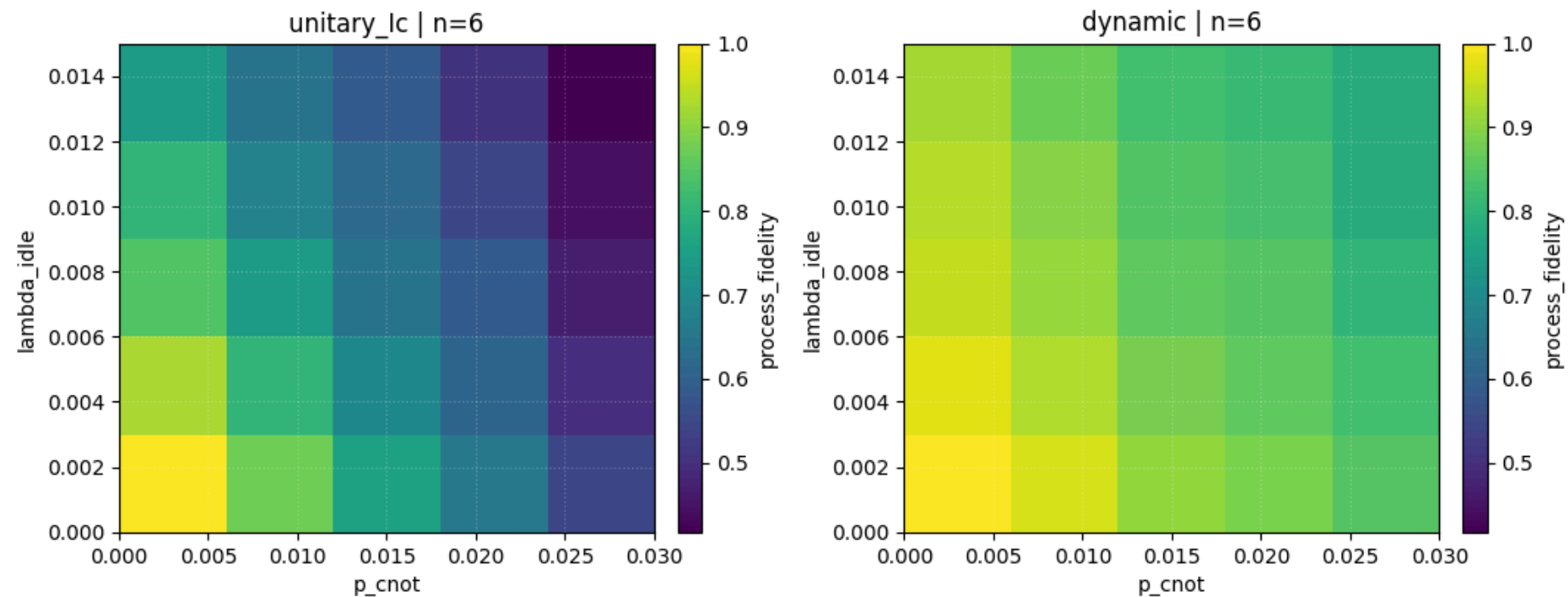
Pairwise noise grid: p_cnot vs p_readout



Pairwise noise grid: p_cnot vs p_readout



Pairwise noise grid: p_cnot vs lambda_idle



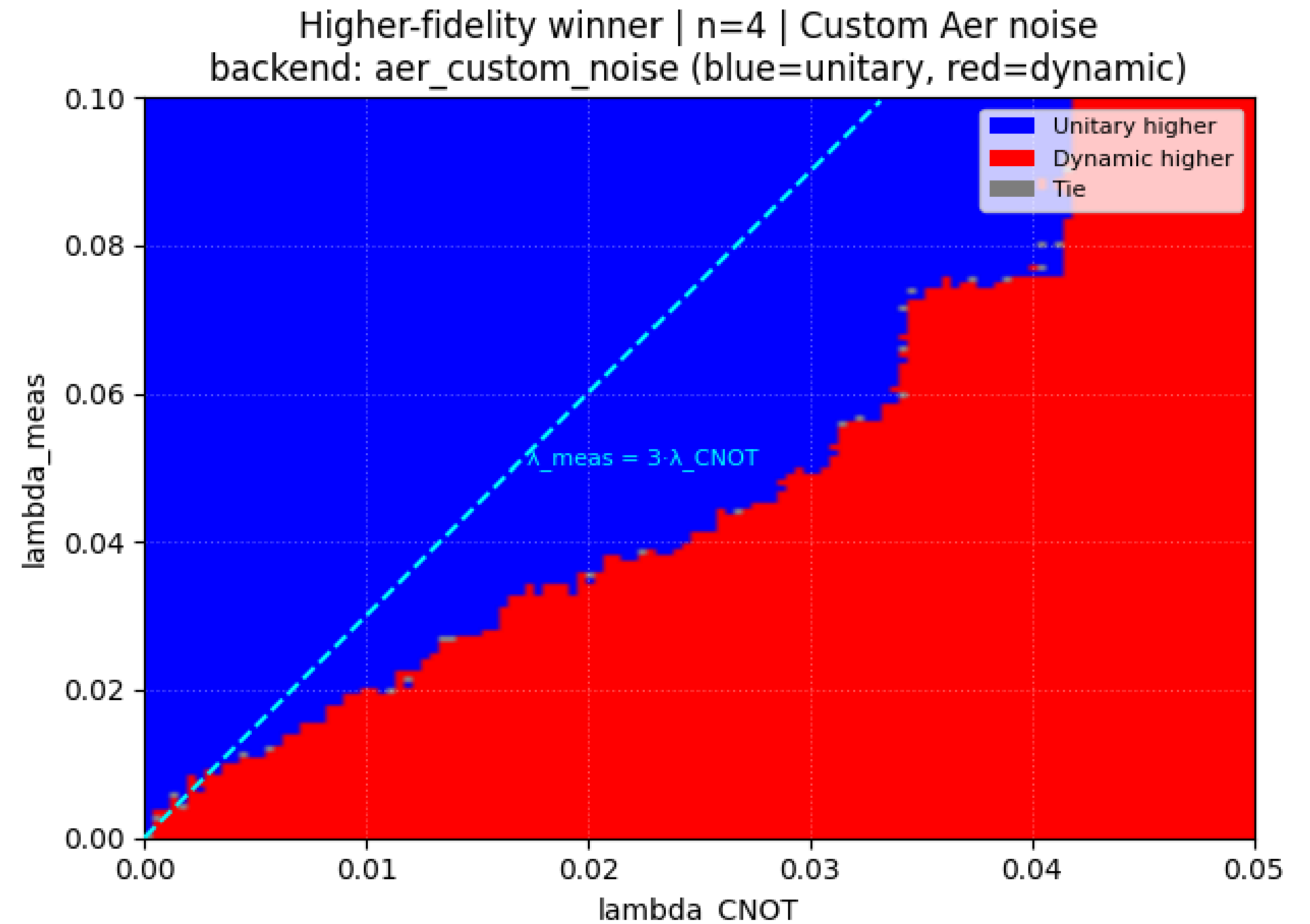
- For unitary circuits, **depolarizing noise** (p_cnot) was dominant.
- For dynamic circuits, **readout error** (p_readout) was dominant.

Error comparison

Focusing on λ_{CNOT} and λ_{meas}

$$\lambda_{meas} < 3\lambda_{CNOT}$$

Tendency observed.

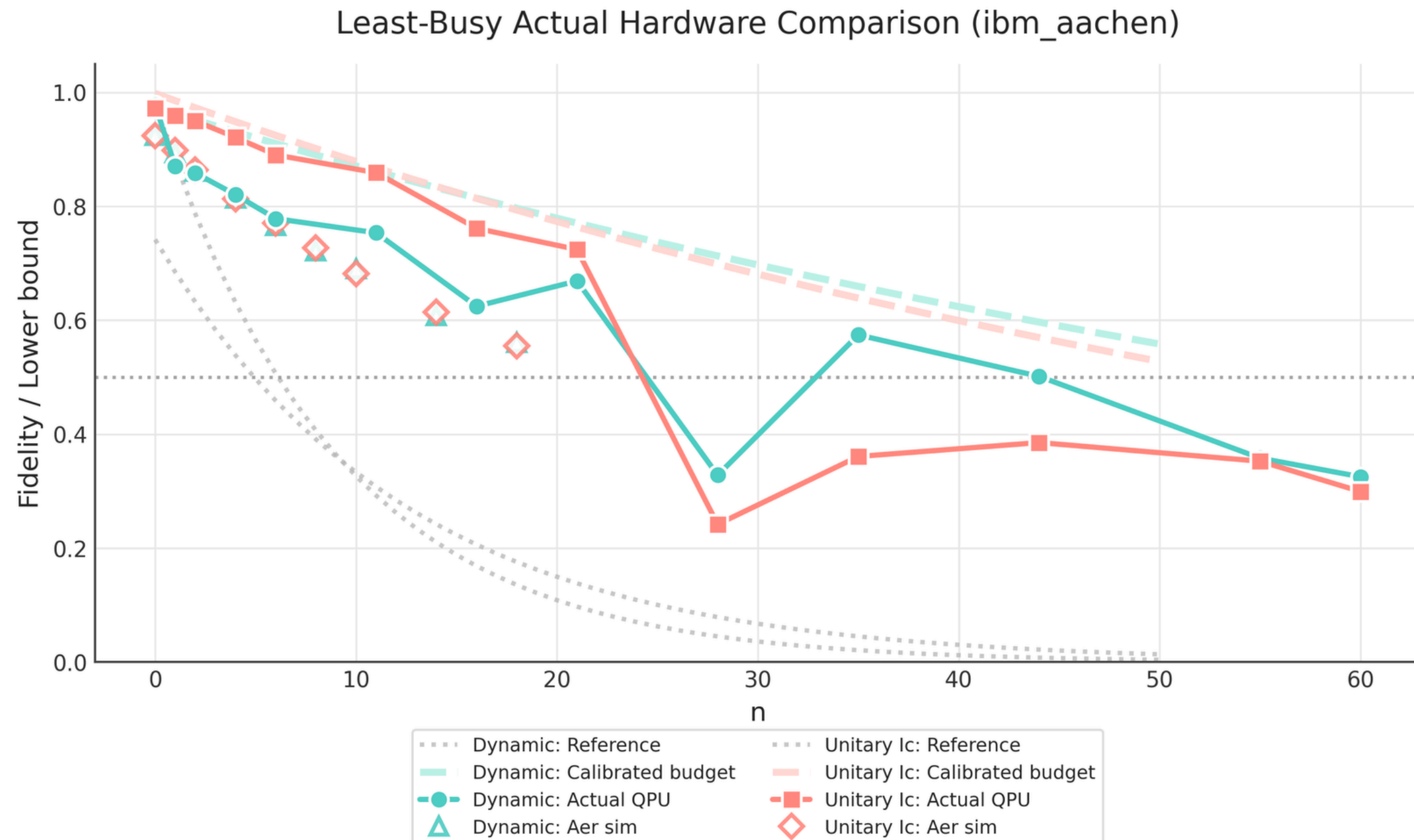


Red (dynamic win), blue (unitary win), gray (tie) in

F_{avg} over $\lambda_{CNOT} - \lambda_{meas}$

Real hardware simulation

- In real hardware simulation, crossover dynamic and unitary circuit observed.
- Due to inconsistent error of qubits, fidelity below lower bound detected.



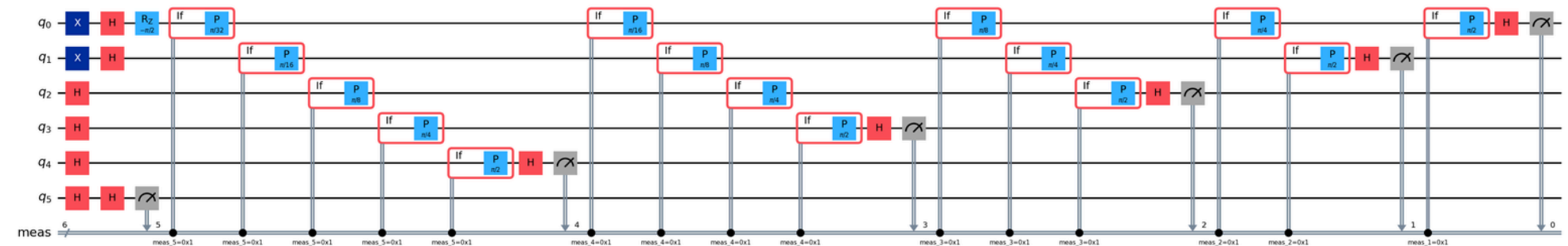
Problem 4

Open Problem : Dynamic circuit inside Larger Algorithms

4.1. Dynamic Quantum Fourier Transform (QFT) & Dynamic Quantum Phase Estimation (QPE)

4.2. Dynamic Multi-Controlled Gates & Dynamic Grover Search
Ancilla Uncomputation using Dynamic Measurements

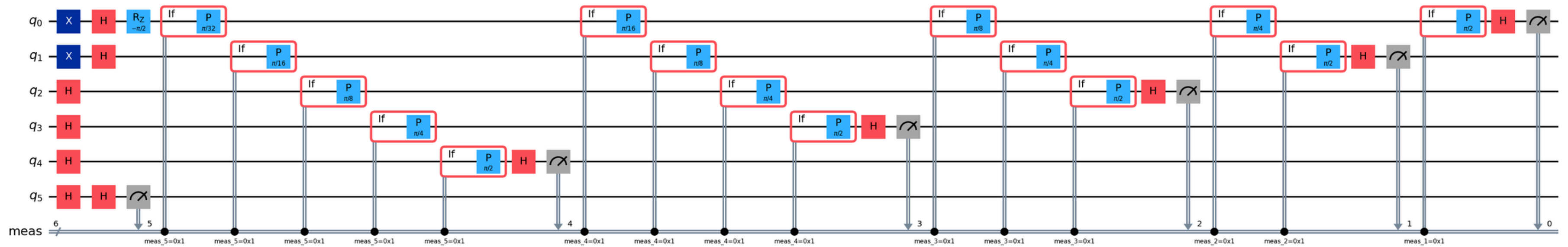
Problem 4.1



Dynamic Quantum Fourier Transform (QFT) & Dynamic Quantum Phase Estimation (QPE)

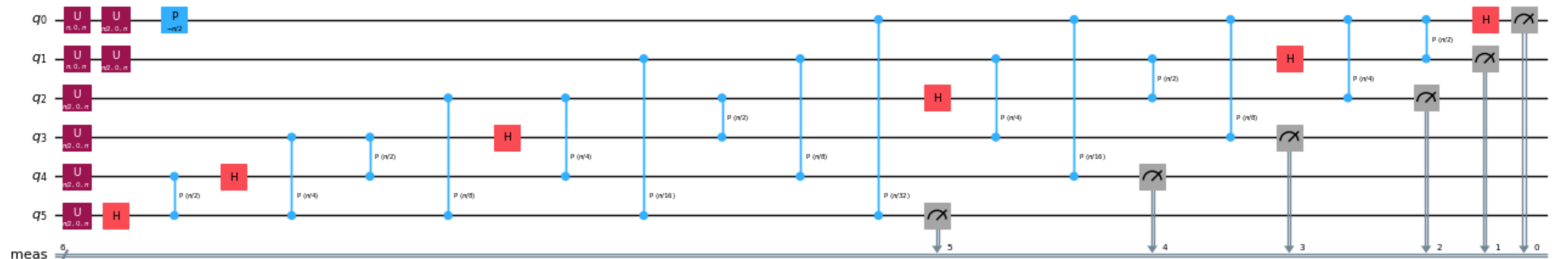
Dynamic Quantum Fourier Transform

Circuit implementation - dynamic



Circuit implementation - unitary

Global Phase: $\pi/4$



Dynamic Quantum Fourier Transform

Fidelity Calculation

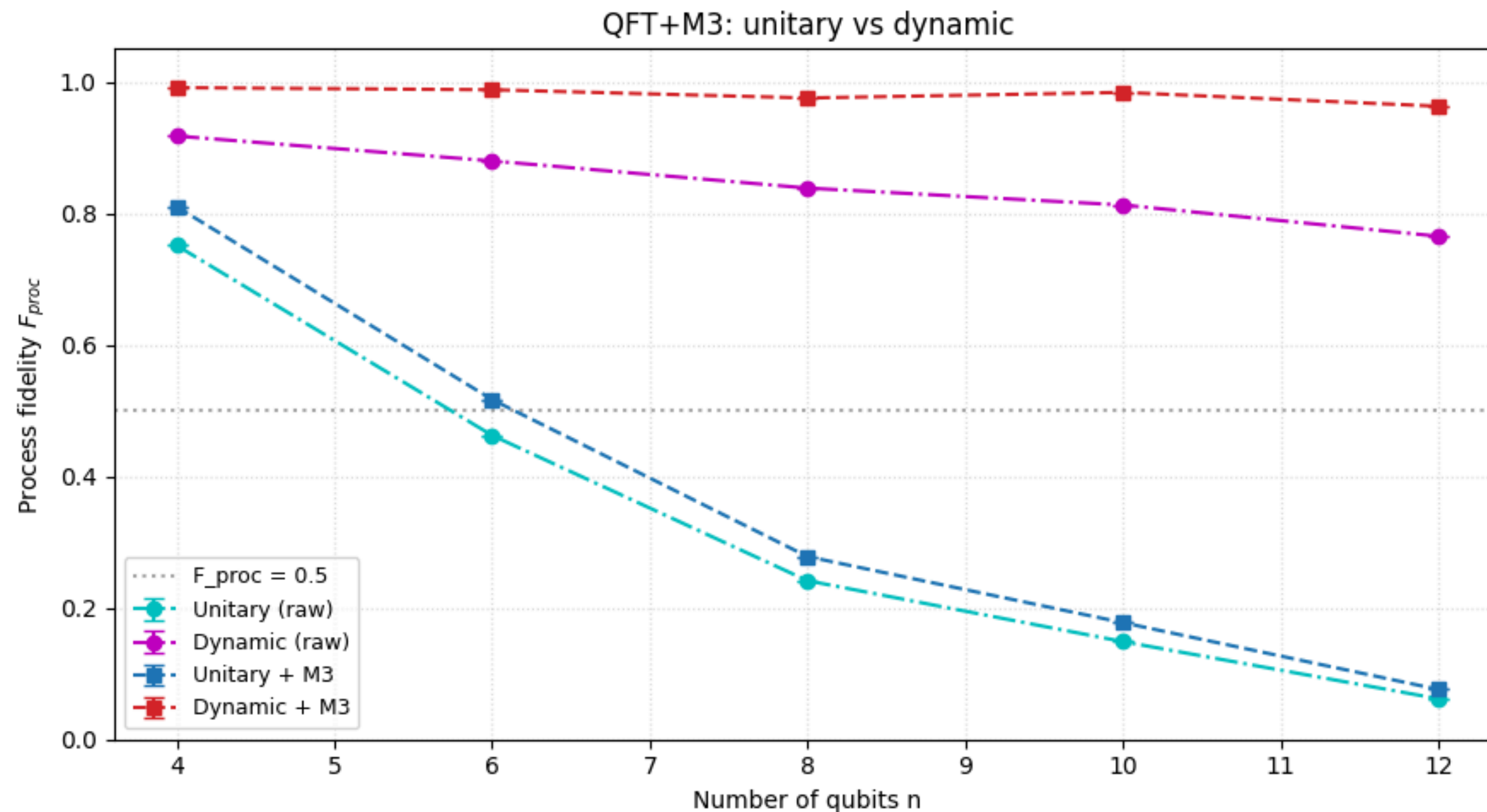
$$F_{\text{proc}} \approx \frac{m}{m-1} \left[\frac{1}{m} \sum_l \sqrt{P(k_l)} \right]^2 - \frac{1}{m(m-1)} \sum_l P(k_l)$$

$$P(k_l) = \text{Pr}(k_l | Q\tilde{F}T(\sigma_{k_l}^*))$$

(Reference 2, “Quantum Fourier Transform Using Dynamic Circuits”, E. Baumer, et al.)

Dynamic Quantum Fourier Transform (QFT)

Noise Simulator Results (fake_fez)



Dynamic is better than unitary version.

Because of the same number of measurement, dynamic implementation would be better that use feed-forward to compress circuit depth.

SHOTS = 1024

Dynamic Quantum Fourier Transform (QFT)

Resource-count comparison

Resource	Unitary QFT+M	Dynamic QFT+M
<i>State prep</i> σ_k^*		
Two-qubit gates	0	0
Single-qubit (X , H , RZ)	$O(n)$	$O(n)$
<i>QFT body</i>		
Hadamard	n	n
Quantum CP (2Q)	$\frac{n(n-1)}{2}$	0
Classical feed-forward P	0	$\frac{n(n-1)}{2}$
Mid-circuit measurements	0	n
Terminal measurement	n	—
<i>Logical</i>		
Body N_{2Q}	$n(n-1)/2$	0
Body circuit depth	$O(n)$	$\Theta(n^2)$
<i>ISA (typical)</i>		
Body N_{2Q}	$\Theta(n^2)$	≈ 0

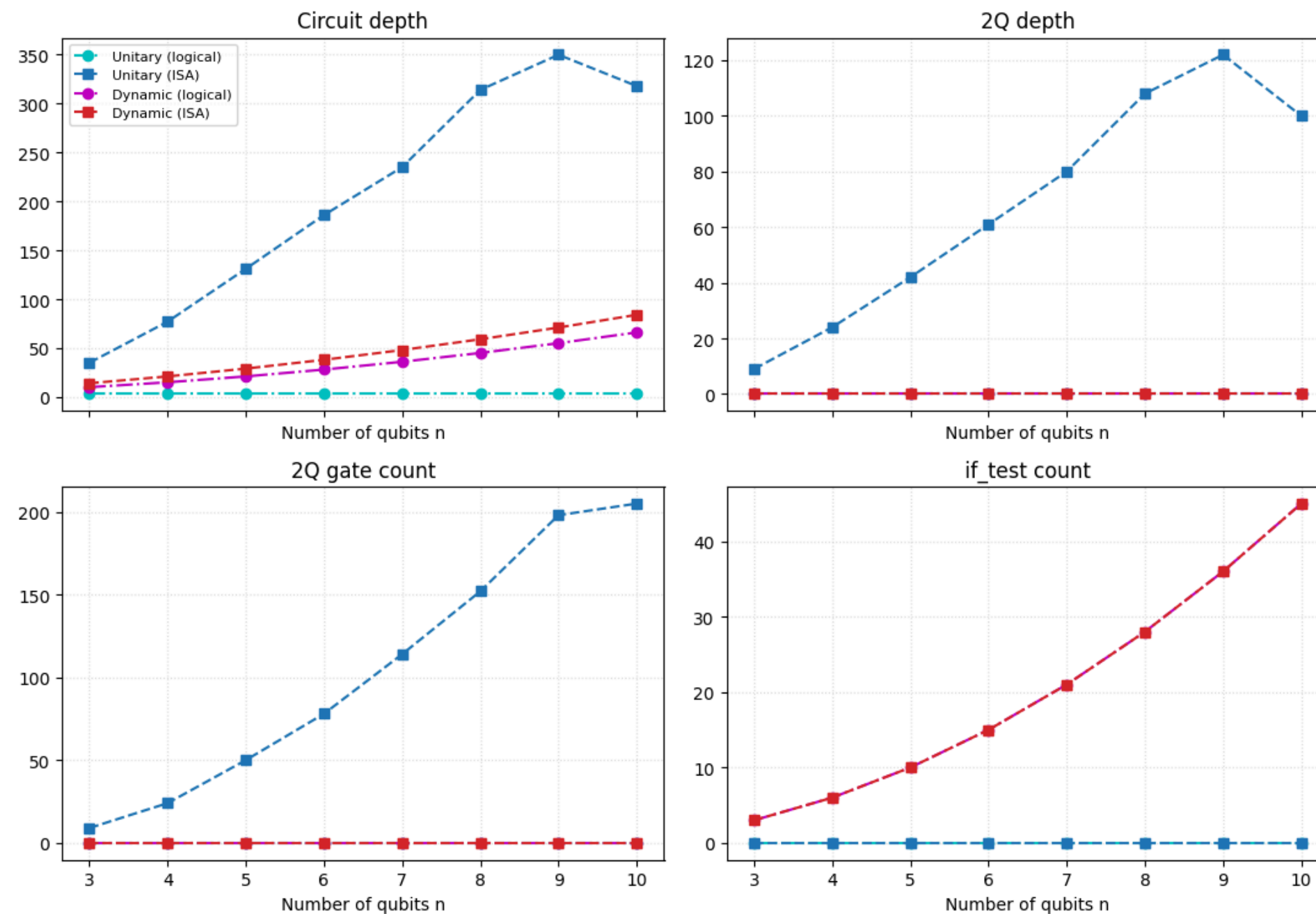
Trade off between controlled gates and classical control.

But in transpile, it makes almost constant 2Q

Dynamic Quantum Fourier Transform (QFT)

Resource-count comparison

QFT+M resource scaling vs qubit count (k=0, fake_fez)

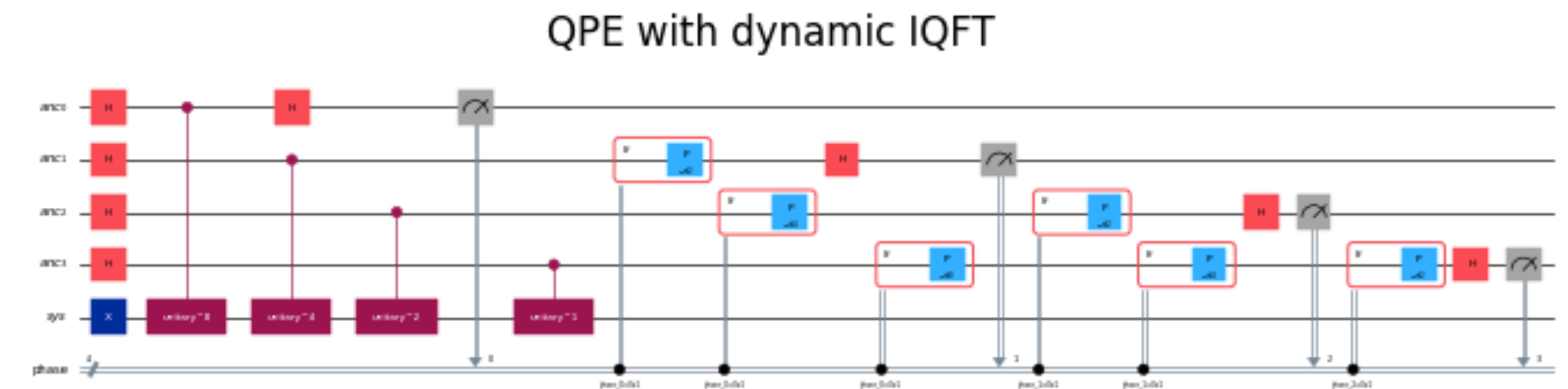
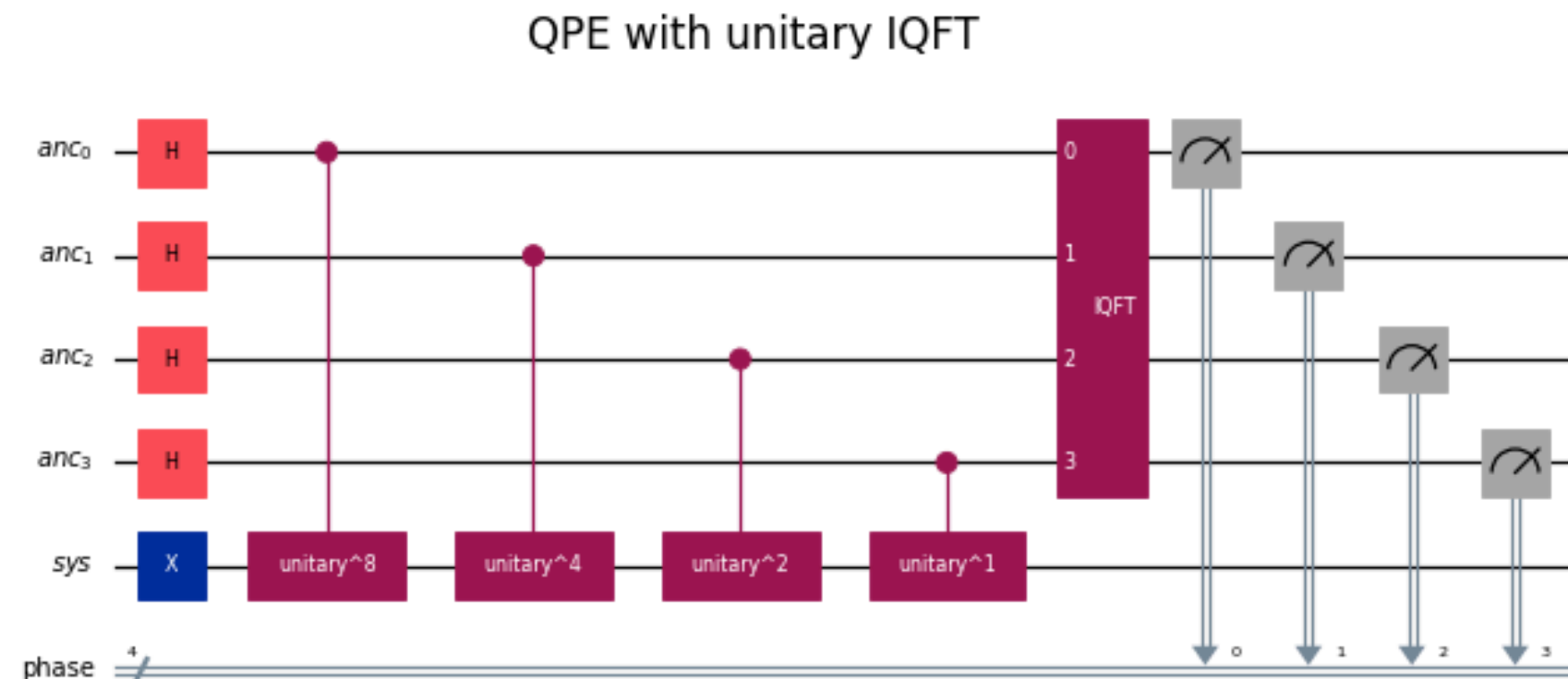


Dynamic has lower circuit depth and 2Q gate than unitary after transpiling.

cf) the decline at n=10 might be due to scheduling and routing process (transpile)

Dynamic Quantum Phase Estimation (QPE)

Comparing Unitary QPE with Dynamic QPE

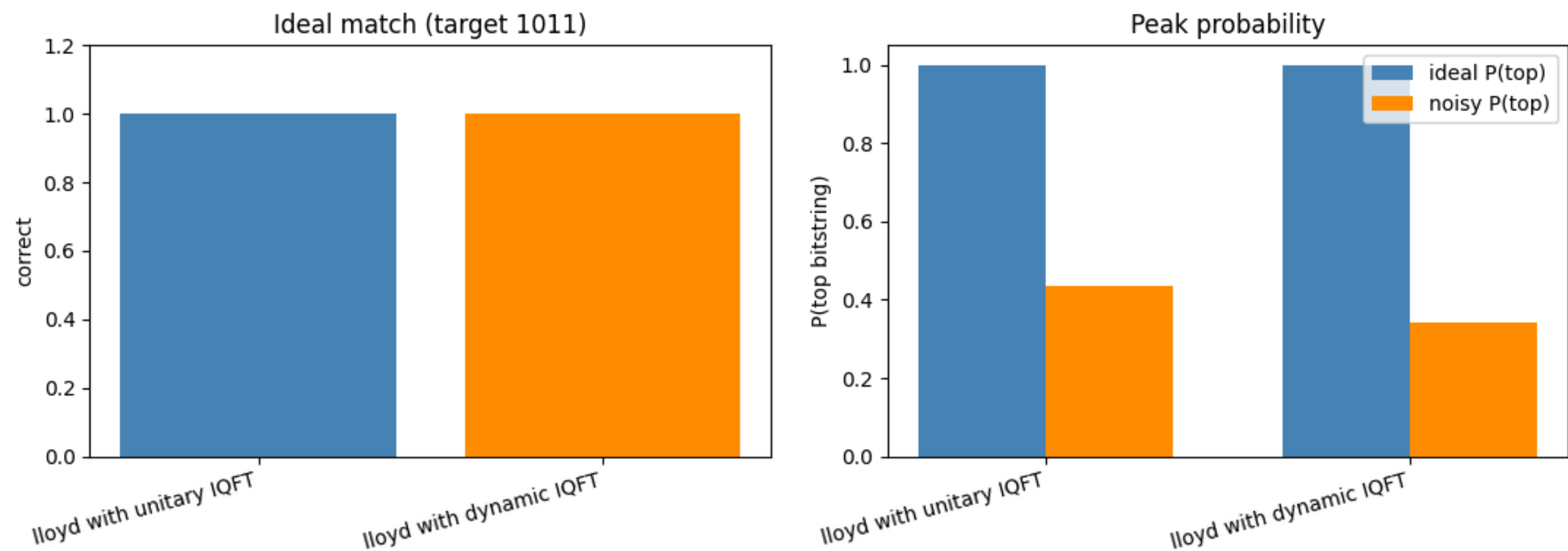


The unitary QPE utilizes the inverse QFT unitary block.

The dynamic QPE utilizes the dynamic QFT implementation.

Dynamic Quantum Phase Estimation (QPE)

Qubits $n=4$, target=1011, noise simulator



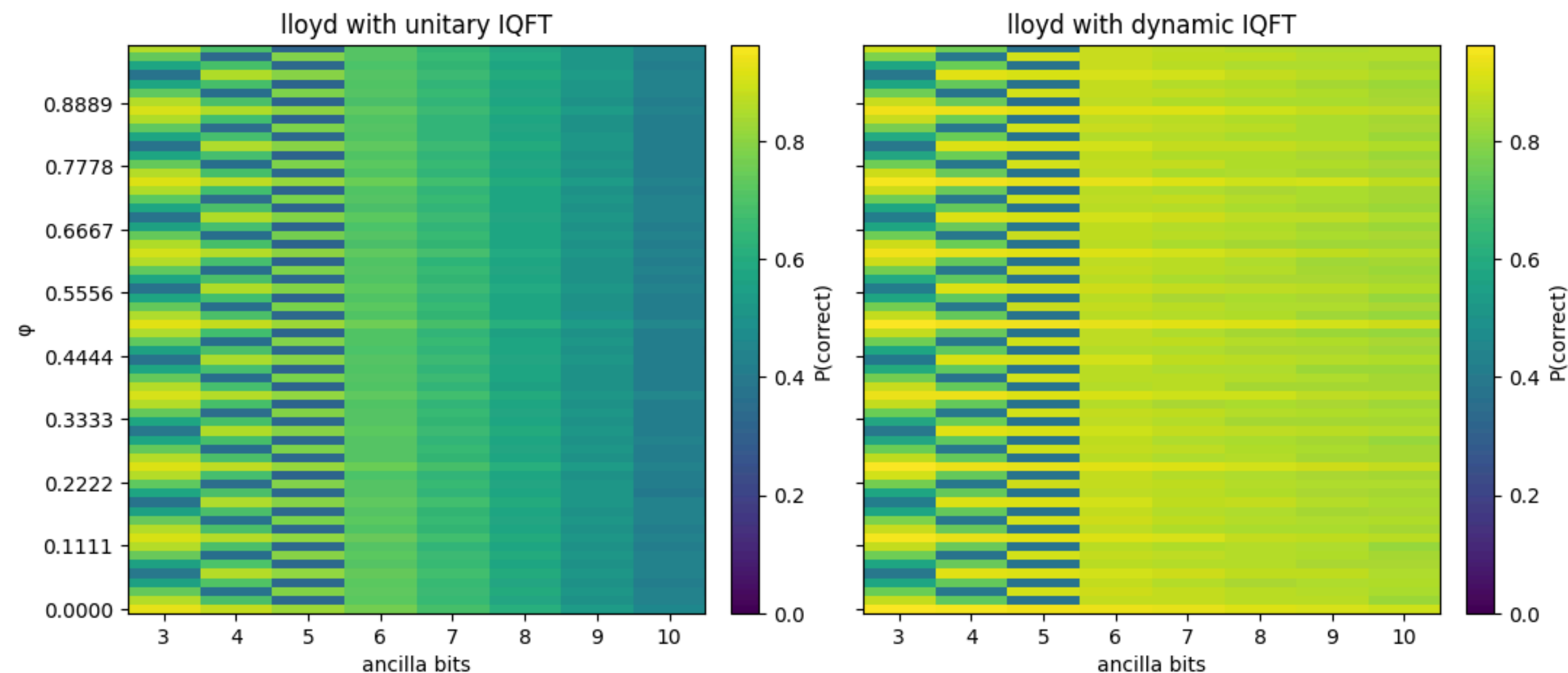
Error Source	Error Rate
One Qubit Error	0.001
Idle Error	0.001
Two Qubit Error	0.01
Reset Error	0.01
Readout Error	0.01

In $n=4$ system, unitary version is better than dynamic one.

Dynamic Quantum Phase Estimation (QPE)

Grid search for noise simulator (noise_model)

Pairwise noisy grid: $\phi \in [0, 1)$ vs ancilla bits (aer_simulator)



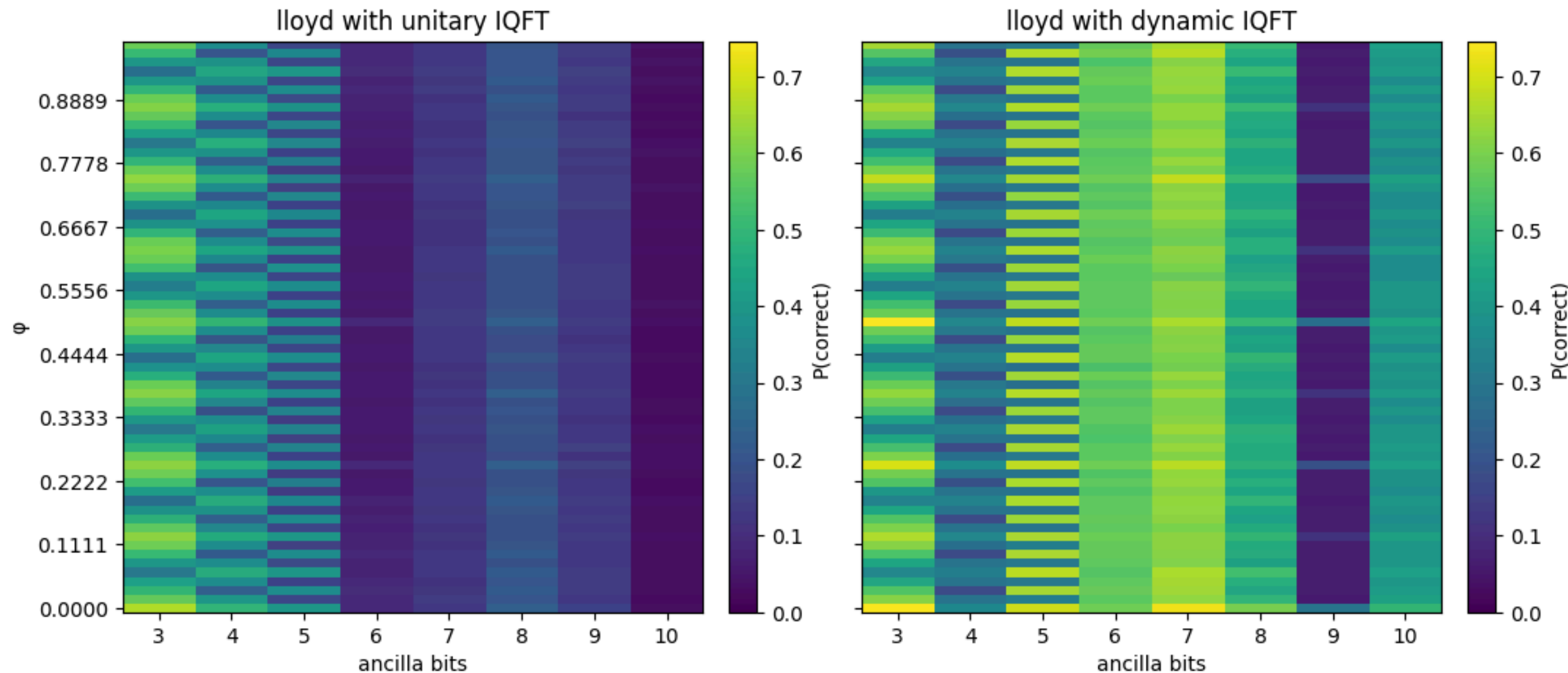
Error Source	Error Rate
One Qubit Error	0.001
Idle Error	0.001
Two Qubit Error	0.01
Reset Error	0.01
Readout Error	0.01
Shots	5000

$$P_{correct} = \frac{\text{num. target}}{\text{num. total shots}}$$

Dynamic Quantum Phase Estimation (QPE)

Grid search for noise simulator (fake_yonsei)

Pairwise noisy grid: $\phi \in [0, 1)$ vs ancilla bits (ibm_yonsei)

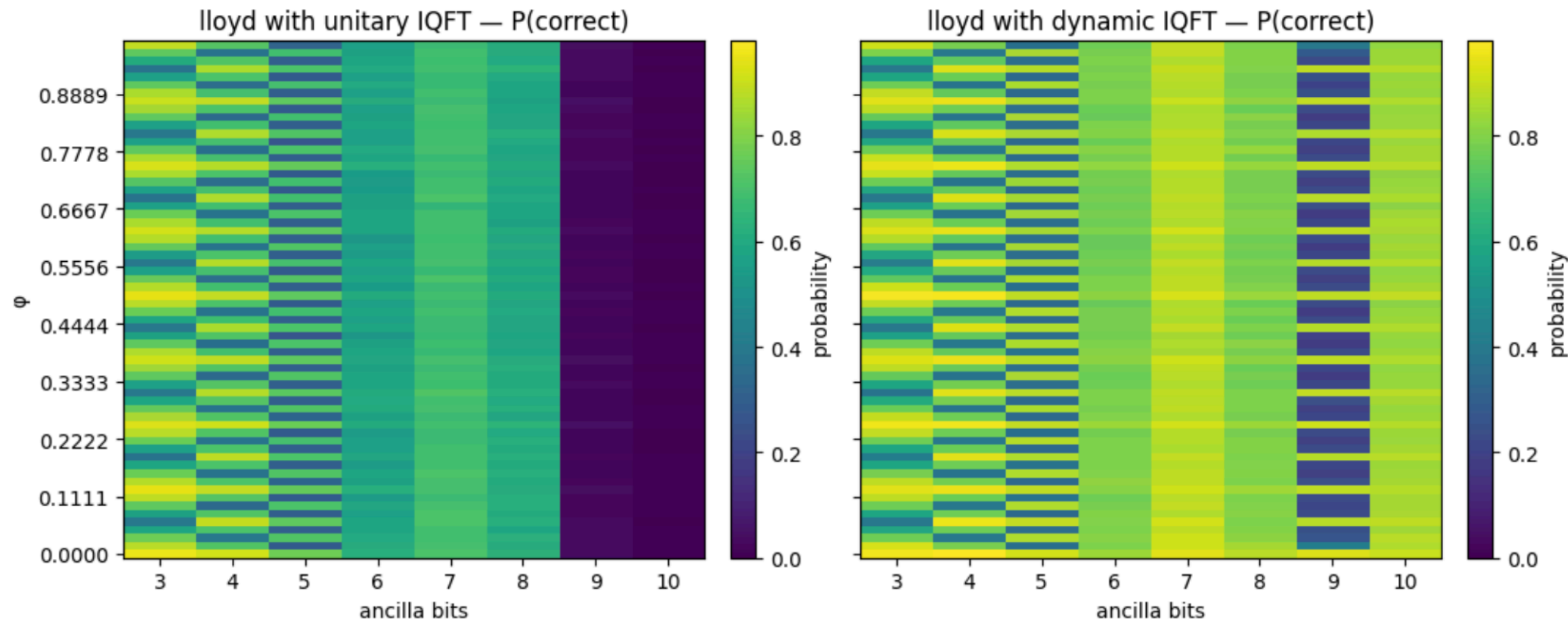


Except for $n=9$ case, dynamic version is more scalable than unitary because dynamic QFT is robust in scaling process.

shots = 5000

Dynamic Quantum Phase Estimation (QPE)

Grid search for noise simulator (fake_fez) shots = 5000



Dynamic Quantum Phase Estimation (QPE)

Resource-count comparison (table)

Resource	Unitary IQFT	Dynamic IQFT
<i>Shared prefix</i>		
Controlled- U (2Q)	n	n
Hadamard on ancilla	n	n
<i>IQFT suffix</i>		
Quantum controlled-phase (2Q)	$\frac{n(n-1)}{2}$	0
Classical <code>if_test</code> + P	0	$\frac{n(n-1)}{2}$
Mid-circuit measurements	n (terminal)	n
<i>Logical</i>		
Total 2Q count	$\Theta(n^2)$	$\Theta(n)$
IQFT circuit depth	$O(n)$	$\Theta(n^2)$
<i>ISA (typical)</i>		
2Q gate count	$\Theta(n^2)$	$\Theta(n)$
2Q depth	$\uparrow$ with n	weak / flat
<code>if_test</code> count	0	$\frac{n(n-1)}{2}$

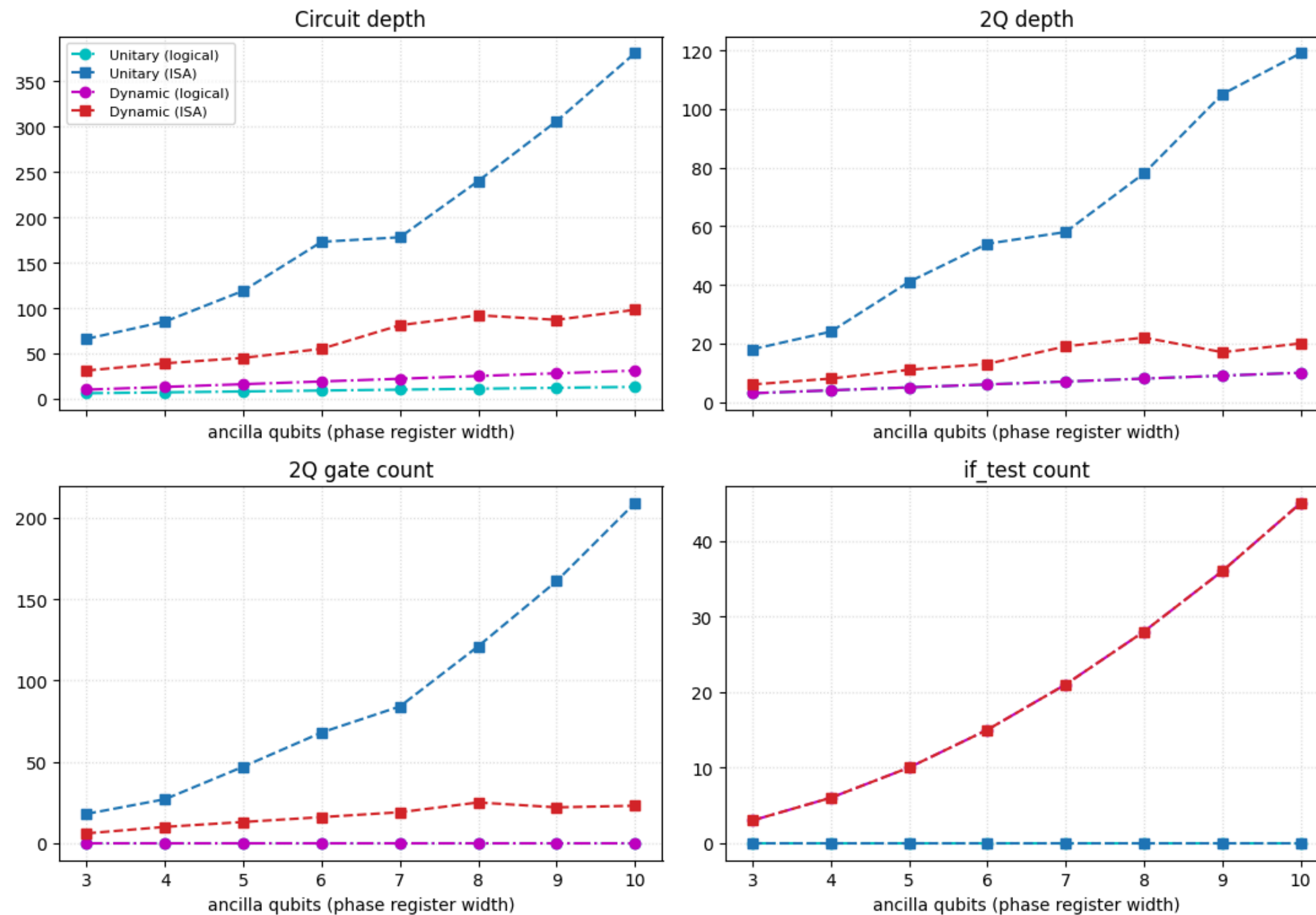
Trade off between controlled gates and classical control (same results from QFT)

Almost const. depth after transpile

Dynamic Quantum Phase Estimation (QPE)

Resource-count comparison (graph)

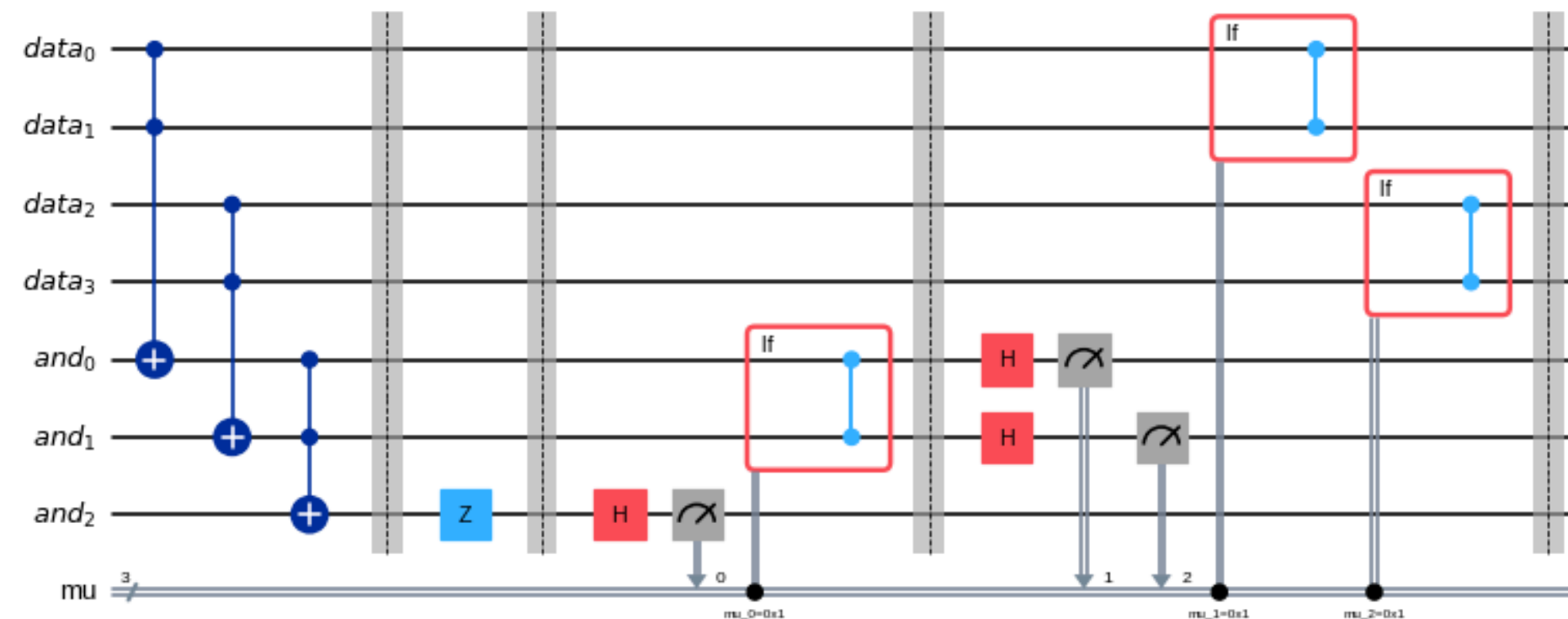
QPE resource scaling vs ancilla qubits ($\phi=0.6875$, ibm_yonsei)



Dynamic has lower circuit depth and 2Q gate than unitary one after transpile.

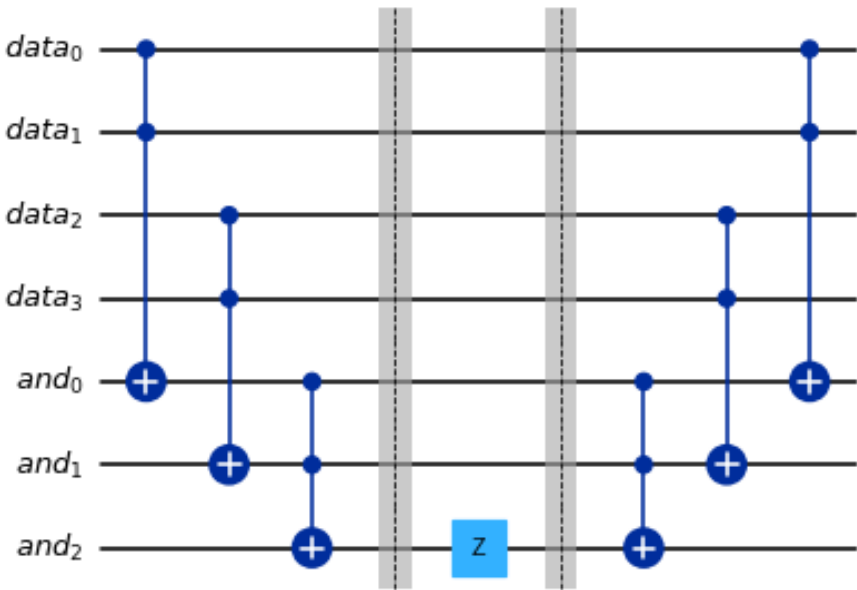
Problem 4.2

Dynamic Multi-Controlled Gates & Dynamic Grover Search
Ancilla Uncomputation using Dynamic Measurements



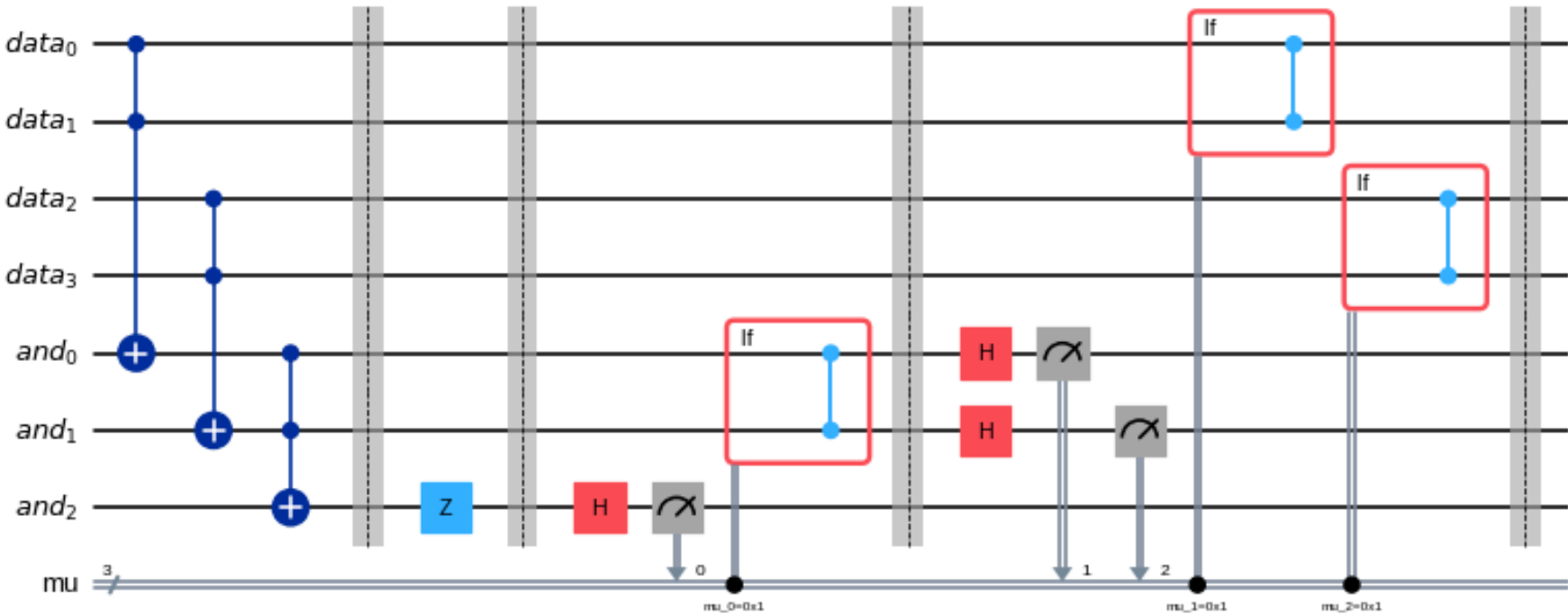
Dynamic Multi-Controlled Gates

$$|c_0, c_1, \dots, c_n\rangle |t\rangle \rightarrow |c_0, c_1, \dots, c_n\rangle |t \oplus c_0 c_1 \dots c_n\rangle$$



Unitary Implementation

$$\text{CCX}(u, v; y)^{-1} = \text{CCX}(u, v; y)$$



Dynamic Circuit

$$\text{CCX}(u, v; y)^{-1} = \text{CZ}(u, v)^m$$

Dynamic Multi-Controlled Gates

Uncomputing the intermediate ancillas of AND operations

$$\sum_{u,v} \alpha_{u,v} |u, v\rangle |uv\rangle_y$$
$$m = \begin{cases} 0, & |+\rangle \\ 1, & |-\rangle \end{cases} \quad \langle + | uv \rangle = \frac{1}{\sqrt{2}}, \quad \langle - | uv \rangle = \frac{(-1)^{uv}}{\sqrt{2}}$$

Measurement by X-basis doesn't give any information about the superposition, but leaves a unit phase instead.
We uncompute this unnecessary phase with the measurement result.

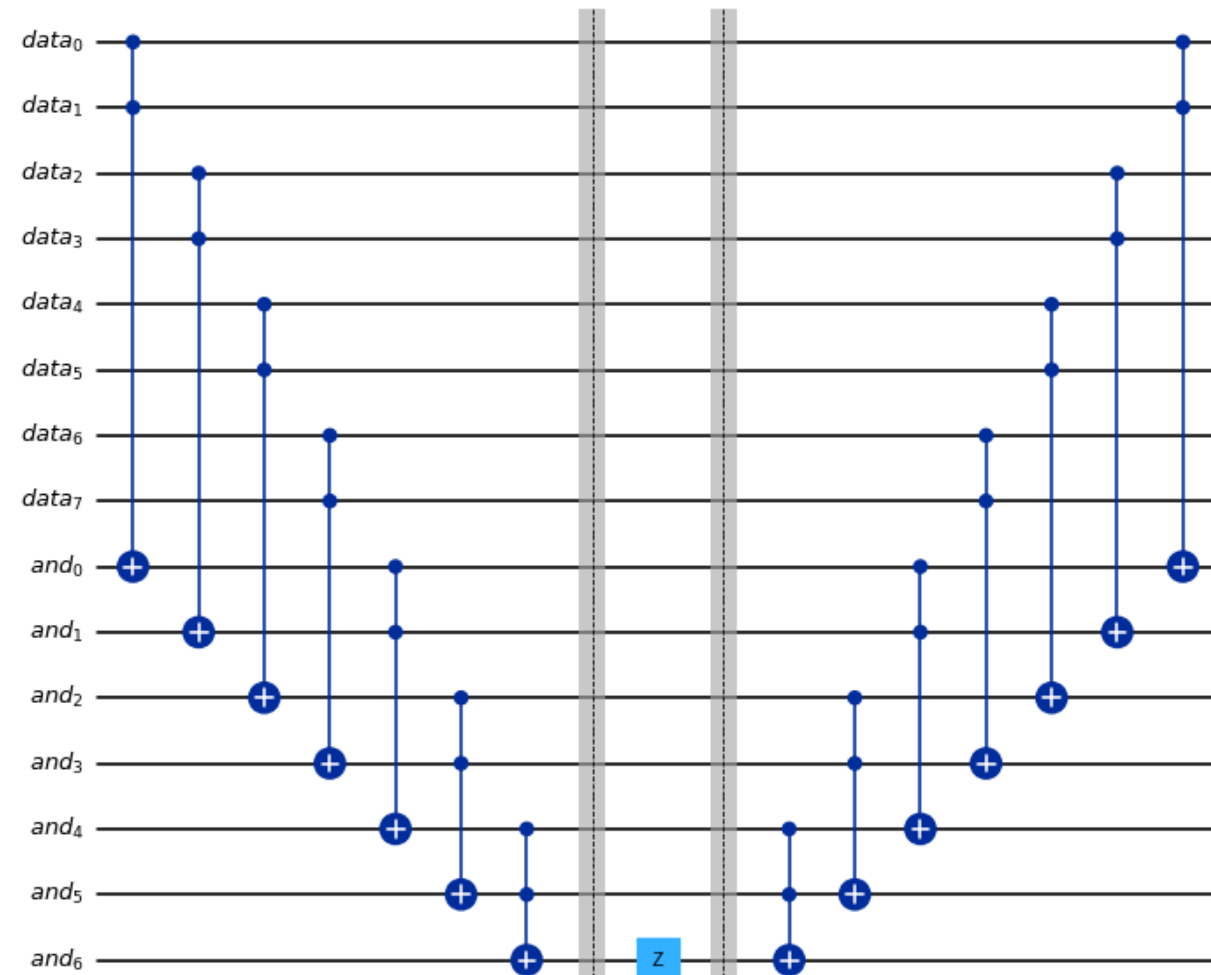
$$\sum_{u,v} \alpha_{u,v} (-1)^{muv} |u, v\rangle$$
$$\text{CCX}(u, v; y)^{-1} = \text{CZ}(u, v)^m$$

$\text{CCX}(u, v; y)^{-1} \rightsquigarrow X\text{-measure } y + \text{conditional CZ}(u, v).$
--

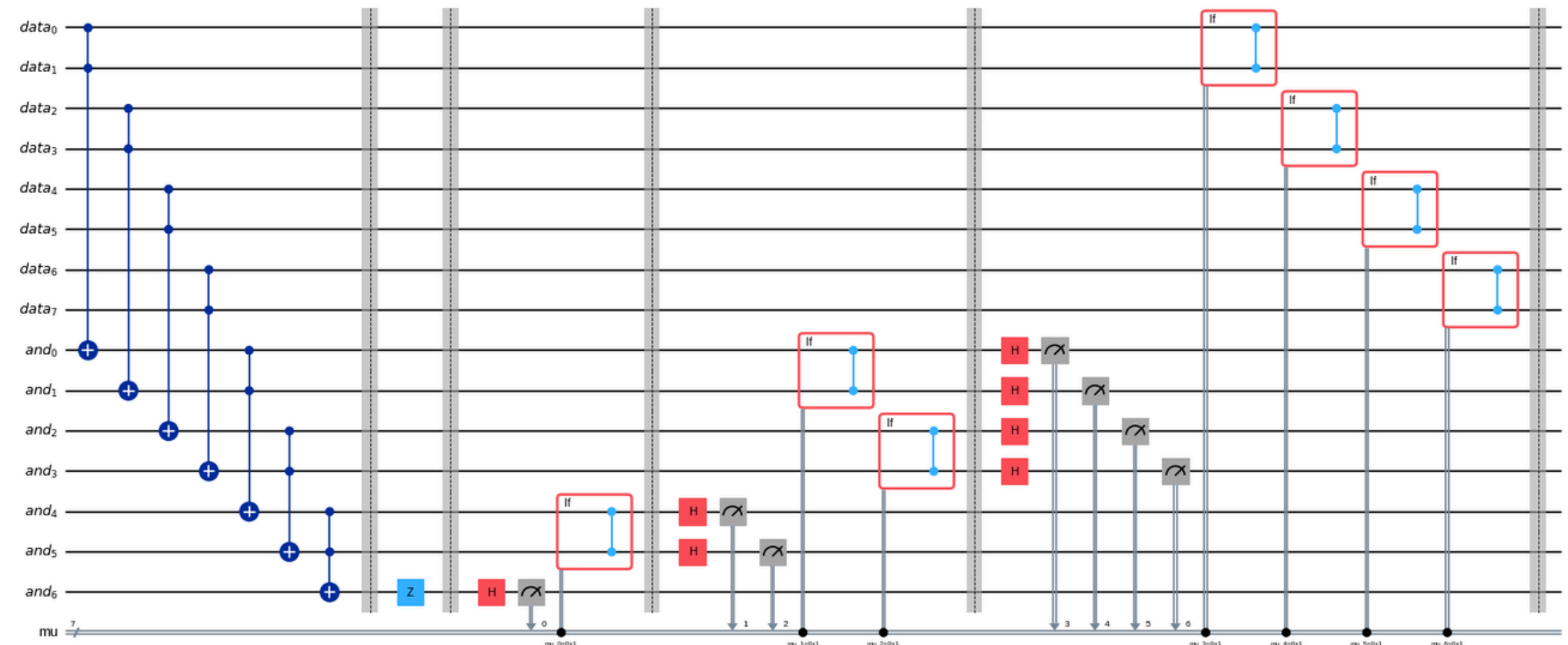
We transformed a complex CCX gate to a simple dynamic CZ gate!

Dynamic Multi-Controlled Gates

control qubits = 8



Unitary Implementation



Dynamic Circuit

This idea of register uncomputation is **powerful** and **general**, we can apply it to various kinds of circuits!!

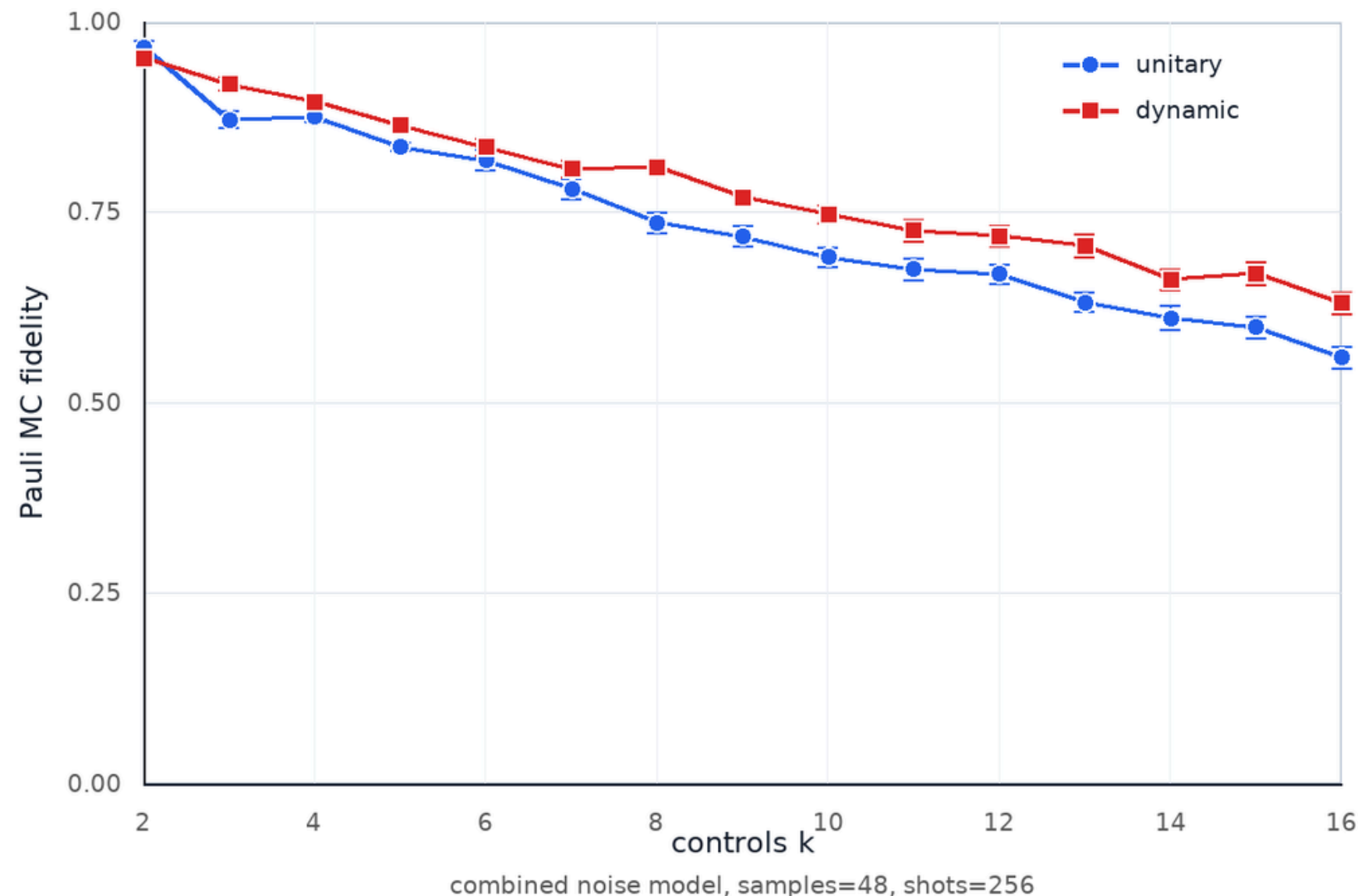
Just **measure the circuit in X-basis and resolve the leftover phase**, which is an easier problem.

Dynamic Multi-Controlled Gates

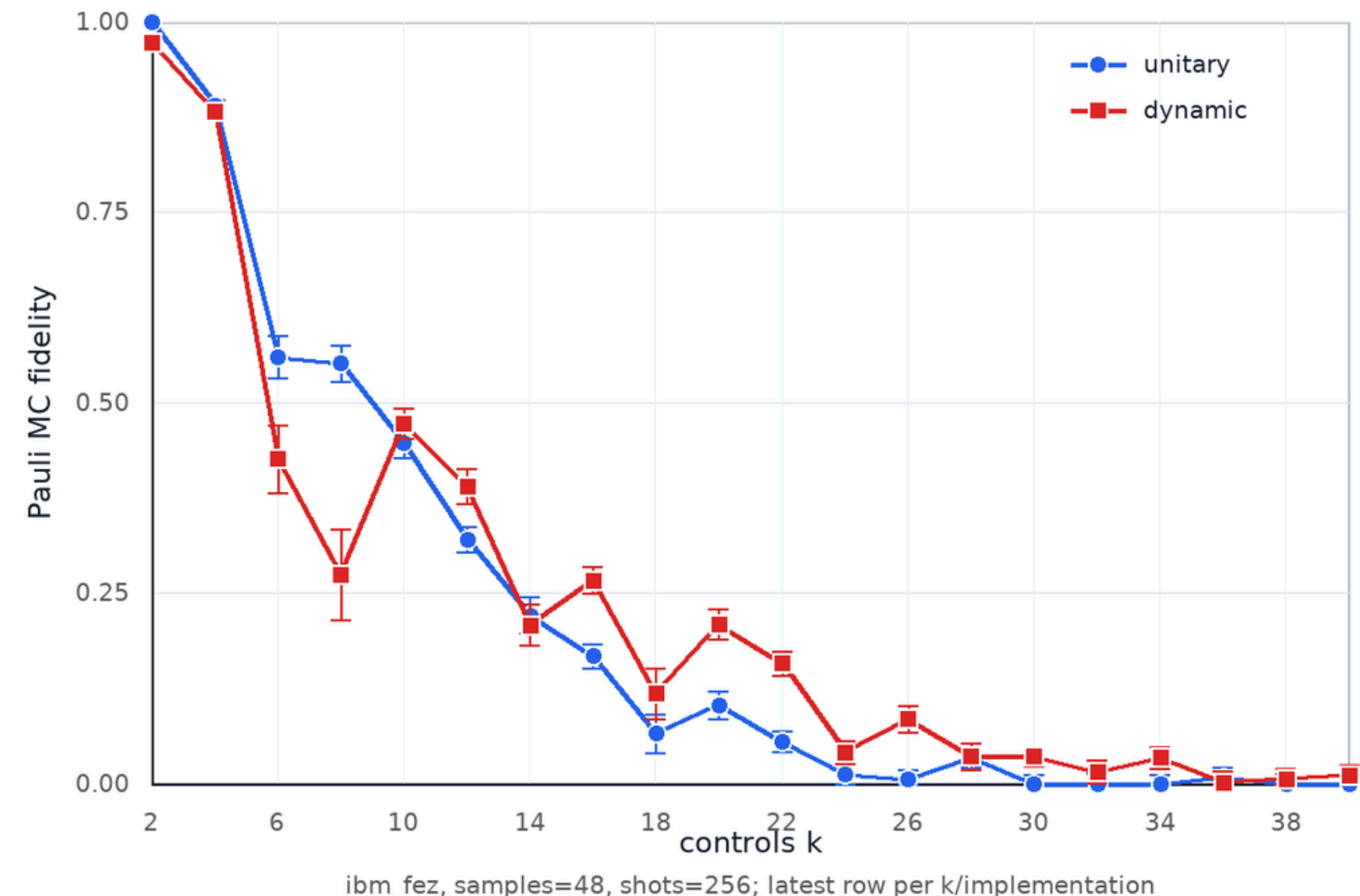
Multi-Controlled Z gate fidelity experiment on noisy model & ibm_fez hardware

MCZ Pauli fidelity comparison

Local noisy simulation



IBM Fez hardware



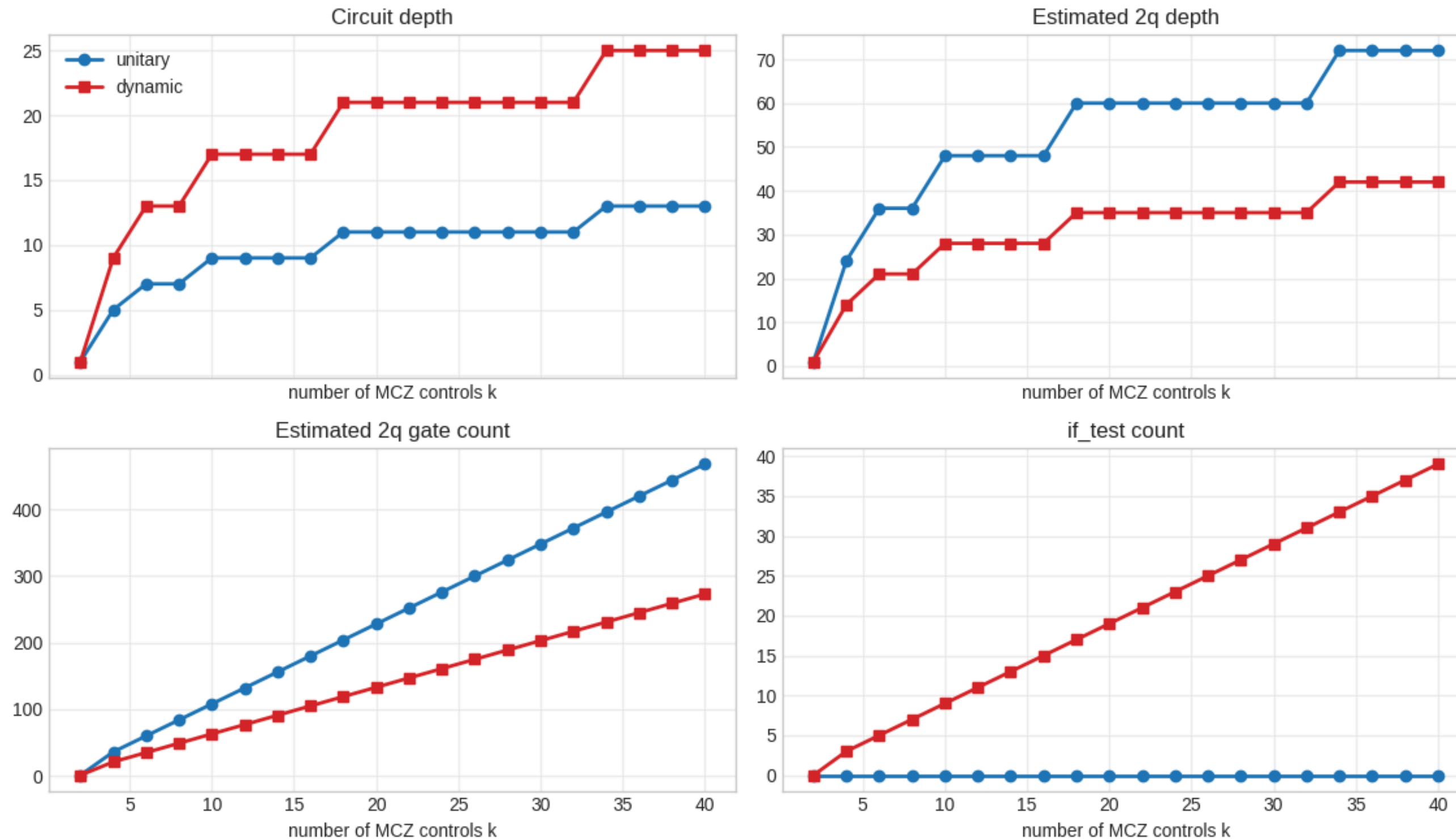
Feed the MCZ through few superposed input states, and calculate the state fidelity of the result using monte carlo pauli stabilizers, just like GHZ fidelity experiment.

Dynamic is better than Unitary!

Dynamic Multi-Controlled Gates

Resource Comparison

Unitary vs dynamic MCZ resource scaling, $k=2..40$

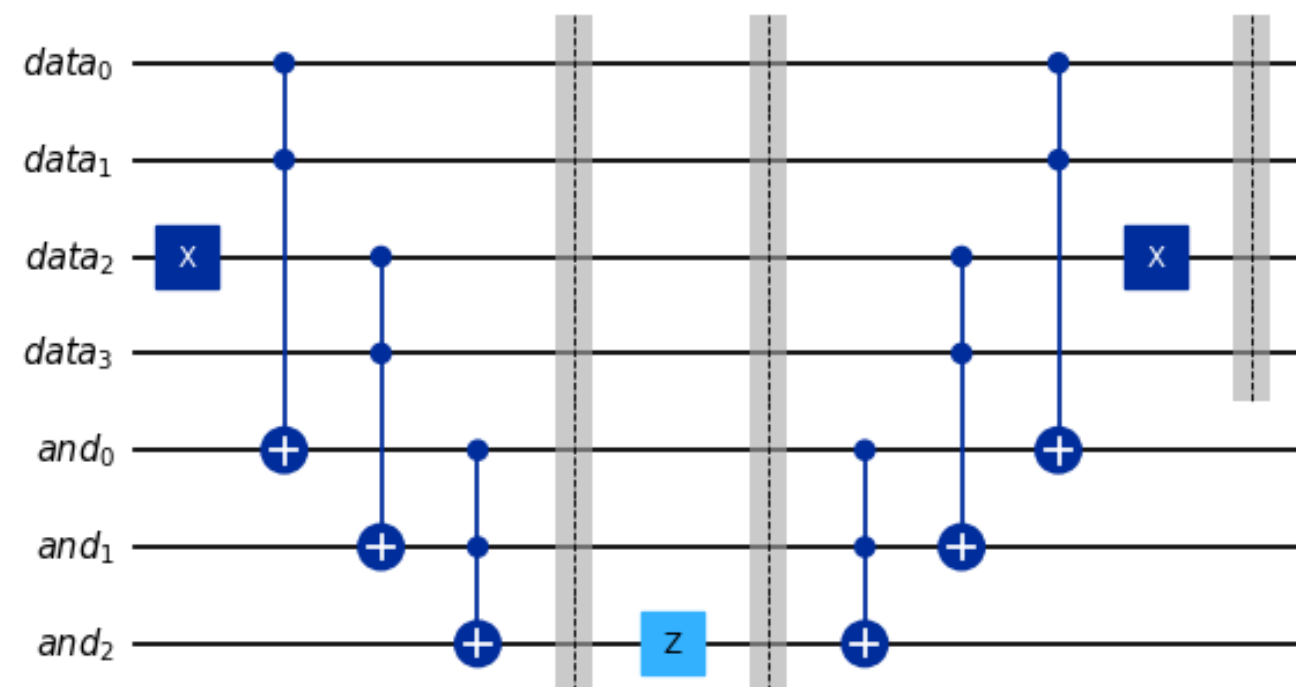
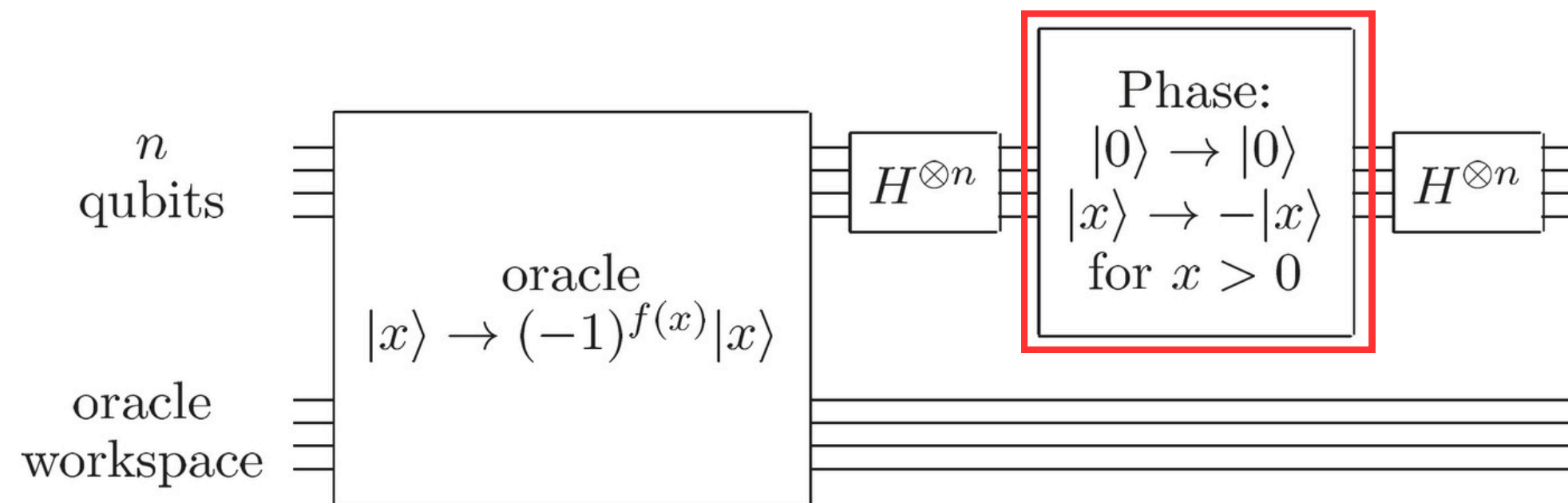


Note that circuit depth counts toffoli gates as ONE gate!

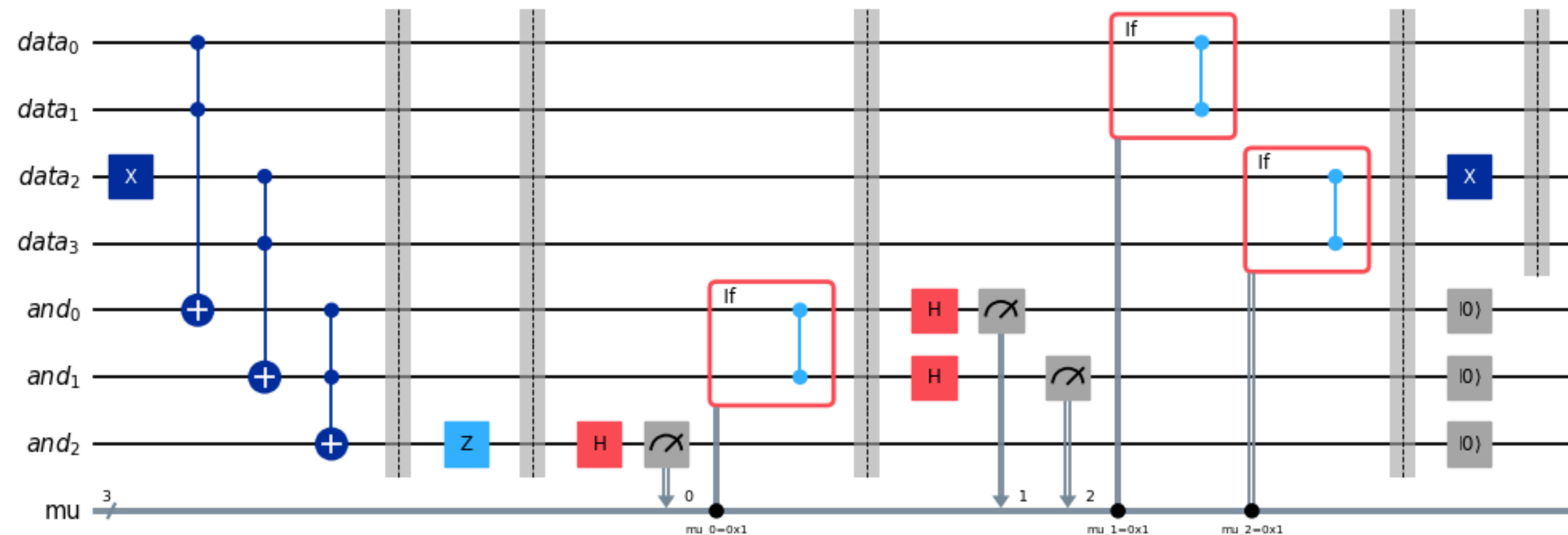
Dynamic is better than Unitary!

Dynamic Grover Search Algorithm

Motivation : MCZ is used in Grover!!



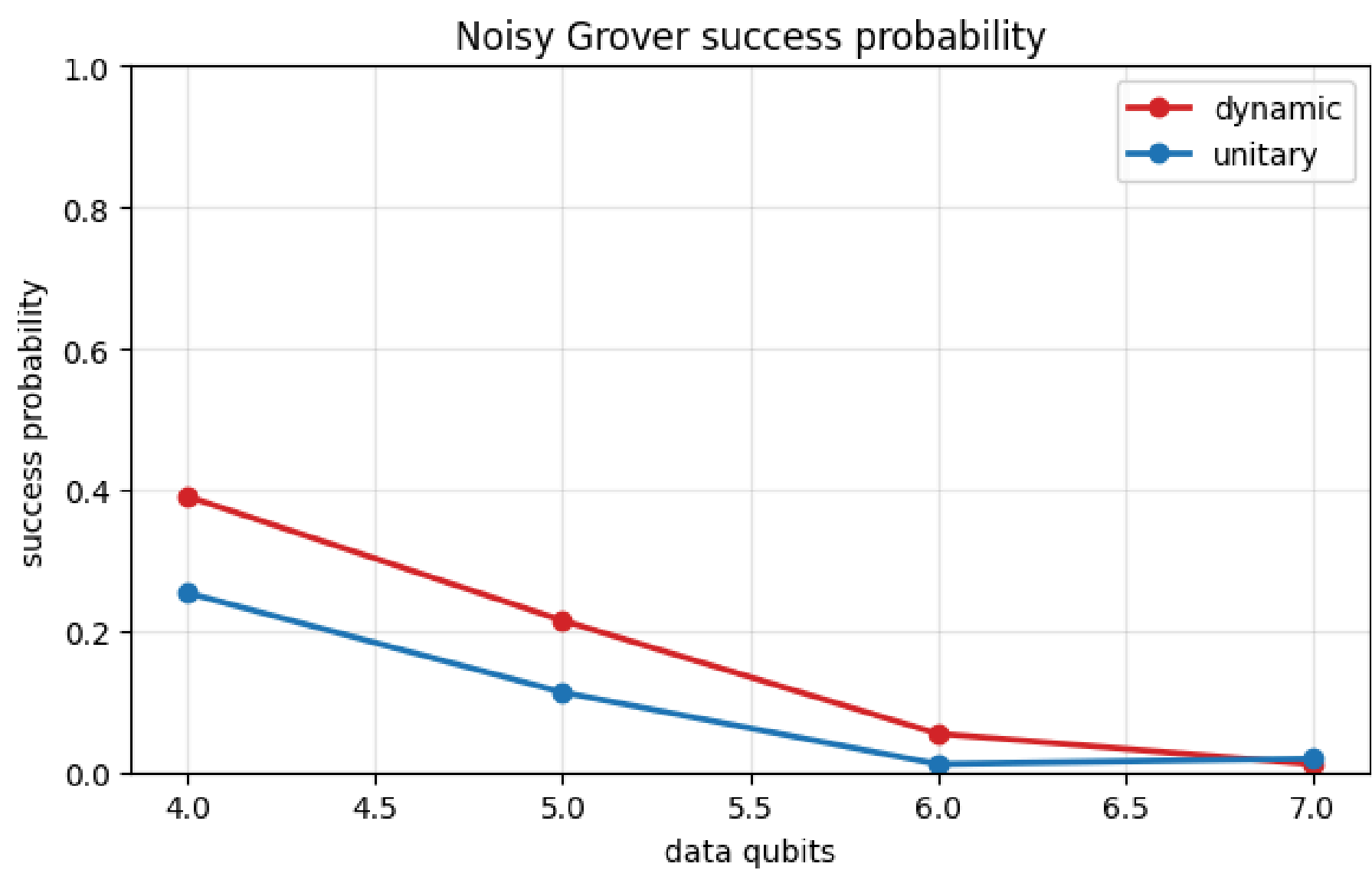
Unitary Grover Oracle



Dynamic Grover Oracle

Dynamic Grover Search Algorithm

Dynamic Grover Search on Noise Model



Dynamic is better than Unitary!

Error Source	Error Rate
One Qubit Error	0.001
Idle Error	0.001
Two Qubit Error	0.01
Reset Error	0.01
Readout Error	0.01

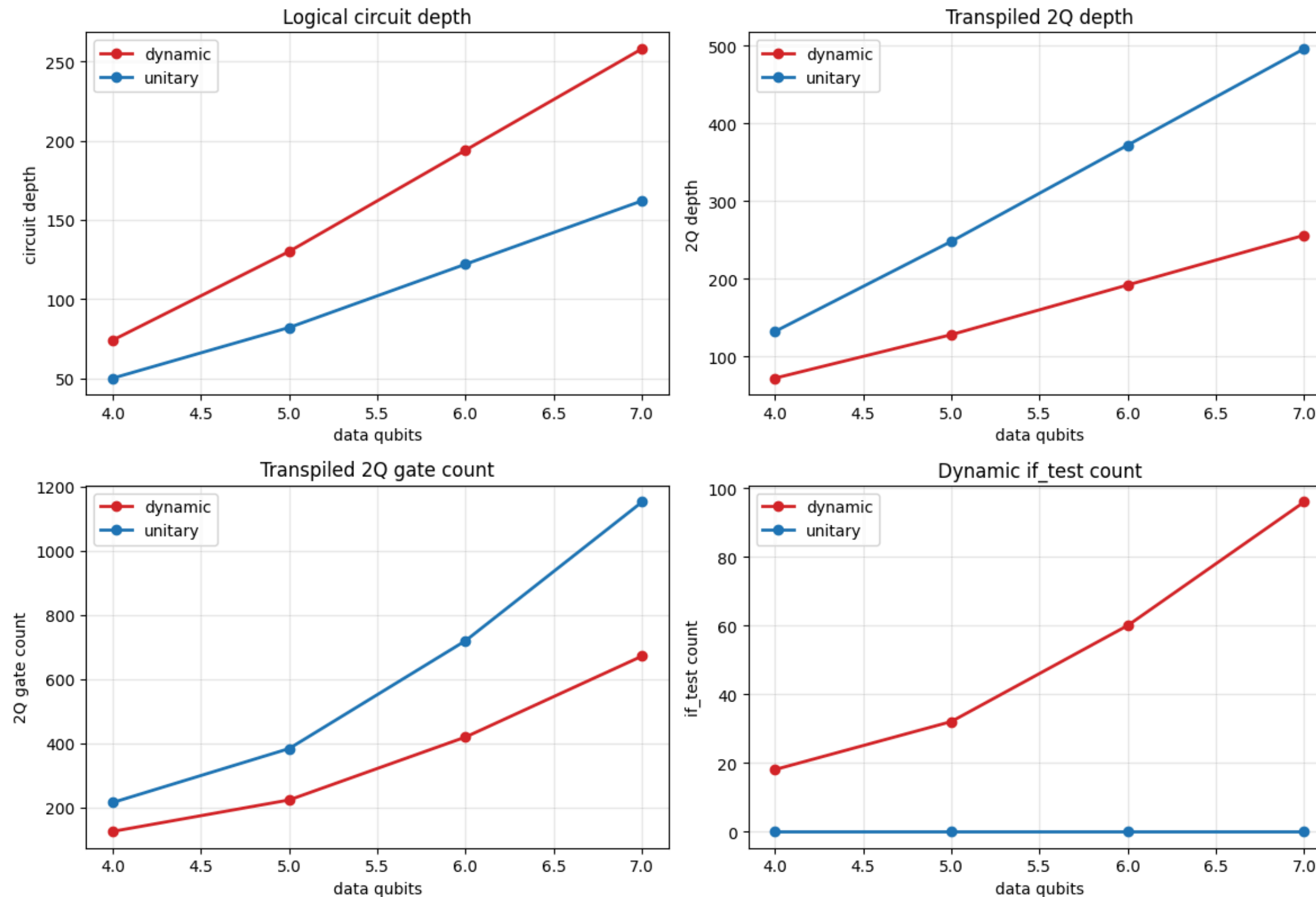
Target bitstring: 1010...1010

$$r_{\text{optimal}} = \text{round} \left(\frac{\pi}{4\theta} - \frac{1}{2} \right), \quad \text{where} \quad \theta = \arcsin \sqrt{\frac{M}{N}}$$
$$\approx \left\lfloor \frac{\pi}{4} \sqrt{N} \right\rfloor$$

$$P_{\text{success}} = \frac{\text{target found}}{\text{total shots} = 256}$$

Dynamic Grover Search Algorithm

Resource Comparison

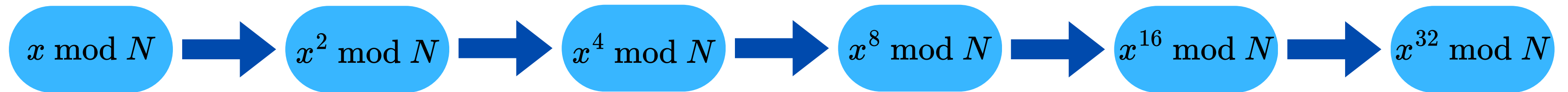


Note that circuit depth counts toffoli gates as ONE gate!

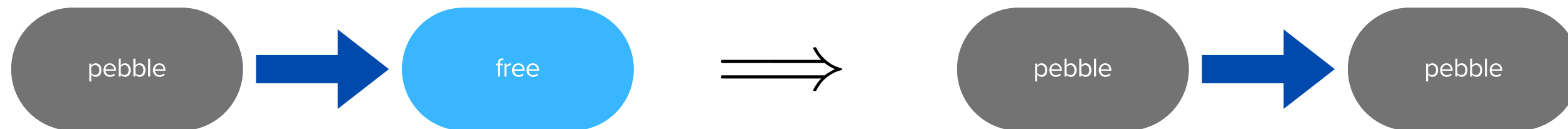
Dynamic is better than Unitary!

Ancilla Reduction using Spooky Pebbling

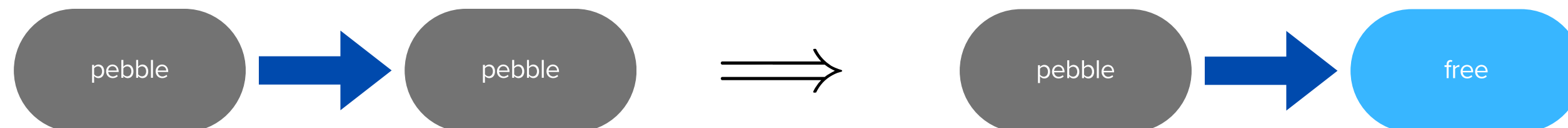
Generalization : We want to save ancilla registers on a irreversible computation graph.



Pebble : compute the next node from the previous node

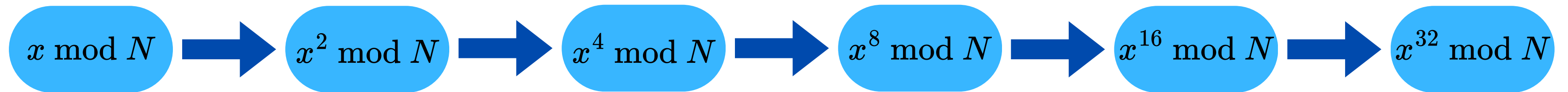


Unpebble : compute the next node from the previous node



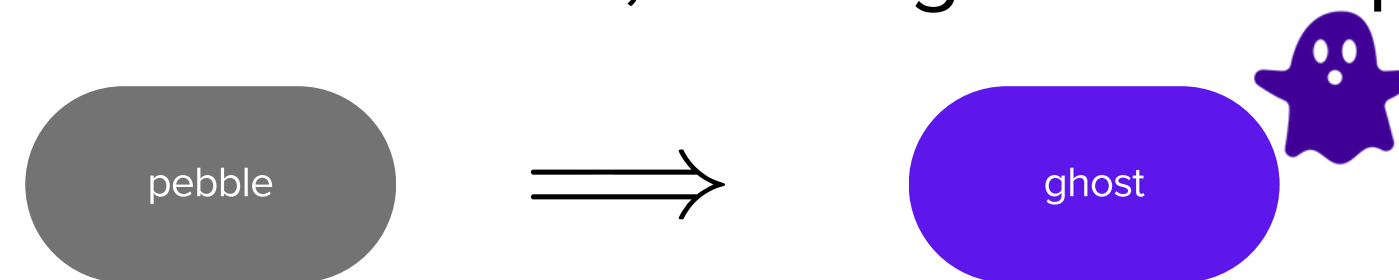
Ancilla Reduction using Spooky Pebbling

Generalization : We want to save ancilla registers on a irreversible computation graph.

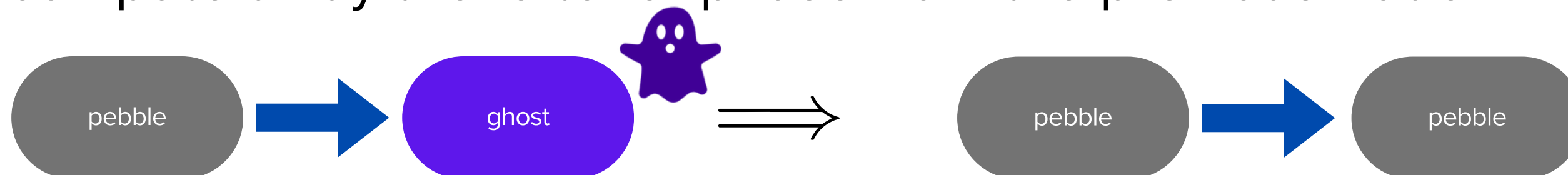


Dynamic Circuit!!

Ghost : measure away a node in X-basis, leaving behind a phase “ghost”

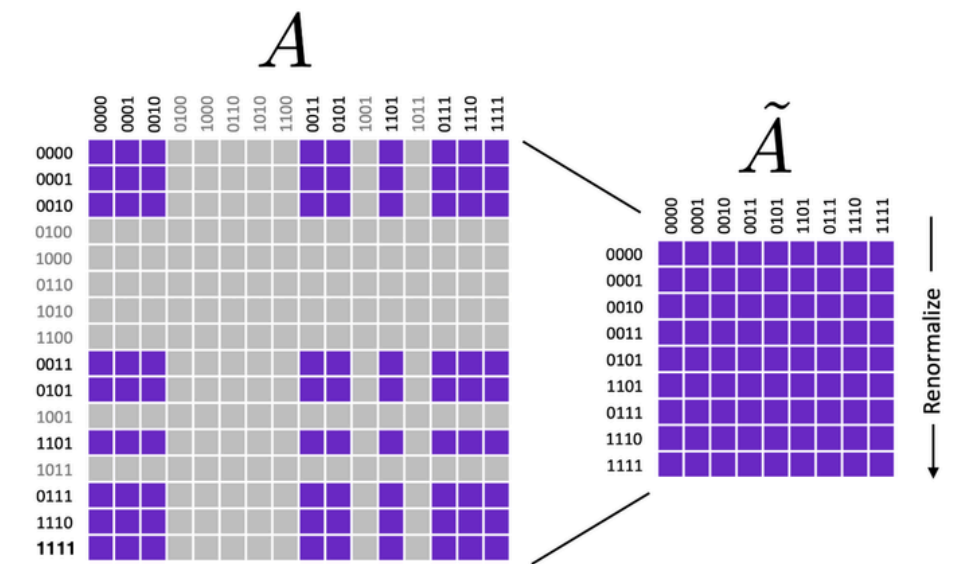


Exorcise : compute away the leftover phase from the previous node



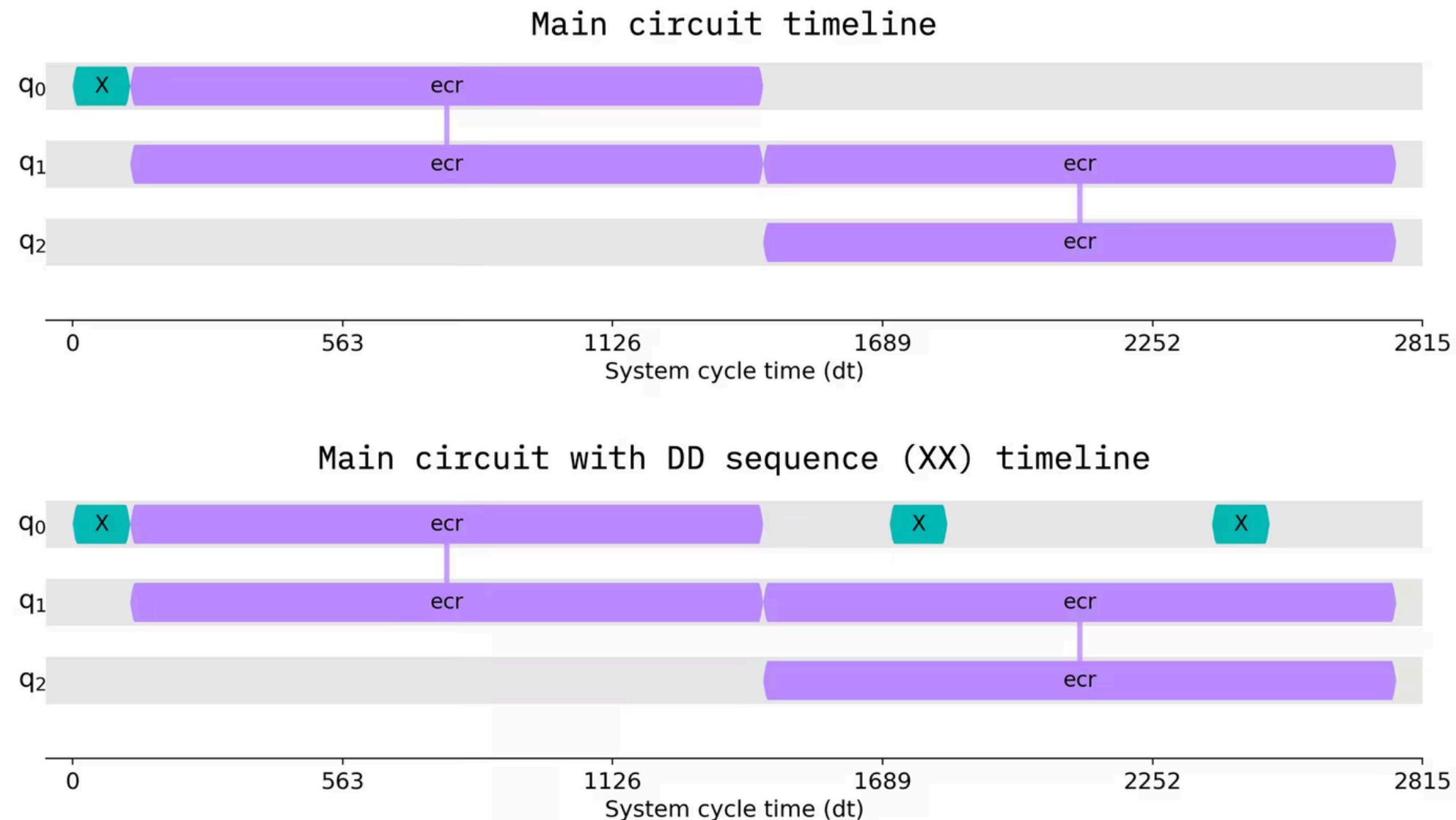
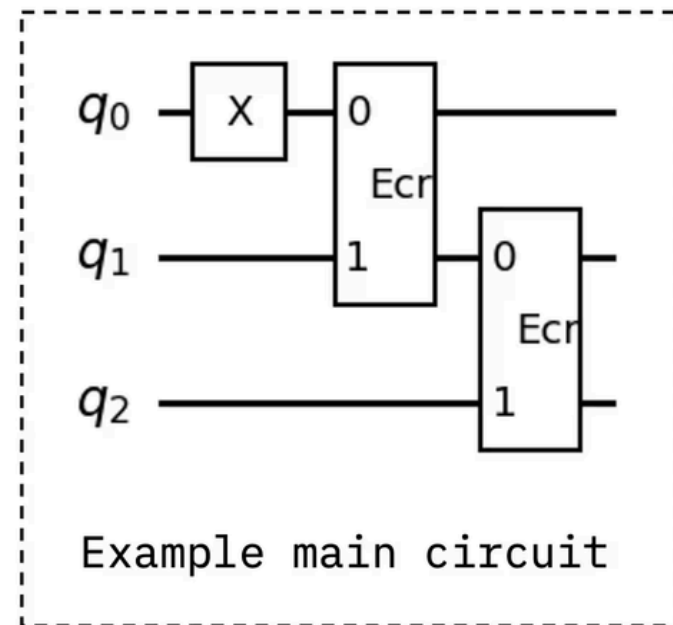
Appendix

Quantum Error Mitigation:
Dynamical Decoupling
Pauli Twirling
M3



Appendix: Quantum Error Mitigation

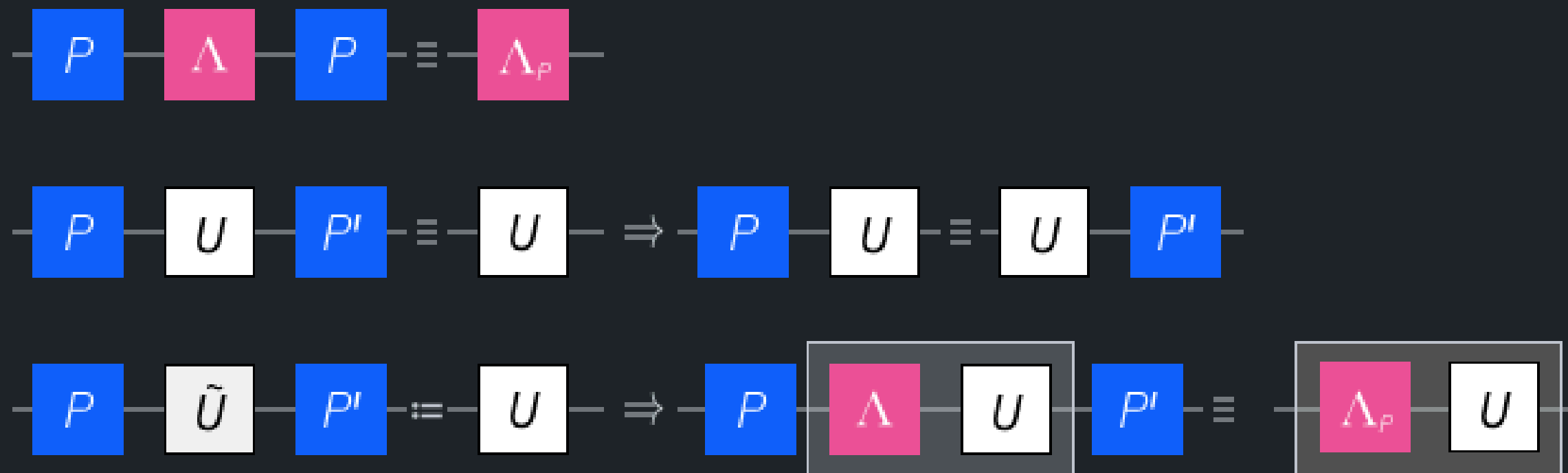
Dynamical Decoupling: Use pulse to mitigate idle error.



Appendix: Quantum Error Mitigation

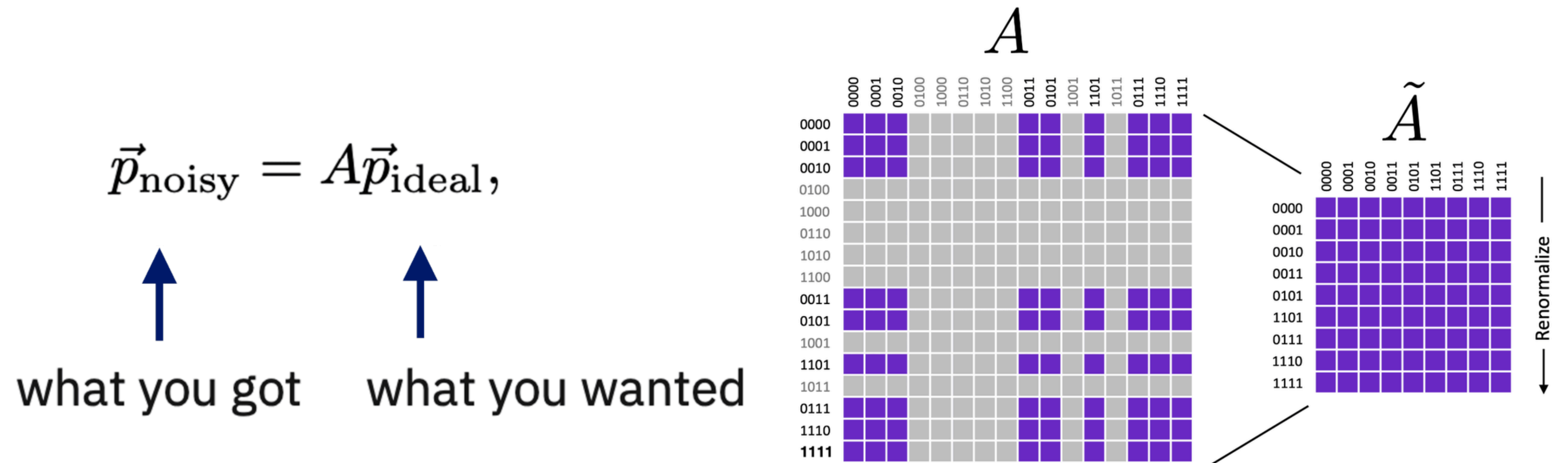
Pauli Twirling:

Conversion of coherent errors into stochastic noise,
averaging out coherent deviations through Pauli gating



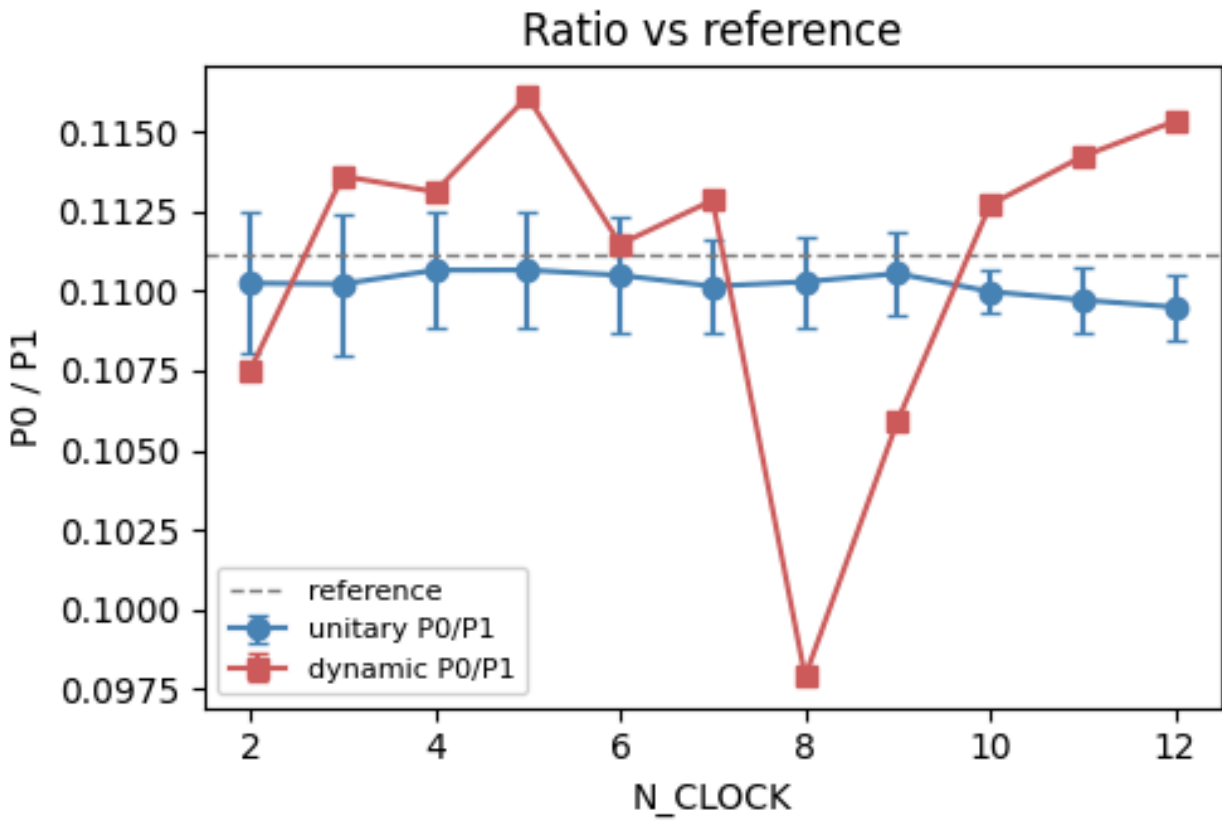
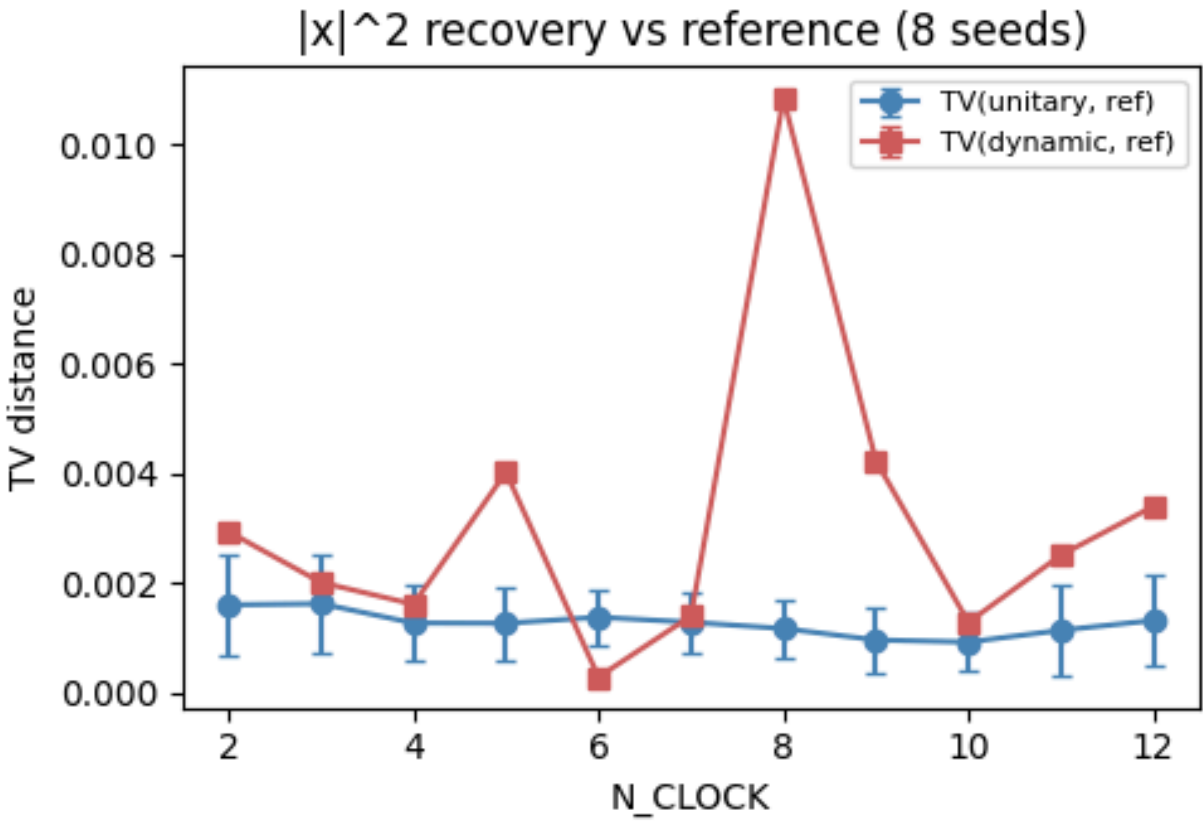
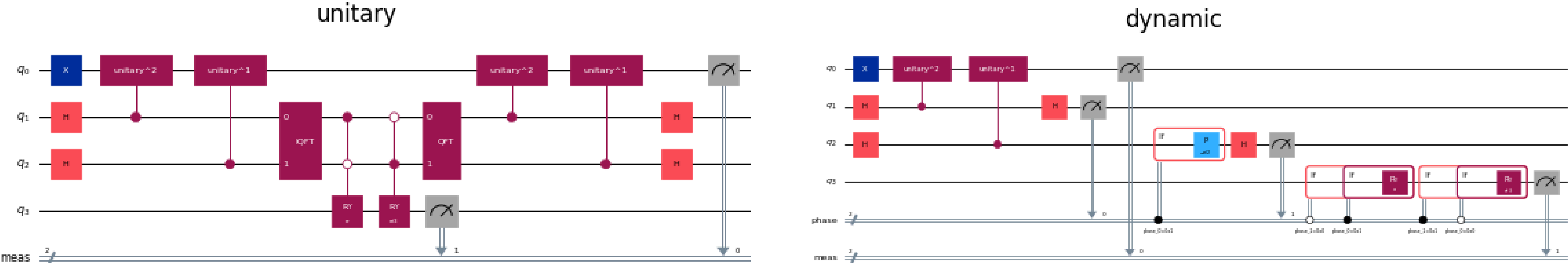
Appendix: Quantum Error Mitigation

M3: To mitigate measurement error,
solve linear systems over the most likely bitstrings, bypassing the full assignment matrix



Future Work

Dynamic Sampling HHL algorithm (unstable, experimental results)



Questions

Hyeongmin Lim



Jaemin Park



Jeongbin Jo



Junghwan Kim



Seokwon Choi





2026
QHackathon

Quantum Joa

THANK YOU

Do you have any questions? Feel free to ask.

Reference

- [1] E. Bäumer, V. Tripathi, D. S. Wang, P. Rall, E. H. Chen, S. Majumder, A. Seif, and Z. K. Minev, “Efficient Long-Range Entanglement Using Dynamic Circuits,” *PRX Quantum* **5**, 030339 (2024).
- [2] E. Bäumer, V. Tripathi, A. Seif, D. A. Lidar, and D. S. Wang, “Quantum Fourier Transform Using Dynamic Circuits,” *Physical Review Letters* **133**, 150602 (2024).
- [3] A. D. Córcoles et al., “Exploiting Dynamic Quantum Circuits in a Quantum Algorithm with Superconducting Qubits,” *Physical Review Letters* **127**, 100501 (2021).
- [4] R. Raussendorf, D. E. Browne, and H. J. Briegel, “Measurement-based Quantum Computation on Cluster States,” *Physical Review A* **68**, 022312 (2003).
- [5] G. D. Kahanamoku-Meyer, S. Ragavan, and K. Van Kirk, “Parallel Spooky Pebbling Makes Regev Factoring More Practical,” arXiv:2510.08432 [quant-ph] (2026).